Lab Assignment 7 — Multiobjective Optimization

37 algorithms · trade-offs · applications | Optimization Techniques Lab · B.Tech CE · Batch B . Roll NO : 241070044

> **In one sentence:** real problems have several goals that fight each other (cheap vs. fast, light vs. stiff), so instead of *one* best answer we compute a *set* of best compromises — the **Pareto front** — and this notebook builds 37 different ways of finding it, from scratch.

Map of the notebook

Section	What you get	Look for
1–2B	Aim, rules, schedule	–
3	Catalogue of all 37 algorithms	one table
4	The 14 test problems P1–P14 written as *min F(x)*	one table with encoding & constraint handling
Task A1	Shared tools: dominance, sorting, crowding, HV, IGD, Spacing + unit tests	`moo_utils unit tests passed`
Task A2.1–A2.37	One cell per algorithm: *docstring (steps) → unit test → run → front plot + metrics*	red dots = found front, black dots = true front
Task A3	Implementation log (HV, IGD, runtime, status per algorithm)	one big table
Task B	Master table: advantages / disadvantages / best & worst use, with numbers from your own runs	grouped by family
Task C	Deep dive on your 3 algorithms: study card, hand trace, sensitivity study	set `ROLL` first
Task D	Head-to-head: P3 & P5 (all methods), P10 M=5, P8 TSP, plus TOPSIS/VIKOR/MGDA/GradNorm	ranked tables + front plots
Task E	Which algorithm for which problem, backed by Task D numbers	one table
Task F	Viva demos (weighted sum vs. dent, VEGA collapse, Deb's rules, elitism)	short printed answers
Export	The two zip files	download starts automatically

Plain-words glossary

Term	Meaning
Objective	A quantity to make small (we turn "maximize f" into "minimize −f").
Dominates	Solution A dominates B if A is at least as good in every objective and strictly better in one.
Pareto front	All solutions that nobody dominates — the best possible trade-offs.
Normalization	Objectives are rescaled to ≈ 0…1 (ideal → 0, nadir → 1) so metrics are comparable across problems.
HV — hypervolume	Area/volume dominated by your front up to a reference point (1.1 after normalization). **Bigger is better.**
IGD	Average distance from the *true* front to your nearest point. **Smaller is better** (convergence + spread). Only where the true front is known.
Spacing	Spread of nearest-neighbour gaps along your front. **Smaller = more evenly spread.**
Budget / seed	Every algorithm gets the same number of function evaluations; a fixed random seed makes results repeatable. Results are mean ± std over several seeds.

Conventions in this notebook

- Every code cell starts with a short comment saying what the section does; there is **no per-line commenting**.
- Algorithms are classes/functions with `initialize → evaluate → step → run`; the docstring lists the lecture steps and default parameters.
- **"unit test passed"** means a small hand-checkable rule of that algorithm was verified (e.g. Deb's rules rank a feasible point above an infeasible one).
- Files written: `results/task_a_log.csv`, `results/task_d_runs.csv` (raw numbers behind the tables).

1. Aim and Learning Outcomes

By the end you can: (1) implement all 37 algorithms and explain each step by step; (2) prove their advantages/disadvantages with **your own** plots and numbers; (3) choose an algorithm for a problem from convexity, number of objectives, variable type, constraints, preferences and budget; (4) measure front quality with HV, IGD and Spacing.

2. Logistics

Item	Used here
Mode	Individual; roll number decides the 3 deep-dive algorithms (Task C)
Language	Python 3.10+ (NumPy, Matplotlib); one language for the whole assignment
Reproducibility	Fixed seeds `SEED0 + run`; stochastic algorithms use several independent runs (mean ± std)
Fair budget	Same number of function evaluations for every algorithm (`BUDGET`; PDF suggests 10,000 → `FULL = True`)

2A. Coding rules

- Only **NumPy, Matplotlib and the standard library** are used for every algorithm, metric and baseline.
- Allowed exception: `scipy.optimize` / `linprog` solve the single-objective or LP *sub-problem* of algorithms 1–7 (the scalarization, weight/ε sweeps, deviation variables and priority logic are our own code). scikit-learn only **loads** the breast-cancer data for P9.
- Shared tools (dominance, sorting, crowding, HV, IGD, Spacing) live together in the **A1** cell (`moo_utils`) and are unit-tested on the car example and Deb's table.

2B. Suggested lab schedule

Session	Build and test	Algorithms
1	Toolkit, problems P1–P14, classical + ranking methods	1–9
2	Early evolutionary methods and NSGA-II	10–14
3	Archive-based and decomposition methods	16–20
4	Constrained method and first metaheuristics	21–25
5	Remaining metaheuristics	26–30
6	Many-objective, indicator, preference and gradient methods	15, 31–37
7	Comparison experiments, report, viva preparation	Tasks B–F

```
# ---- Configuration and imports ----
import os, time, math, itertools, warnings
import numpy as np
import matplotlib
try: get_ipython(); IN_NB = True                    # inside Jupyter/Colab figures are shown inline
except NameError: IN_NB = False
if os.environ.get("QUICK") == "1" and not IN_NB: matplotlib.use("Agg")   # headless smoke-test only
import matplotlib.pyplot as plt
from scipy.optimize import minimize, linprog      # allowed ONLY inside algorithms 1-7
from IPython.display import Markdown, display
warnings.filterwarnings("ignore")

QUICK_TEST = False                                 # True -> ~1-minute dry run (tiny budget, 2 seeds)
QUICK  = QUICK_TEST or os.environ.get("QUICK") == "1"   # smoke-test switch
FULL   = True                                     # True -> 10,000 evals and 10 seeds (slow)
BUDGET = 1200 if QUICK else 5000                   # evaluations per run (2-3 objectives)
BUDGET_MANY = 2500 if QUICK else 10000             # evaluations for many-objective runs (M >= 5)
RUNS   = 2 if QUICK else 5                         # independent runs for mean ± std
if FULL: BUDGET, BUDGET_MANY, RUNS = 10000, 10000, 10
SEED0  = 2026                                      # master seed
PEN    = 10.0                                      # penalty weight on constraint violation
plt.rcParams.update({"figure.dpi": 80, "axes.grid": True})
print(f"BUDGET={BUDGET}, BUDGET_MANY={BUDGET_MANY}, RUNS={RUNS}, SEED0={SEED0}")
```

```
BUDGET=10000, BUDGET_MANY=10000, RUNS=10, SEED0=2026
```

3. The Algorithm Catalogue (all 37)

Registry used for the implementation log (A3), the master table (B) and the deep dive (C).

```
# ---- Catalogue: (number, name, family, test problems) ----
CAT = [
 (1,"Weighted Sum","Classical",["P1","P3","P4"]), (2,"ε-Constraint","Classical",["P2","P3","P5"]),
 (3,"Benson's Method","Classical",["P1","P7"]), (4,"Weight Metric (p-norm, Tchebycheff)","Classical",["P1","P3"]),
 (5,"Value Function","Classical",["P4","P14"]), (6,"Goal Programming","Classical",["P11","P7"]),
 (7,"Lexicographic","Classical",["P11","P7"]), (8,"TOPSIS","Ranking/MCDM",["P12","P4"]),
 (9,"VIKOR","Ranking/MCDM",["P12","P4"]), (10,"VEGA","Non-elitist EA",["P2","P3"]),
 (11,"MOGA","Non-elitist EA",["P3","P5"]), (12,"NSGA","Non-elitist EA",["P5","P7"]),
 (13,"NPGA","Non-elitist EA",["P3","P5"]), (14,"NSGA-II","Elitist EA",["P3","P5","P6"]),
 (15,"NSGA-III","Elitist many-obj",["P10_3","P10_5"]), (16,"SPEA2","Elitist EA",["P4","P5"]),
 (17,"PESA-II","Elitist EA",["P3","P5"]), (18,"MOEA/D","Decomposition",["P3","P10_3"]),
 (19,"MOEA/D-DE","Decomposition",["P3","P5"]), (20,"MOEA/D-ACO","Decomposition",["P8"]),
 (21,"Constrained NSGA-II","Constrained",["P5","P6","P7"]), (22,"MOPSO","Swarm",["P4","P5"]),
 (23,"MOGWO","Swarm",["P5","P9"]), (24,"MOABC","Swarm",["P3","P6"]), (25,"MOACO","Swarm",["P8"]),
 (26,"MOSA","Physics-inspired",["P3","P5"]), (27,"MODE","Evolutionary",["P3","P5"]),
 (28,"MO-TLBO","Human-inspired",["P3","P5"]), (29,"MOBA","Swarm",["P3","P4"]),
 (30,"MOMA","Hybrid",["P5","P9"]), (31,"RVEA","Many-objective",["P10_3","P10_5"]),
 (32,"IBEA","Indicator-based",["P3","P14"]), (33,"SMS-EMOA","Indicator-based",["P3","P14"]),
 (34,"HypE","Indicator-based",["P10_3"]), (35,"R-NSGA-II","Preference-based",["P5","P14"]),
 (36,"MGDA","Gradient-based",["P13"]), (37,"GradNorm","Gradient-based",["P13"])]
```

```
NAME = {c[0]: c[1] for c in CAT}; FAMILY = {c[0]: c[2] for c in CAT}; TESTS = {c[0]: c[3] for c in CAT}

def table(rows, cols):
    """Render a markdown table (no pandas needed)."""
    s = "| " + " | ".join(cols) + " |\n|" + "---|" * len(cols) + "\n"
    for r in rows: s += "| " + " | ".join(str(v).replace("|", "/") for v in r) + " |\n"
    display(Markdown(s))
table([(c[0], c[1], c[2], ", ".join(c[3])) for c in CAT], ["#", "Algorithm", "Family", "Test problems"])
```

#	Algorithm	Family	Test problems
1	Weighted Sum	Classical	P1, P3, P4
2	ε-Constraint	Classical	P2, P3, P5
3	Benson's Method	Classical	P1, P7
4	Weight Metric (p-norm, Tchebycheff)	Classical	P1, P3
5	Value Function	Classical	P4, P14
6	Goal Programming	Classical	P11, P7
7	Lexicographic	Classical	P11, P7
8	TOPSIS	Ranking/MCDM	P12, P4
9	VIKOR	Ranking/MCDM	P12, P4
10	VEGA	Non-elitist EA	P2, P3
11	MOGA	Non-elitist EA	P3, P5
12	NSGA	Non-elitist EA	P5, P7
13	NPGA	Non-elitist EA	P3, P5
14	NSGA-II	Elitist EA	P3, P5, P6
15	NSGA-III	Elitist many-obj	P10_3, P10_5
16	SPEA2	Elitist EA	P4, P5
17	PESA-II	Elitist EA	P3, P5
18	MOEA/D	Decomposition	P3, P10_3
19	MOEA/D-DE	Decomposition	P3, P5
20	MOEA/D-ACO	Decomposition	P8
21	Constrained NSGA-II	Constrained	P5, P6, P7
22	MOPSO	Swarm	P4, P5
23	MOGWO	Swarm	P5, P9
24	MOABC	Swarm	P3, P6
25	MOACO	Swarm	P8
26	MOSA	Physics-inspired	P3, P5
27	MODE	Evolutionary	P3, P5
28	MO-TLBO	Human-inspired	P3, P5
29	MOBA	Swarm	P3, P4
30	MOMA	Hybrid	P5, P9
31	RVEA	Many-objective	P10_3, P10_5
32	IBEA	Indicator-based	P3, P14
33	SMS-EMOA	Indicator-based	P3, P14
34	HypE	Indicator-based	P10_3
35	R-NSGA-II	Preference-based	P5, P14
36	MGDA	Gradient-based	P13
37	GradNorm	Gradient-based	P13

4. Problem Bank (P1–P14) — formulations

All problems are written as **min F(x) s.t. constraints**; maximizations use Max f ≡ Min −f. Constraint violation CV ≥ 0 is returned by every problem. Objectives are normalized with ideal/nadir (taken from a dense reference front); HV reference point = 1.1 in normalized space. Termination: fixed evaluation budget.

P	Standard form	Encoding	Constraint handling	Notes
P1	min $(x-3)^2$, $(x-7)^2$, $x\in[1,10]$	real x	bounds	PS = [3,7]
P2	min 100/x, x^2, $x\in[1,10]$	real / integer x	bounds	every x Pareto-optimal
P3	ZDT2, n=10	real in $[0,1]^{10}$	bounds	non-convex front $f_2=1-f_1^2$
P4	min $w^T\Sigma w$, $-\mu^T w$, $\Sigma w=1$	raw $x\in[0,1]^4$, $w=x/\Sigma x$	**repair by normalization** (always feasible, no penalty tuning)	convex front
P5	min mass, deflection; $b\cdot h^2 \geq 2.4e-5$	real (b,h)	penalty (and Deb rules in alg. 21)	
P6	Binh–Korn	real (x_1,x_2)	penalty (Deb rules in alg. 21)	2 nonlinear constraints
P7	max $0.8x_1$, $1.2x_2$; $x_1+x_2\leq 90$	real	penalty	linear front
P8	bi-objective TSP, 15 cities, seed 7	permutation	always valid	distance & time
P9	min error(5-NN, 5-fold), #features	0/1 mask (30)	≥1 feature (CV=1 otherwise)	300-sample subset for speed
P10	DTLZ2, M=3 and 5, n=M+9	real	bounds	true front $\Sigma f^2=1$
P11	max profit, units; LP constraints	real ≥0	LP rows	goals / priorities
P12	5 suppliers × 4 criteria	data table	–	MCDM ranking
P13	(a) toy $\|\|x-a\|\|^2$,$\|\|x-b\|\|^2$; (b) 2-head network	real	–	gradients
P14	min weight, max runtime	real (x_1,x_2)	bounds	reference point (1.8 kg, 12 h)

5. Tasks

Task A — Build all 37 algorithms

A1. Shared toolkit (`moo_utils`)

```
# ---- moo_utils: dominance, sorting, crowding, metrics ----
def dom_matrix(F, CV=None):
    """D[i,j]=True if i dominates j; with CV given, Deb's three feasibility rules are applied."""
    D = (F[:, None, :] <= F[None, :, :]).all(2) & (F[:, None, :] < F[None, :, :]).any(2)
    if CV is not None:
        fe = CV <= 1e-12; a, b = fe[:, None], fe[None, :]
        D = np.where(a & b, D, np.where(a & ~b, True, np.where(~a & b, False, CV[:, None] < CV[None, :])))
    return D

def dominates(a, b): return bool(np.all(a <= b) and np.any(a < b))

def nds(F, CV=None):
    """Fast non-dominated sorting -> list of index arrays (front 0 is the best)."""
    D = dom_matrix(F, CV); cnt = D.sum(0).astype(int); left = np.ones(len(F), bool); fronts = []
    while left.any():
        cur = np.where(left & (cnt == 0))[0]
        if len(cur) == 0: cur = np.where(left)[0]
        fronts.append(cur); left[cur] = False; cnt = cnt - D[cur].sum(0)
    return fronts

def crowding(F):
    """Crowding distance of one front (boundary points get inf)."""
    n, m = F.shape; d = np.zeros(n)
    if n <= 2: return np.full(n, np.inf)
    for j in range(m):
        o = np.argsort(F[:, j]); r = F[o[-1], j] - F[o[0], j]; d[o[0]] = d[o[-1]] = np.inf
        if r > 0: d[o[1:-1]] += (F[o[2:], j] - F[o[:-2], j]) / r
    return d

def cdist(A, B): return np.sqrt(((A[:, None, :] - B[None, :, :]) ** 2).sum(2))

def nd2d(F):
    """Non-dominated subset of a 2-objective set (sort + running minimum)."""
    F = F[np.lexsort((F[:, 1], F[:, 0]))]; cm = np.minimum.accumulate(F[:, 1])
    return F[F[:, 1] < np.r_[np.inf, cm[:-1]]]

def hv2d(F, ref):
    """Exact 2-D hypervolume (minimization) by sweeping sorted points."""
    F = F[(F < ref).all(1)]
    if len(F) == 0: return 0.0
    F = F[np.argsort(F[:, 0])]; h = 0.0; best = ref[1]
    for a, b in F:
        if b < best: h += (ref[0] - a) * (best - b); best = b
    return h
```

```
def hv_mc(F, ref, n=20000, seed=0):
    """Monte-Carlo hypervolume for 3+ objectives (fixed sampling seed)."""
    F = F[(F < ref).all(1)]
    if len(F) == 0: return 0.0
    lo = F.min(0); S = np.random.default_rng(seed).uniform(lo, ref, (n, len(ref))); cov = np.zeros(n, bool)
    for i in range(0, len(F), 50): cov |= (F[i:i + 50, None, :] <= S[None]).all(2).any(0)
    return cov.mean() * np.prod(ref - lo)

def hv(F, ref): return 0.0 if len(F) == 0 else (hv2d(F, ref) if F.shape[1] == 2 else hv_mc(F, ref))
def igd(PF, F): return np.inf if len(F) == 0 else cdist(PF, F).min(1).mean()
def spacing(F):
    """Std of nearest-neighbour distances (smaller = more even)."""
    if len(F) < 3: return np.nan
    d = cdist(F, F); np.fill_diagonal(d, np.inf); return d.min(1).std()
def thin(F, n=400):
    if len(F) <= n: return F
    F = F[np.argsort(F[:, 0])]; return F[np.linspace(0, len(F) - 1, n).astype(int)]
```

```
# ---- Unit tests for moo_utils (hand-checkable) ----
# Car example: (price k$, 0-100 time s) — cheap/slow vs expensive/fast; both minimized
cars = np.array([[20, 12], [30, 8], [25, 10], [35, 9], [28, 11], [40, 6]], float)
fr = nds(cars); assert set(fr[0]) == {0, 1, 2, 5} and set(fr[1]) == {3, 4}       # (35,9) and (28,11) are dominated
assert dominates(np.array([1, 1]), np.array([2, 2])) and not dominates(np.array([1, 3]), np.array([2, 2]))
c = crowding(np.array([[0, 4], [1, 2], [2, 1], [4, 0]], float)); assert np.isinf(c[0]) and np.isinf(c[3]) and abs(c[1] - (2/4 +
assert abs(hv2d(np.array([[1, 3], [2, 2], [3, 1]], float), np.array([4, 4.])) - 6.0) < 1e-9  # staircase area = 6
assert abs(hv_mc(np.array([[0.5, 0.5, 0.5]]), np.ones(3), 40000) - 0.125) < 0.01
# Deb's three rules: feasible beats infeasible; both feasible -> dominance; both infeasible -> smaller CV
Ft = np.array([[5., 5.], [1., 1.], [2., 2.], [9., 9.]]); CVt = np.array([0, 3, 1, 0.])
D = dom_matrix(Ft, CVt)
assert D[0, 1] and not D[1, 0]        # rule 1: feasible > infeasible (even if infeasible has better F)
assert D[0, 3] and not D[3, 0]        # rule 2: both feasible -> usual dominance
assert D[2, 1] and not D[1, 2]        # rule 3: both infeasible -> smaller violation wins
assert abs(igd(np.array([[0, 0], [1, 1.]]), np.array([[0, 0.]])) - np.sqrt(2) / 2) < 1e-9
print("moo_utils unit tests passed")
```

```
moo_utils unit tests passed
```

```
# ---- Problem class and the 14 problems ----
class Problem:
    """name, n vars, bounds, m objectives, kind in {real,int,bin,perm}; f(X)->(F raw minimized, CV>=0)."""
    def __init__(s, name, n, lb, ub, m, fun, kind="real", g=None, pf=None, ideal=None, nadir=None, pref=None, lin=None):
        s.name, s.n, s.m, s.fun, s.kind, s.g, s.lin, s.pref = name, n, m, fun, kind, g, lin, pref
        s.lb = np.full(n, lb, float) if np.isscalar(lb) else np.array(lb, float)
        s.ub = np.full(n, ub, float) if np.isscalar(ub) else np.array(ub, float)
        s.pf = None if pf is None else thin(pf, 500)
        s.ideal = np.array(ideal if ideal is not None else pf.min(0), float)
        s.nadir = np.array(nadir if nadir is not None else pf.max(0), float)
        s.ref = np.full(m, 1.1); s.pf_n = None if s.pf is None else s.norm(s.pf)
    def f(s, X): return s.fun(np.atleast_2d(X))
    def norm(s, F): return (F - s.ideal) / (s.nadir - s.ideal)
    def denorm(s, Fn): return Fn * (s.nadir - s.ideal) + s.ideal
    def init(s, rng, N):
        if s.kind == "bin": return (rng.random((N, s.n)) < 0.5).astype(float)
        if s.kind == "perm": return np.array([rng.permutation(s.n) for _ in range(N)])
        return s.repair(s.lb + rng.random((N, s.n)) * (s.ub - s.lb))
    def repair(s, X):
        if s.kind == "perm": return X
        if s.kind == "bin":
            X = (X > 0.5).astype(float); z = X.sum(1) == 0; X[z, 0] = 1; return X
        X = np.clip(X, s.lb, s.ub); return np.round(X) if s.kind == "int" else X

Z = lambda X: np.zeros(len(X))
# P1 — factory speed
P1 = Problem("P1", 1, 1, 10, 2, lambda X: (np.c_[(X[:, 0] - 3) ** 2, (X[:, 0] - 7) ** 2], Z(X)),
             pf=(lambda x: np.c_[(x - 3) ** 2, (x - 7) ** 2])(np.linspace(3, 7, 500)[:, None])[:, :, 0] if False else
                np.c_[(np.linspace(3, 7, 500) - 3) ** 2, (np.linspace(3, 7, 500) - 7) ** 2])
# P2 — fleet sizing (real and integer)
_p2 = lambda X: (np.c_[100 / X[:, 0], X[:, 0] ** 2], Z(X))
_p2i = lambda X: _p2(np.round(X))
P2 = Problem("P2", 1, 1, 10, 2, _p2, pf=np.c_[100 / np.linspace(1, 10, 500), np.linspace(1, 10, 500) ** 2])
P2i = Problem("P2i", 1, 1, 10, 2, _p2i, kind="int", pf=np.c_[100 / np.arange(1, 11), np.arange(1, 11) ** 2.], ideal=[10, 1], na
# P3 — ZDT2 (n=10)
def _zdt2(X):
    f1 = X[:, 0]; g = 1 + 9 * X[:, 1:].mean(1); return np.c_[f1, g * (1 - (f1 / g) ** 2)], Z(X)
_t = np.linspace(0, 1, 500); P3 = Problem("P3", 10, 0, 1, 2, _zdt2, pf=np.c_[_t, 1 - _t ** 2])
```

```
# P4 — portfolio (repair by normalization)
MU = np.array([.08, .12, .15, .10])
SIG = np.array([[.010, .002, .001, .003], [.002, .040, .006, .004], [.001, .006, .090, .005], [.003, .004, .005, .020]])
def _p4(X):
    s = X.sum(1, keepdims=True); W = np.where(s > 1e-9, X / np.maximum(s, 1e-9), 0.25)
    return np.c_[np.einsum("ij,jk,ik->i", W, SIG, W), -(W @ MU)], Z(X)
_r = np.random.default_rng(0)
_W = np.vstack([_r.dirichlet(np.full(4, a), 100000) for a in (0.2, 1, 4)] + [np.eye(4)])
_pf4 = nd2d(_p4(_W)[0]); P4 = Problem("P4", 4, 0, 1, 2, _p4, pf=_pf4)
# P5 — cantilever beam (stress limit as constraint)
def _p5(X):
    b, h = X[:, 0], X[:, 1]; F = np.c_[7850 * b * h, 4 * 1000 / (200e9 * b * h ** 3)]
    return F, np.maximum(0, (2.4e-5 - b * h ** 2) / 2.4e-5)
_B, _H = np.meshgrid(np.linspace(.01, .1, 500), np.linspace(.02, .2, 500)); _XY = np.c_[_B.ravel(), _H.ravel()]
_F5, _C5 = _p5(_XY); _pf5 = nd2d(_F5[_C5 <= 0])
P5 = Problem("P5", 2, [.01, .02], [.1, .2], 2, _p5, g=lambda x: np.array([(x[0] * x[1] ** 2 - 2.4e-5) * 1e5]), pf=_pf5,
             pref=np.array([20.0, 5e-4]))
# P6 — Binh-Korn
def _p6(X):
    x1, x2 = X[:, 0], X[:, 1]; F = np.c_[4 * x1 ** 2 + 4 * x2 ** 2, (x1 - 5) ** 2 + (x2 - 5) ** 2]
    return F, np.maximum(0, (x1 - 5) ** 2 + x2 ** 2 - 25) + np.maximum(0, 7.7 - (x1 - 8) ** 2 - (x2 + 3) ** 2)
_A, _Bb = np.meshgrid(np.linspace(0, 5, 700), np.linspace(0, 3, 420)); _XY = np.c_[_A.ravel(), _Bb.ravel()]
_F6, _C6 = _p6(_XY); _pf6 = nd2d(_F6[_C6 <= 0])
P6 = Problem("P6", 2, [0, 0], [5, 3], 2, _p6, pf=_pf6,
             g=lambda x: np.array([25 - (x[0] - 5) ** 2 - x[1] ** 2, (x[0] - 8) ** 2 + (x[1] + 3) ** 2 - 7.7]))
# P7 — water allocation (maximize -> minimize negatives)
_p7 = lambda X: (np.c_[-0.8 * X[:, 0], -1.2 * X[:, 1]], np.maximum(0, X.sum(1) - 90) / 90)
_x1 = np.linspace(10, 70, 300); _pf7 = np.c_[-0.8 * _x1, -1.2 * (90 - _x1)]
P7 = Problem("P7", 2, [10, 20], [90, 90], 2, _p7, pf=_pf7, g=lambda x: np.array([90 - x[0] - x[1]]),
             pref=np.array([-40., -50.]), lin=dict(C=np.array([[.8, 0], [0, 1.2]]), A=np.array([[1., 1.]]), b=np.array([90.]),
# P8 — bi-objective TSP (15 cities, numpy seed 7)
np.random.seed(7)
CITY = np.random.uniform(0, 100, (15, 2)); _U = np.triu(np.random.uniform(0.5, 2.0, (15, 15)), 1); _U = _U + _U.T
DIST = cdist(CITY, CITY); TIME = DIST * _U
def _p8(X):
    X = X.astype(int); nxt = np.roll(X, -1, 1); return np.c_[DIST[X, nxt].sum(1), TIME[X, nxt].sum(1)], Z(X)
def nn_tour(C, s0=0):
    t = [s0]; left = set(range(len(C))) - {s0}
    while left: j = min(left, key=lambda k: C[t[-1], k]); t.append(j); left.remove(j)
    return np.array(t)
def two_opt(C, t):
    t = t.copy(); n = len(t); imp = True
    while imp:
        imp = False
        for i in range(1, n - 1):
            for j in range(i + 1, n):
                a, b, c, d = t[i - 1], t[i], t[j], t[(j + 1) % n]
                if C[a, c] + C[b, d] < C[a, b] + C[c, d] - 1e-12: t[i:j + 1] = t[i:j + 1][::-1].copy(); imp = True
    return t
_rg = np.random.default_rng(1)
def best_tour(C): return min((two_opt(C, _rg.permutation(15)) for _ in range(30)), key=lambda t: C[t, np.roll(t, -1)].sum())
T_D, T_T = best_tour(DIST), best_tour(TIME); _e1, _e2 = _p8(T_D[None])[0][0], _p8(T_T[None])[0][0]
P8 = Problem("P8", 15, 0, 14, 2, _p8, kind="perm", ideal=[_e1[0], _e2[1]], nadir=[_e2[0], _e1[1]])
P8.D, P8.Tm = DIST, TIME
# P9 — feature selection (binary): 5-NN + 5-fold CV coded by hand; data loaded with scikit-learn
from sklearn.datasets import load_breast_cancer
_d = load_breast_cancer(); _rs = np.random.default_rng(3); _ix = _rs.permutation(len(_d.target))[:300]
_Xd = (_d.data[_ix] - _d.data[_ix].mean(0)) / _d.data[_ix].std(0); _yd = _d.target[_ix]
_T9 = ((_Xd[:, None, :] - _Xd[None, :, :]) ** 2).astype(np.float32)          # per-feature squared differences
_fold = np.arange(300) % 5; _same = _fold[:, None] == _fold[None, :]
def knn_cv_error(mask):
    """5-fold CV error of 5-NN restricted to the selected features."""
    Dm = _T9 @ mask.astype(np.float32); Dm[_same] = np.inf                  # neighbours only from other folds
    nb = np.argpartition(Dm, 5, axis=1)[:, :5]; pred = (_yd[nb].sum(1) >= 3).astype(int)
    return 1 - (pred == _yd).mean()
def _p9(X):
    X = (X > 0.5).astype(float); k = X.sum(1); F = np.zeros((len(X), 2)); cv = (k == 0).astype(float)
    for i in range(len(X)): F[i] = (1.0, 0) if k[i] == 0 else (knn_cv_error(X[i]), k[i])
    return F, cv
P9 = Problem("P9", 30, 0, 1, 2, _p9, kind="bin", ideal=[0, 1], nadir=[0.4, 30])
BASE_ERR = knn_cv_error(np.ones(30))
# P10 — DTLZ2
def make_dtlz2(M):
    def fun(X):
        g = ((X[:, M - 1:] - .5) ** 2).sum(1); c, s_ = np.cos(X * np.pi / 2), np.sin(X * np.pi / 2); F = np.empty((len(X), M))
        for m in range(M): F[:, m] = (1 + g) * np.prod(c[:, :M - 1 - m], axis=1) * (s_[:, M - 1 - m] if m > 0 else 1)
        return F, np.zeros(len(X))
```

```
    z = np.abs(np.random.default_rng(5).normal(size=(2500, M))); z /= np.linalg.norm(z, axis=1, keepdims=True)
    return Problem(f"P10_{M}", M + 9, 0, 1, M, fun, pf=z, ideal=np.zeros(M), nadir=np.ones(M))
P10_3, P10_5 = make_dtlz2(3), make_dtlz2(5)
# P11 — production planning (max profit, units)
_p11 = lambda X: (np.c_[-(4 * X[:, 0] + 6 * X[:, 1]), -(X[:, 0] + X[:, 1])],
                  np.maximum(0, 2 * X[:, 0] + 3 * X[:, 1] - 1200) / 1200 + np.maximum(0, 4 * X[:, 0] + 2 * X[:, 1] - 1600) / 16
P11 = Problem("P11", 2, 0, 600, 2, _p11, ideal=[-2400, -500], nadir=[0, 0], pf=np.array([[-2400., -500.]]) if False else None,
              g=lambda x: np.array([1200 - 2 * x[0] - 3 * x[1], 1600 - 4 * x[0] - 2 * x[1]]),
              lin=dict(C=np.array([[4., 6.], [1., 1.]]), A=np.array([[2., 3.], [4., 2.]]), b=np.array([1200., 1600.]), lb=[0, 0
# P12 — supplier selection data (cost min, quality max, delivery min, reliability max)
P12 = dict(names=["S1", "S2", "S3", "S4", "S5"], w=np.array([.3, .3, .2, .2]), benefit=np.array([False, True, False, True]),
           M=np.array([[120, 8, 10, 92], [100, 7, 14, 88], [140, 9, 7, 95], [90, 6, 18, 80], [130, 8, 9, 90]], float))
# P14 — laptop design with a preference
_p14 = lambda X: (np.c_[.4 + .05 * X[:, 0] + .012 * X[:, 1], -(.2 * X[:, 1] / (.5 + .03 * X[:, 0]))], Z(X))
_x2 = np.linspace(40, 90, 300); P14 = Problem("P14", 2, [13, 40], [17, 90], 2, _p14, pf=_p14(np.c_[np.full(300, 13.), _x2])[0],
                                          pref=np.array([1.8, -12.]))
PROBS = {"P1": P1, "P2": P2, "P2i": P2i, "P3": P3, "P4": P4, "P5": P5, "P6": P6, "P7": P7, "P8": P8, "P9": P9,
         "P10_3": P10_3, "P10_5": P10_5, "P11": P11, "P14": P14}
```

```
# ---- Sanity checks from the PDF (verified by hand) ----
assert np.allclose(P1.f(np.array([5.]))[0], [[4, 4]]) and np.allclose(P2.f(np.array([2.]))[0], [[50, 4]])
assert np.allclose(P3.f(np.r_[.5, np.zeros(9)])[0], [[.5, .75]])
assert np.allclose(P4.f(np.array([0, 0, 1, 0.]))[0], [[.09, -.15]])
F5, C5 = P5.f(np.array([.01, .05])); assert np.allclose(F5, [[3.925, .016]]) and C5[0] == 0
assert np.allclose(P6.f(np.array([0, 0.]))[0], [[0, 50]]) and np.allclose(P6.f(np.array([5, 3.]))[0], [[136, 4]])
assert np.allclose(P7.f(np.array([10, 80.]))[0], [[-8, -96]]) and np.allclose(P14.f(np.array([13, 40.]))[0][0, 0], 1.53)
assert abs(-P14.f(np.array([13, 40.]))[0][0, 1] - 9.0) < 0.02
assert np.allclose((P10_3.pf ** 2).sum(1), 1)
rr = np.random.default_rng(0); Xr = P5.init(rr, 20000); print("P5 infeasible fraction of random designs: %.2f" % (P5.f(Xr)[1] >
print("P8 cities (seed 7):\n", CITY.round(2).tolist()); print("P9 all-ones baseline CV error: %.3f" % BASE_ERR)
print("All problem sanity checks passed")
```

```
P5 infeasible fraction of random designs: 0.03
P8 cities (seed 7):
 [[7.63, 77.99], [43.84, 72.35], [97.8, 53.85], [50.11, 7.21], [26.84, 49.99], [67.92, 80.37], [38.09, 6.59], [28.81, 90.96], [2
P9 all-ones baseline CV error: 0.027
All problem sanity checks passed
```

A2. The 37 implementations — helpers

Each algorithm cell contains: the implementation (docstring lists the lecture steps), a unit test of its key rule, and a fixed-seed run with a front plot and HV / Spacing / IGD (IGD only where the true front is known). Results go to the implementation log (A3).

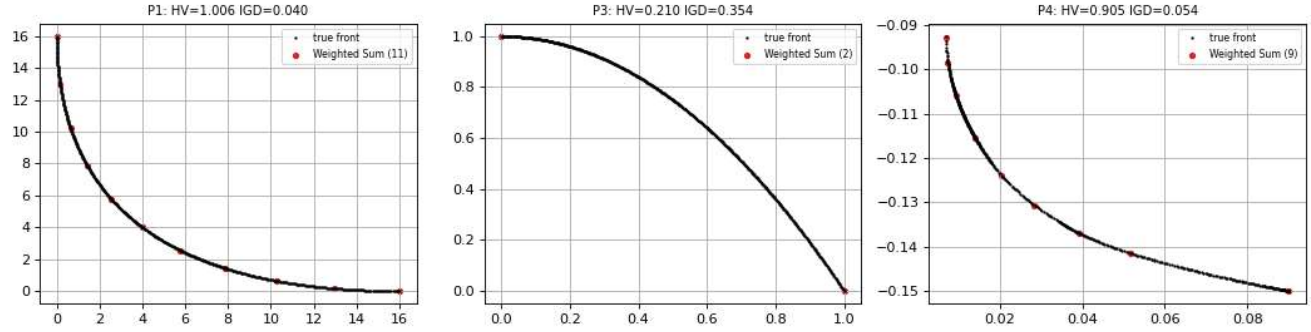
```
# ---- Common runner / logging helpers ----
LOG = []
def finalize(p, X):
    """Evaluate a decision set, keep feasible non-dominated points -> normalized objectives."""
    if X is None or len(X) == 0: return np.empty((0, p.m))
    F, CV = p.f(np.asarray(X)); Fn = p.norm(F[CV <= 1e-9])
    if len(Fn) == 0: return Fn
    Fn = Fn[nds(Fn)[0]]; return np.unique(Fn.round(12), axis=0)

def metrics(p, Fn):
    return hv(Fn, p.ref), spacing(Fn), (igd(p.pf_n, Fn) if p.pf_n is not None else np.nan)

def A2(idx, runner, probs=None, unit=None, status="works"):
    """Run the unit test, run the algorithm on its test problems with a fixed seed, plot and log metrics."""
    if unit: unit(); print(f"[{idx}] {NAME[idx]}: unit test passed")
    probs = probs or TESTS[idx]; fig, axs = plt.subplots(1, len(probs), figsize=(4.4 * len(probs), 3.5), squeeze=False)
    for ax, pn in zip(axs[0], probs):
        P = PROBS[pn]; b = BUDGET_MANY if P.m > 2 else BUDGET; t = time.time(); Fn = runner(P, SEED0, b); rt = time.time() - t
        h, sp, ig = metrics(P, Fn); R = P.denorm(Fn) if len(Fn) else Fn
        LOG.append(dict(idx=idx, name=NAME[idx], prob=pn, budget=b, hv=h, sp=sp, igd=ig, rt=rt, status=status, n=len(Fn)))
        if P.pf is not None: ax.plot(P.pf[:, 0], P.pf[:, 1], "k.", ms=2, label="true front")
        if len(R): ax.scatter(R[:, 0], R[:, 1], s=14, c="C3", label=f"{NAME[idx]} ({len(R)})")
        ax.set_title(f"{pn}: HV={h:.3f} IGD={ig:.3f}" if not np.isnan(ig) else f"{pn}: HV={h:.3f}", fontsize=9); ax.legend(font
    plt.tight_layout(); plt.show()

def cls_runner(cls, **kw):
    """Runner for population-based algorithms (defined later)."""
    return lambda P, seed, b: cls(P, budget=b, seed=seed, **kw).run().result()
```

A2.1 Weighted Sum

Scalarize `w·f` with fixed weights and sweep the weights; one solve per weight. Cannot reach non-convex (dented) front regions.

```
# ---- Shared single-objective solver for classical methods (scipy allowed here) ----
def _solve(p, fun, rng, cons=(), starts=4, x0=None, aug=0):
    """Multi-start SLSQP of fun(y); y = [x, aug extra vars]; honours p.g (>=0) and bounds."""
    n = p.n; cs = list(cons)
    if p.g: cs.append({"type": "ineq", "fun": lambda y: p.g(y[:n])})
    bnds = list(zip(p.lb, p.ub)) + [(0, 10)] * aug; best = None
    for k in range(starts):
        y0 = p.lb + rng.random(n) * (p.ub - p.lb)
        if x0 is not None and k == 0: y0 = np.asarray(x0, float)
        r = minimize(fun, np.r_[y0, np.ones(aug)], method="SLSQP", bounds=bnds, constraints=cs, options={"maxiter": 150})
        if p.f(r.x[:n])[1][0] <= 1e-6 and np.isfinite(r.fun) and (best is None or r.fun < best.fun): best = r
    return None if best is None else best.x[:n]
_fn = lambda p, x: p.norm(p.f(x)[0])[0]          # normalized objective vector of one point

def weighted_sum(p, seed=0, n_w=11):
    """Steps: 1) pick w (sum 1)  2) minimize w·f  3) repeat for n_w weights. Returns decision vectors."""
    rng = np.random.default_rng(seed); out = []
    for w in np.linspace(0, 1, n_w):
        x = _solve(p, lambda x, wv=np.r_[w, 1 - w]: wv @ _fn(p, x), rng)
        if x is not None: out.append(x)
    return np.array(out)

def ut_1():
    X = weighted_sum(P1, 0, 3); F = P1.f(X)[0]
    assert abs(F[0, 1]) < 1e-3 and abs(F[-1, 0]) < 1e-3      # w=(0,1) -> x=7 (f2=0) ; w=(1,0) -> x=3 (f1=0)
    X = weighted_sum(P3, 0, 11); Fn = finalize(P3, X)
    assert ((Fn[:, 0] < 0.05) | (Fn[:, 0] > 0.95)).all()     # ZDT2 (non-convex): only the end points are found
A2(1, lambda P, s, b: finalize(P, weighted_sum(P, s)), unit=ut_1)
```

```
[1] Weighted Sum: unit test passed
```

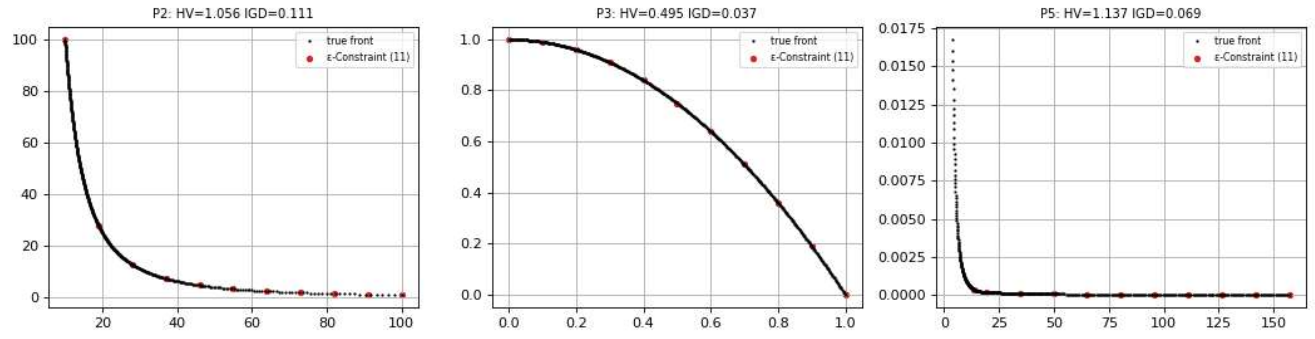


A2.2 ε-Constraint

Minimize f_2 subject to $f_1 \le \varepsilon$; sweep ε between the ideal and nadir of f_1. Reaches non-convex fronts; needs ε bounds.

```
def eps_constraint(p, seed=0, n_eps=11):
    """Steps: 1) choose eps in [ideal,nadir] of f1  2) min f2 s.t. f1<=eps  3) sweep eps."""
    rng = np.random.default_rng(seed); out = []
    for e in np.linspace(0, 1, n_eps):
        x = _solve(p, lambda x: _fn(p, x)[1], rng, cons=[{"type": "ineq", "fun": lambda x, e=e: e - _fn(p, x)[0]}])
        if x is not None: out.append(x)
    return np.array(out)

def ut_2():
    X = eps_constraint(P1, 0, 3); F = P1.f(X)[0]
    assert abs(F[0, 0]) < 1e-3                               # eps=0 forces f1 = 0 (x = 3)
    Fn = finalize(P3, eps_constraint(P3, 0, 11)); assert ((Fn[:, 0] > 0.05) & (Fn[:, 0] < 0.95)).any()   # reaches the middle c
A2(2, lambda P, s, b: finalize(P, eps_constraint(P, s)), unit=ut_2)
```

```
[2] ε-Constraint: unit test passed
```

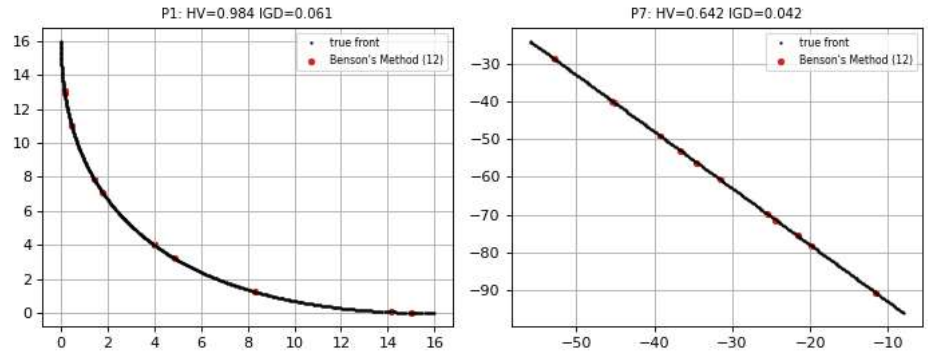


A2.3 Benson's Method

From a feasible reference point x_0 maximize the total improvement Σl_i subject to $f(x)+l = f(x_0)$, $l \geq 0 \rightarrow$ a Pareto point that dominates x_0.

```
def benson(p, seed=0, n_start=12):
    """Steps: 1) feasible x0  2) max sum(l) s.t. f(x)+l=f(x0), l>=0 (equivalently min sum f s.t. f<=f(x0))  3) repeat for new x
    rng = np.random.default_rng(seed); out = []; X0 = []
    while len(X0) < n_start:
        x = p.init(rng, 1)[0]
        if p.f(x)[1][0] <= 0: X0.append(x)
    for x0 in X0:
        F0 = _fn(p, x0)
        x = _solve(p, lambda x: _fn(p, x).sum(), rng, cons=[{"type": "ineq", "fun": lambda x, F0=F0: F0 - _fn(p, x)}], starts=2
        if x is not None: out.append(x)
    return np.array(out)

def ut_3():
    X = benson(P1, 1, 5)
    for x in X: pass
    rng = np.random.default_rng(1); x0 = np.array([1.5]); F0 = _fn(P1, x0)
    x = _solve(P1, lambda x: _fn(P1, x).sum(), rng, cons=[{"type": "ineq", "fun": lambda x: F0 - _fn(P1, x)}], x0=x0)
    assert (_fn(P1, x) <= F0 + 1e-6).all()                  # result is at least as good as x0 in every objective
A2(3, lambda P, s, b: finalize(P, benson(P, s)), unit=ut_3)
```

```
[3] Benson's Method: unit test passed
```



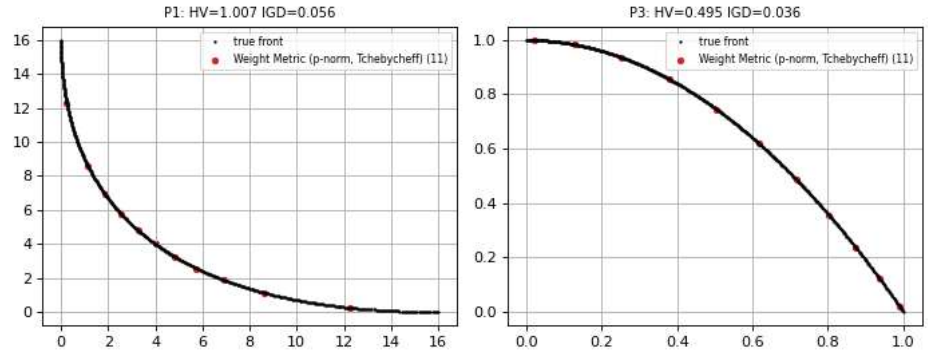
A2.4 Weight Metric (p-norm, Tchebycheff)

Minimize the weighted distance to the ideal point: p = 1, 2 or ∞ (Tchebycheff via an epigraph variable t).

```
def weight_metric(p, seed=0, pnorm=2, n_w=11):
    """Steps: 1) weights w  2) min (sum w|f-z|^p)^(1/p), or min t s.t. w f <= t for p=inf  3) sweep w."""
    rng = np.random.default_rng(seed); out = []; n = p.n
    for w in np.linspace(0.02, 0.98, n_w):
        wv = np.r_[w, 1 - w]
        if np.isinf(pnorm):
            x = _solve(p, lambda y: y[-1], rng, cons=[{"type": "ineq", "fun": lambda y, wv=wv: y[n] - wv * _fn(p, y[:n])}], aug
        else:
            x = _solve(p, lambda x, wv=wv: np.sum(wv * np.abs(_fn(p, x)) ** pnorm) ** (1 / pnorm), rng)
        if x is not None: out.append(x)
    return np.array(out)
```

```
def ut_4():
    x = weight_metric(P1, 0, np.inf, 3)[1]                    # w = 0.5: equal weights -> balanced point x = 5
    assert abs(x[0] - 5) < 0.05
    Fn = finalize(P3, weight_metric(P3, 0, np.inf, 9)); assert ((Fn[:, 0] > 0.05) & (Fn[:, 0] < 0.95)).any()   # Tchebycheff re
A2(4, lambda P, s, b: finalize(P, weight_metric(P, s, np.inf)), unit=ut_4)
```

```
[4] Weight Metric (p-norm, Tchebycheff): unit test passed
```

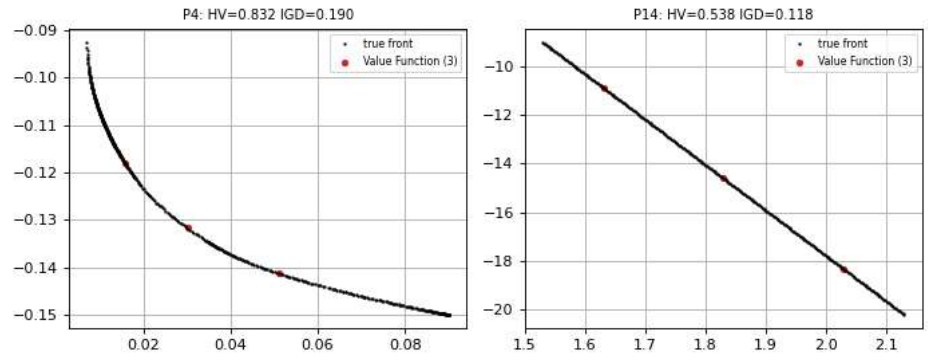


A2.5 Value Function

The decision-maker's utility U(f) is maximized directly: here U = (1.05−f_1)^a (1.05−f_2)^(1−a) on normalized objectives; `a` is the preference input.

```
def value_function(p, seed=0, alphas=(0.2, 0.5, 0.8)):
    """Steps: 1) elicit utility U(f)  2) maximize U over feasible x  3) one solution per preference a."""
    rng = np.random.default_rng(seed); out = []
    for a in alphas:
        U = lambda x, a=a: -(max(1.05 - _fn(p, x)[0], 1e-9) ** a) * (max(1.05 - _fn(p, x)[1], 1e-9) ** (1 - a))
        x = _solve(p, U, rng)
        if x is not None: out.append(x)
    return np.array(out)

def ut_5():
    F = P1.f(value_function(P1, 0, (0.2, 0.5, 0.8)))[0]
    assert F[0, 0] > F[1, 0] > F[2, 0] - 1e-9 and abs(F[1, 0] - 4) < 0.3    # larger a weights f1 more -> smaller f1; a=.5 -> x
def run_5(P, s, b):
    X = value_function(P, s); print(f"  {P.name}: preference a=(0.2,0.5,0.8) -> objectives", P.f(X)[0].round(3).tolist()); retu
A2(5, run_5, unit=ut_5)
```

```
[5] Value Function: unit test passed
  P4: preference a=(0.2,0.5,0.8) -> objectives [[0.051, -0.141], [0.03, -0.132], [0.016, -0.118]]
  P14: preference a=(0.2,0.5,0.8) -> objectives [[2.028, -18.315], [1.83, -14.606], [1.632, -10.9]]
```



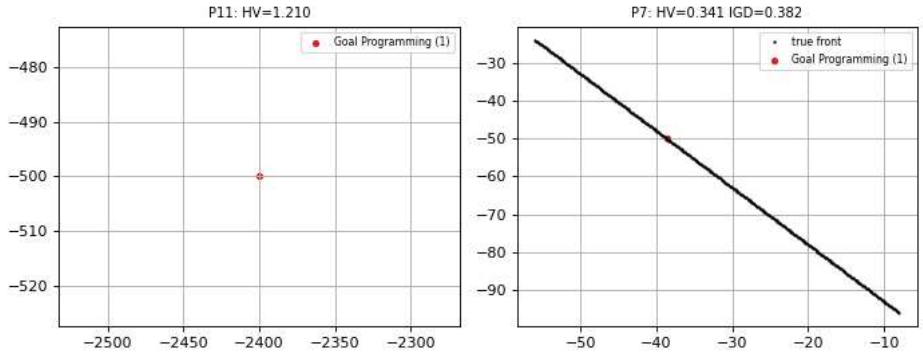
A2.6 Goal Programming

Targets T_i with deviation variables d^-, d^+: `f_i(x) + d_i^- - d_i^+ = T_i`; minimize the weighted unwanted deviations (an LP here).

```
def goal_programming(lin, targets, w, senses=(">=", ">=")):
    """Steps: 1) targets  2) add d-,d+  3) minimize sum w*d (d- for '>=' goals, d+ for '<=')  4) solve LP."""
    C, A, b = lin["C"], lin["A"], lin["b"]; m, n = C.shape
    c = np.r_[np.zeros(n), [w[i] if senses[i] in (">=", "=") else 0 for i in range(m)], [w[i] if senses[i] in ("<=", "=") else
    r = linprog(c, A_ub=np.hstack([A, np.zeros((len(A), 2 * m))]), b_ub=b, A_eq=np.hstack([C, np.eye(m), -np.eye(m)]), b_eq=tar
                bounds=list(zip(lin["lb"], lin["ub"])) + [(0, None)] * (2 * m), method="highs")
    return r.x[:n], r.x[n:n + m], r.x[n + m:]
```

```
def ut_6():
    x, dm, dp = goal_programming(P11.lin, [2400, 600], [1, 1])
    assert abs(dm[1] - 100) < 1e-6 and dm[0] < 1e-6          # units goal 600 unreachable -> shortfall exactly 100; profit met
def run_6(P, s, b):
    lin = P.lin; T = {"P11": [2400, 600], "P7": [40, 50]}[P.name]; sols = []
    for lab, w in [("unit weights", [1, 1]), ("normalized 1/T", [1 / T[0], 1 / T[1]])]:
        x, dm, dp = goal_programming(lin, T, w); sols.append(x); print(f"  {P.name} {lab}: x={x.round(2)}, shortfall d-={dm.rou
    return finalize(P, np.array(sols))
A2(6, run_6, unit=ut_6)
```

```
[6] Goal Programming: unit test passed
  P11 unit weights: x=[300. 200.], shortfall d-=[  0. 100.]
  P11 normalized 1/T: x=[300. 200.], shortfall d-=[  0. 100.]
  P7 unit weights: x=[48.33 41.67], shortfall d-=[1.33 0.  ]
  P7 normalized 1/T: x=[48.33 41.67], shortfall d-=[1.33 0.  ]
```

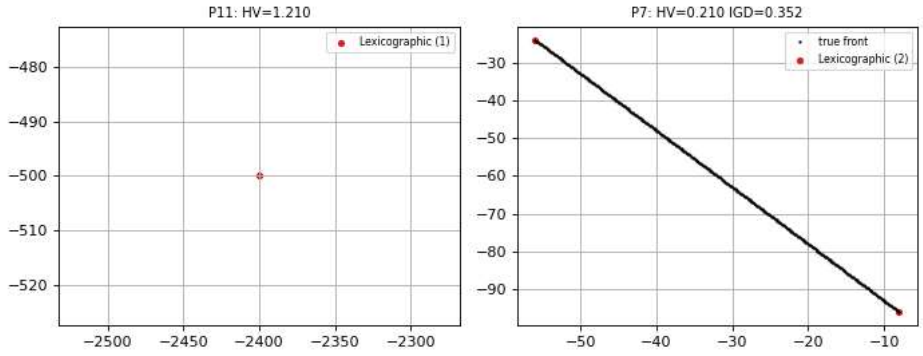


A2.7 Lexicographic

Optimize objectives in strict priority order; each stage fixes the previous optimum (with tolerance). The second stage breaks ties left by the first.

```
def lexicographic(lin, order, tol=0.0):
    """Steps: 1) optimize highest priority  2) constrain it to its optimum (tolerance tol)  3) optimize next priority."""
    C, A, b = lin["C"], lin["A"], lin["b"]; Au, bu = A.copy(), b.copy(); bnds = list(zip(lin["lb"], lin["ub"]))
    for k in order:
        r = linprog(-C[k], A_ub=Au, b_ub=bu, bounds=bnds, method="highs"); val = -r.fun
        Au = np.vstack([Au, -C[k]]); bu = np.r_[bu, -(val * (1 - tol) - 1e-7)]
    return r.x

def ut_7():
    x = lexicographic(P11.lin, [0, 1]); assert np.allclose(x, [300, 200], atol=1e-3)     # profit tie on segment broken by unit
    x1 = linprog(-P11.lin["C"][0], A_ub=P11.lin["A"], b_ub=P11.lin["b"], bounds=[(0, None)] * 2).x
    assert abs(P11.lin["C"][0] @ x1 - 2400) < 1e-6          # stage 1 alone gives 2400 but not a unique point
def run_7(P, s, b):
    sols = [lexicographic(P.lin, [0, 1]), lexicographic(P.lin, [1, 0])]
    for o, x in zip(("f1>f2", "f2>f1"), sols): print(f"  {P.name} priority {o}: x={x.round(2)}")
    return finalize(P, np.array(sols))
A2(7, run_7, unit=ut_7)
```

```
[7] Lexicographic: unit test passed
  P11 priority f1>f2: x=[300. 200.]
  P11 priority f2>f1: x=[300. 200.]
  P7 priority f1>f2: x=[70. 20.]
  P7 priority f2>f1: x=[10. 80.]
```



A2.8 TOPSIS and A2.9 VIKOR (ranking methods, not optimizers)

TOPSIS ranks by closeness to the ideal and anti-ideal; VIKOR ranks by Q = v·S* + (1−v)·R* (group utility vs. worst regret).

```
def topsis(M, w, benefit):
    """Steps: normalize columns (vector) -> weight -> ideal/anti-ideal -> distances -> closeness C = d-/(d+ + d-)."""
    V = M / np.sqrt((M ** 2).sum(0)) * w
    best = np.where(benefit, V.max(0), V.min(0)); worst = np.where(benefit, V.min(0), V.max(0))
    dp, dm = np.linalg.norm(V - best, axis=1), np.linalg.norm(V - worst, axis=1); return dm / (dp + dm + 1e-18)

def vikor(M, w, benefit, v=0.5):
    """Steps: best/worst per criterion -> S (sum), R (max) of weighted regrets -> Q = v S* + (1-v) R*; smaller Q is better."""
    fs = np.where(benefit, M.max(0), M.min(0)); fm = np.where(benefit, M.min(0), M.max(0)); r = w * (fs - M) / (fs - fm + 1e-18
    S, R = r.sum(1), r.max(1); nz = lambda a: (a - a.min()) / (a.max() - a.min() + 1e-18)
    return v * nz(S) + (1 - v) * nz(R)

def ut_8():
    M = np.array([[1., 10.], [2., 5.], [3., 1.]]); c = topsis(M, np.array([.5, .5]), np.array([False, True]))
    assert c.argmax() == 0 and ((c >= 0) & (c <= 1)).all()   # alt 0 dominates (cheapest and best) -> closeness highest
    assert vikor(M, np.array([.5, .5]), np.array([False, True])).argmin() == 0
ut_8()
Mx, w0, bf, nm = P12["M"], P12["w"], P12["benefit"], P12["names"]
Fmin = np.where(bf, -Mx, Mx); pareto = nds(Fmin)[0]; print("P12 Pareto-optimal suppliers:", [nm[i] for i in pareto])
w1 = w0 + np.array([.1, -.1, 0, 0])                          # shift one weight by 0.1
rank = lambda sc, hi: [nm[i] for i in (np.argsort(-sc) if hi else np.argsort(sc))]
rows = [("TOPSIS w", rank(topsis(Mx, w0, bf), True)), ("TOPSIS w shifted", rank(topsis(Mx, w1, bf), True))]
for v in (.25, .5, .75): rows += [(f"VIKOR v={v}", rank(vikor(Mx, w0, bf, v), False)), (f"VIKOR v={v} shifted", rank(vikor(Mx,
table([(a, " > ".join(b)) for a, b in rows], ["Method", "Ranking (best first)"])
# Apply to the reference front of P4 (risk min, return max); NSGA-II front is used again in Task D
F4 = P4.pf[::8]; M4 = np.c_[F4[:, 0], -F4[:, 1]]; c4 = topsis(M4, np.array([.5, .5]), np.array([False, True]))
print("P4 TOPSIS top-3 (risk, return):", M4[np.argsort(-c4)[:3]].round(3).tolist())
LOG += [dict(idx=8, name="TOPSIS", prob="P12/P4", budget=0, hv=np.nan, sp=np.nan, igd=np.nan, rt=0, status="works", n=5),
        dict(idx=9, name="VIKOR", prob="P12/P4", budget=0, hv=np.nan, sp=np.nan, igd=np.nan, rt=0, status="works", n=5)]
```

```
P12 Pareto-optimal suppliers: ['S1', 'S2', 'S3', 'S4', 'S5']
```

Method	Ranking (best first)
TOPSIS w	S3 > S1 > S5 > S2 > S4
TOPSIS w shifted	S1 > S2 > S3 > S5 > S4
VIKOR v=0.25	S1 > S2 > S5 > S3 > S4
VIKOR v=0.25 shifted	S2 > S1 > S4 > S5 > S3
VIKOR v=0.5	S1 > S2 > S5 > S3 > S4
VIKOR v=0.5 shifted	S2 > S1 > S3 > S5 > S4
VIKOR v=0.75	S1 > S3 > S2 > S5 > S4
VIKOR v=0.75 shifted	S1 > S2 > S3 > S5 > S4

```
P4 TOPSIS top-3 (risk, return): [[0.013, 0.113], [0.013, 0.114], [0.013, 0.114]]
```

A2 (shared base for algorithms 10–35)

Variation operators (SBX + polynomial mutation, order crossover + inversion, uniform crossover + bit-flip), the `MOEA` base class (`initialize → evaluate → step → run`), an external archive with grid density, and NSGA-II survivor selection. Constraints are handled by a penalty on normalized objectives unless the algorithm says otherwise (alg. 21 uses Deb's rules).

```
def sbx(rng, A, B, lb, ub, eta=15, pc=0.9):
    """Simulated binary crossover -> two children."""
    u = rng.random(A.shape); be = np.where(u <= .5, (2 * u) ** (1 / (eta + 1)), (1 / (2 * (1 - u + 1e-12))) ** (1 / (eta + 1)))
    c1 = .5 * ((1 + be) * A + (1 - be) * B); c2 = .5 * ((1 - be) * A + (1 + be) * B)
    sw = rng.random(A.shape) < .5; c1, c2 = np.where(sw, c2, c1), np.where(sw, c1, c2); m = rng.random((len(A), 1)) < pc
    return np.where(m, c1, A), np.where(m, c2, B)

def polymut(rng, X, lb, ub, eta=20, pm=None):
    """Polynomial mutation (per-gene probability 1/n)."""
    pm = 1 / X.shape[1] if pm is None else pm; u = rng.random(X.shape); mu = rng.random(X.shape) < pm
    d = np.where(u < .5, (2 * u) ** (1 / (eta + 1)) - 1, 1 - (2 * (1 - u)) ** (1 / (eta + 1)))
    return np.clip(X + mu * d * (ub - lb), lb, ub)

def ox(rng, a, b):
    """Order crossover for permutations."""
```

```
        n = len(a); i, j = sorted(rng.choice(n, 2, replace=False)); c = -np.ones(n, int); c[i:j + 1] = a[i:j + 1]
        seg = set(a[i:j + 1].tolist()); order = [g for g in np.r_[b[j + 1:], b[:j + 1]] if g not in seg]
        for k, g in zip(list(range(j + 1, n)) + list(range(0, i)), order): c[k] = g
        return c

    def invert(rng, x):
        i, j = sorted(rng.choice(len(x), 2, replace=False)); y = x.copy(); y[i:j + 1] = y[i:j + 1][::-1]; return y

    def das_dennis(M, p):
        """Uniform reference points on the unit simplex (C(p+M-1, M-1) points)."""
        def rec(M, p):
            if M == 1: return [[p]]
            return [[i] + r for i in range(p + 1) for r in rec(M - 1, p - i)]
        return np.array(rec(M, p), float) / p

    def tcheb(F, lam, z): return np.max(np.maximum(lam, 1e-6) * np.abs(F - z), axis=-1)

    class MOEA:
        """Base: initialize() -> evaluate() -> step() -> run(); result() = feasible non-dominated set (normalized objectives)."""
        Ndef = 100; swarm = False
        def __init__(s, prob, budget=5000, seed=0, N=None, **kw):
            s.p, s.budget, s.rng, s.N, s.evals, s.gen = prob, budget, np.random.default_rng(seed), N or s.Ndef, 0, 0
            for k, v in kw.items(): setattr(s, k, v)
            s.relax = s.swarm and prob.kind == "bin"          # swarm methods work in [0,1]^n and threshold at 0.5 for binary probl
            s.initialize()
        def evaluate(s, X):
            s.evals += len(X); F, CV = s.p.f((X > .5).astype(float) if s.relax else X); s.last_cv = CV
            return s.p.norm(F) + PEN * CV[:, None]
        def dec(s, X): return (X > .5).astype(float) if s.relax else X
        def newpop(s, n): return s.rng.random((n, s.p.n)) if s.relax else s.p.init(s.rng, n)
        def repair(s, X): return np.clip(X, 0, 1) if s.relax else s.p.repair(X)
        def cross(s, A, B):
            k = s.p.kind
            if s.relax or k in ("real", "int"): C1, C2 = sbx(s.rng, A, B, s.p.lb, s.p.ub)
            elif k == "bin": m = s.rng.random(A.shape) < .5; C1, C2 = np.where(m, A, B), np.where(m, B, A)
            else: C1 = np.array([ox(s.rng, a, b) for a, b in zip(A, B)]); C2 = np.array([ox(s.rng, b, a) for a, b in zip(A, B)])
            return s.repair(C1), s.repair(C2)
        def mutate(s, X):
            k = s.p.kind
            if s.relax or k in ("real", "int"): Y = polymut(s.rng, X, s.p.lb, s.p.ub)
            elif k == "bin": Y = np.where(s.rng.random(X.shape) < 1 / X.shape[1], 1 - X, X)
            else: Y = np.array([invert(s.rng, x) if s.rng.random() < .3 else x for x in X])
            return s.repair(Y)
        def offspring(s, P):
            """Pair consecutive parents, crossover, mutate -> same number of children."""
            n = len(P); P = np.vstack([P, P[:1]]) if n % 2 else P; C1, C2 = s.cross(P[0::2], P[1::2]); return s.mutate(np.vstack([C
        def neighbor(s, X):
            """Small random move (used by local search / annealing)."""
            k = s.p.kind
            if s.relax or k in ("real", "int"): return s.repair(X + s.rng.normal(0, .05, X.shape) * (s.p.ub - s.p.lb))
            if k == "bin":
                Y = X.copy(); j = s.rng.integers(0, X.shape[1], len(X)); Y[np.arange(len(X)), j] = 1 - Y[np.arange(len(X)), j]; ret
            return np.array([invert(s.rng, x) for x in X])
        def tourn(s, pop_n, n, better):
            i, j = s.rng.integers(0, pop_n, (2, n)); return np.where(better(i, j), i, j)
        def run(s):
            while s.evals < s.budget:
                e = s.evals; s.step(); s.gen += 1
                if s.evals == e: break
            return s
        def final(s): return s.dec(s.X)
        def result(s): return finalize(s.p, s.final())

    class Archive:
        """External archive of non-dominated points with grid-density truncation."""
        def __init__(s, cap=100, div=10, rng=None): s.cap, s.div, s.rng, s.X, s.F = cap, div, rng or np.random.default_rng(0), None
        def add(s, X, F):
            XX, FF = (X, F) if s.X is None else (np.vstack([s.X, X]), np.vstack([s.F, F]))
            _, u = np.unique(FF.round(10), axis=0, return_index=True); XX, FF = XX[u], FF[u]
            f0 = nds(FF)[0]; XX, FF = XX[f0], FF[f0]
            while len(FF) > s.cap:
                c = s.counts(FF); k = s.rng.choice(np.where(c == c.max())[0]); XX, FF = np.delete(XX, k, 0), np.delete(FF, k, 0)
            s.X, s.F = XX, FF
        def cell(s, F):
            lo, hi = F.min(0), F.max(0); cid = ((F - lo) / (hi - lo + 1e-12) * s.div).astype(int).clip(0, s.div - 1)
            return np.ravel_multi_index(cid.T, (s.div,) * F.shape[1])
        def counts(s, F=None):
```

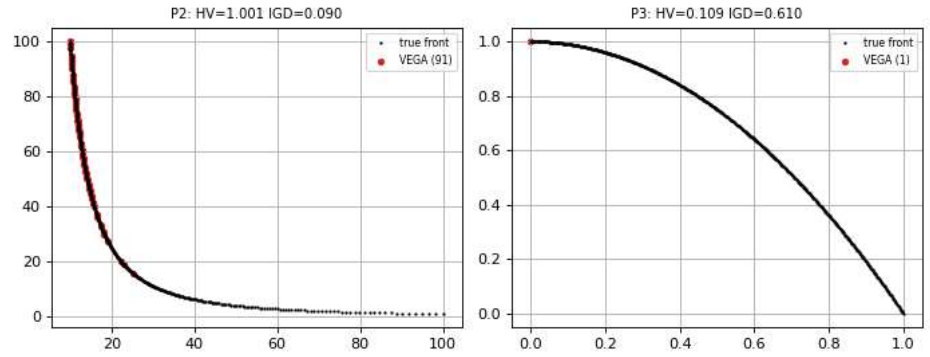
```
        F = s.F if F is None else F; key = s.cell(F); return np.bincount(key)[key]
    def pick(s, k, rng):
        """Leader selection favouring sparse grid cells (roulette on 1/count)."""
        p = 1.0 / s.counts(); return rng.choice(len(p), k, p=p / p.sum())
    def region_pick(s, k, rng):
        """PESA-II: binary tournament between regions (less crowded wins), then a random member."""
        key = s.cell(s.F); regs = np.unique(key); cnt = {r: (key == r).sum() for r in regs}; out = []
        for _ in range(k):
            a, b = rng.choice(regs, 2); r = a if cnt[a] <= cnt[b] else b; out.append(rng.choice(np.where(key == r)[0]))
        return np.array(out)

def env_select(F, CV, n):
    """NSGA-II survivor selection: fill by fronts, break the last front by crowding. Returns (selected, rank, crowd)."""
    fr = nds(F, CV); rank = np.zeros(len(F), int); crowd = np.zeros(len(F)); sel = []
    for k, f in enumerate(fr): rank[f] = k; crowd[f] = crowding(F[f])
    for f in fr:
        if len(sel) + len(f) <= n: sel += list(f)
        else: sel += list(f[np.argsort(-crowd[f])][:n - len(sel)])
        if len(sel) >= n: break
    return np.array(sel), rank, crowd
```

A2.10 VEGA

One sub-population per objective, each selected by that objective alone, then shuffled and mated. No Pareto ranking, no elitism.

```
class VEGA(MOEA):
    """Steps: 1) split into M sub-populations, sub-population m selected by tournament on f_m  2) shuffle  3) crossover+mutatic
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def subpops(s):
        M = s.F.shape[1]; out = []
        for m in range(M):
            i, j = s.rng.integers(0, s.N, (2, s.N // M)); out.append(np.where(s.F[i, m] < s.F[j, m], i, j))
        return out
    def step(s):
        pool = np.concatenate(s.subpops()); pool = np.r_[pool, s.rng.integers(0, s.N, s.N - len(pool))]
        s.X = s.offspring(s.X[s.rng.permutation(pool)]); s.F = s.evaluate(s.X)

def ut_10():
    v = VEGA(P3, budget=10, seed=1, N=60)
    for m, sub in enumerate(v.subpops()): assert v.F[sub, m].mean() < v.F[:, m].mean()    # sub-population m is biased towards
A2(10, cls_runner(VEGA), unit=ut_10)
```

```
[10] VEGA: unit test passed
```



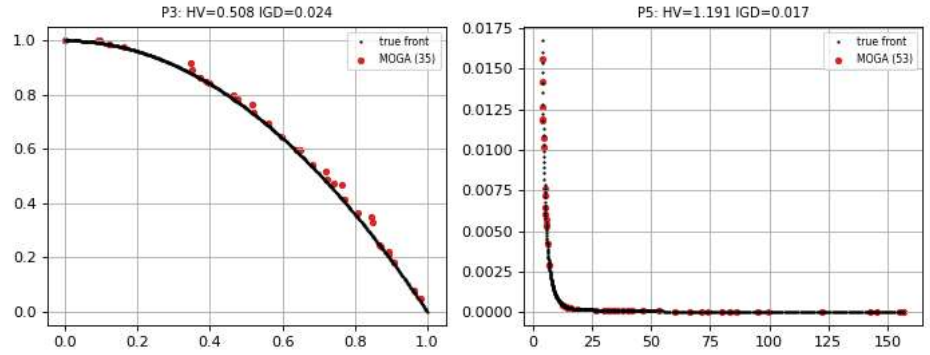
A2.11 MOGA

Rank = 1 + number of dominators; fitness interpolated from rank, averaged per rank, then divided by the niche count (fitness sharing).

```
def moga_fitness(F, sigma):
    N = len(F); rank = 1 + dom_matrix(F).sum(0); o = np.argsort(rank, kind="stable"); pos = np.empty(N); pos[o] = N - np.arange
    for r in np.unique(rank): m = rank == r; fit[m] = pos[m].mean()
    return fit / np.clip(1 - cdist(F, F) / sigma, 0, 1).sum(1), rank

class MOGA(MOEA):
    """Steps: 1) rank=1+#dominators  2) rank-based fitness averaged per rank  3) sharing  4) roulette selection  5) crossover+m
    sigma = 0.1
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def step(s):
        sf, _ = moga_fitness(s.F, s.sigma); par = s.rng.choice(s.N, s.N, p=sf / sf.sum())
```

```
        s.X = s.offspring(s.X[par]); s.F = s.evaluate(s.X)

def ut_11():
    F = np.array([[1, 5], [2, 3], [4, 1], [4, 4.], [5, 5.]]); sf, r = moga_fitness(F, 0.5)
    assert (r[:3] == 1).all() and r[3] > 1 and r[4] > r[3]          # rank = 1 + dominators
    F2 = np.array([[1, 3], [1, 3.0001], [3, 1.]]); sf2, _ = moga_fitness(F2, 1.0); assert sf2[2] > sf2[0]   # crowded twins get
A2(11, cls_runner(MOGA), unit=ut_11)
```

```
[11] MOGA: unit test passed
```

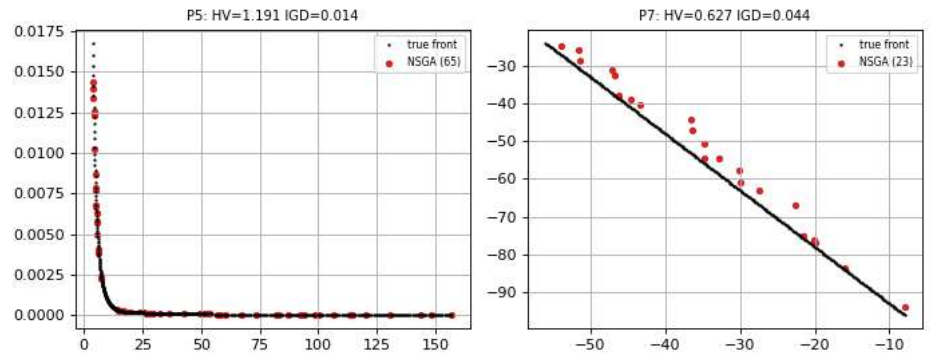


A2.12 NSGA

Non-dominated fronts get dummy fitness (front 1 highest); sharing is applied inside each front; the next front starts slightly below the previous minimum.

```
def nsga_fitness(F, sigma):
    fit = np.zeros(len(F)); cur = float(len(F))
    for f in nds(F):
        sf = cur / np.clip(1 - cdist(F[f], F[f]) / sigma, 0, 1).sum(1); fit[f] = sf; cur = sf.min() * 0.9
    return fit

class NSGA(MOEA):
    """Steps: 1) sort into fronts  2) dummy fitness per front  3) sharing within front  4) roulette  5) crossover+mutation. sig
    sigma = 0.1
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def step(s):
        fit = nsga_fitness(s.F, s.sigma); par = s.rng.choice(s.N, s.N, p=fit / fit.sum()); s.X = s.offspring(s.X[par]); s.F = s

def ut_12():
    F = np.array([[1, 4], [2, 3], [3, 2], [4, 1], [3, 4.], [4, 3.]]); fr = nds(F); fit = nsga_fitness(F, 0.3)
    assert fit[fr[0]].min() > fit[fr[1]].max()                                          # every front-1 point beats every fro
A2(12, cls_runner(NSGA), unit=ut_12)
```

```
[12] NSGA: unit test passed
```



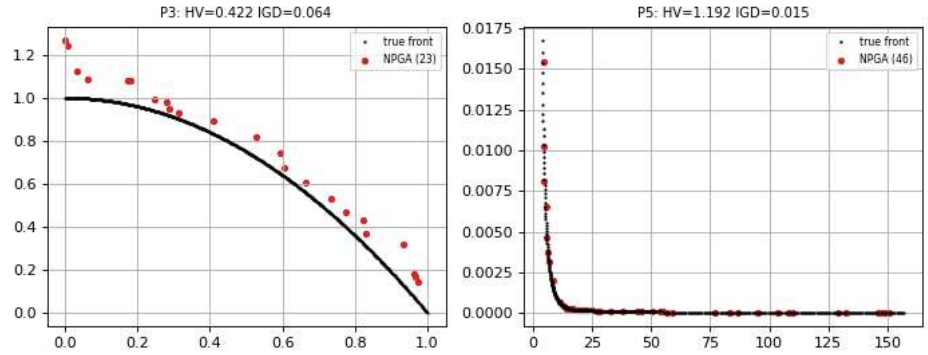
A2.13 NPGA

Pareto tournament: two candidates are compared against a random comparison set; if exactly one is non-dominated by the set it wins, otherwise the smaller niche count wins.

```
def npga_winner(a, b, Ds, niche):
    da, db = Ds[:, a].any(), Ds[:, b].any()
    return (b if da else a) if da != db else (a if niche[a] <= niche[b] else b)
```

```
class NPGA(MOEA):
    """Steps: 1) pick 2 candidates + comparison set (10% of N)  2) dominance vs set  3) tie -> niche count  4) crossover+mutati
    sigma = 0.1; tdom = 0.1
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def step(s):
        D = dom_matrix(s.F); niche = np.clip(1 - cdist(s.F, s.F) / s.sigma, 0, 1).sum(1); k = max(2, int(s.tdom * s.N)); par =
        for _ in range(s.N):
            a, b = s.rng.integers(0, s.N, 2); par.append(npga_winner(a, b, D[s.rng.integers(0, s.N, k)], niche))
        s.X = s.offspring(s.X[np.array(par)]); s.F = s.evaluate(s.X)

def ut_13():
    D = dom_matrix(np.array([[1, 1], [2, 2], [3, 3.]])); assert npga_winner(0, 2, D[[1]], np.zeros(3)) == 0   # b dominated by
    assert npga_winner(1, 2, np.zeros((1, 3), bool), np.array([0, 5, 1.])) == 2                                # tie -> smaller n
A2(13, cls_runner(NPGA), unit=ut_13)
```

```
[13] NPGA: unit test passed
```

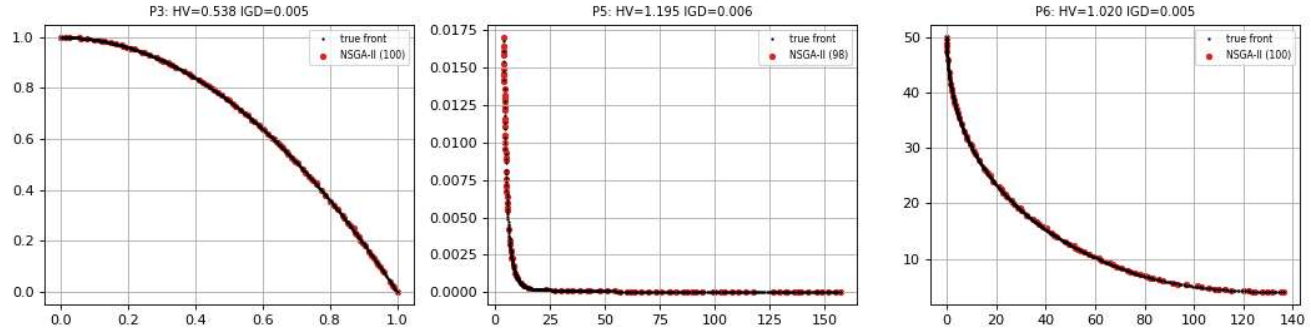


A2.14 NSGA-II

Parents + offspring are sorted into fronts (fast non-dominated sort); the last front is cut by crowding distance. Binary tournament uses (rank, crowding).

```
class NSGA2(MOEA):
    """Steps: 1) tournament on (rank, crowding)  2) crossover+mutation  3) merge parents+offspring  4) fast non-dominated sort
    use_cv = False
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.CV = s.last_cv
    def survive(s, F, CV, n): return env_select(F, CV, n)
    def step(s):
        cv = s.CV if s.use_cv else None; _, rank, crowd = s.survive(s.F, cv, len(s.F))
        par = s.tourn(len(s.X), s.N, lambda i, j: (rank[i] < rank[j]) | ((rank[i] == rank[j]) & (crowd[i] > crowd[j])))
        off = s.offspring(s.X[par]); Fo = s.evaluate(off); XU, FU, CVU = np.vstack([s.X, off]), np.vstack([s.F, Fo]), np.r_[s.C
        sel, _, _ = s.survive(FU, CVU if s.use_cv else None, s.N); s.X, s.F, s.CV = XU[sel], FU[sel], CVU[sel]

def ut_14():
    F = np.array([[0, 4], [1, 2.5], [2, 2], [2.1, 1.9], [4, 0], [5, 5.]]); sel, rank, crowd = env_select(F, None, 3)
    assert 0 in sel and 4 in sel and 5 not in sel                                          # boundary points (inf crowding) surviv
A2(14, cls_runner(NSGA2), unit=ut_14)
```

```
[14] NSGA-II: unit test passed
```

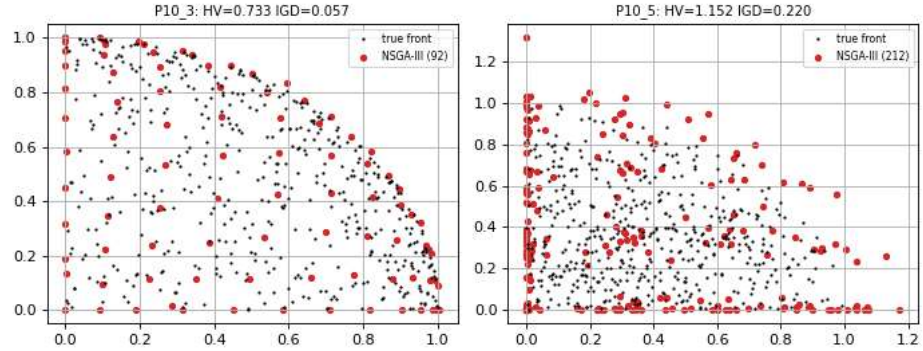


A2.15 NSGA-III

Crowding distance is replaced by reference points: normalize objectives (ideal, intercepts), associate each point with its nearest reference line, and fill the last front by niche counts.

```
def associate(Zn, W):
    """Perpendicular distance of each point to each reference line -> (nearest line, distance)."""
    Wn = W / np.linalg.norm(W, axis=1, keepdims=True); proj = Zn @ Wn.T
    d = np.sqrt(np.maximum((Zn ** 2).sum(1)[:, None] - proj ** 2, 0)); return d.argmin(1), d.min(1)

class NSGA3(MOEA):
    """Steps: 1) random mating, offspring  2) sort union  3) normalize (ideal, extreme points, intercepts)  4) associate to ref
    div = None
    def initialize(s):
        M = s.p.m; s.W = das_dennis(M, s.div or {2: 99, 3: 12, 5: 6}.get(M, 4)); s.N = int(np.ceil(len(s.W) / 4) * 4)
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def niching(s, FU, St, Fl, K):
        S = np.r_[St, Fl].astype(int); Z = FU[S] - FU[S].min(0); M = Z.shape[1]; E = []
        for j in range(M): w = np.full(M, 1e-6); w[j] = 1; E.append(Z[np.argmin((Z / w).max(1))])
        try:
            a = 1 / np.linalg.solve(np.array(E), np.ones(M)); ok = np.all(a > 1e-6) and np.all(np.isfinite(a))
        except np.linalg.LinAlgError: ok = False
        if not ok: a = Z.max(0)
        ref, d = associate(Z / np.maximum(a, 1e-9), s.W); nS = len(St); rho = np.bincount(ref[:nS], minlength=len(s.W)).astype(
        mr, md = ref[nS:], d[nS:]; left = np.ones(len(Fl), bool); chosen = []
        while len(chosen) < K:
            cnt = np.bincount(mr[left], minlength=len(s.W)); rr = np.where(cnt > 0, rho, np.inf); j = s.rng.choice(np.where(rr
            mem = np.where(left & (mr == j))[0]; pick = mem[np.argmin(md[mem])] if rho[j] == 0 else s.rng.choice(mem)
            chosen.append(Fl[pick]); left[pick] = False; rho[j] += 1
        return np.array(chosen, int)
    def step(s):
        off = s.offspring(s.X[s.rng.permutation(s.N)]); Fo = s.evaluate(off); XU, FU = np.vstack([s.X, off]), np.vstack([s.F, F
        while k < len(fr) and len(sel) + len(fr[k]) <= s.N: sel += list(fr[k]); k += 1
        if len(sel) < s.N: sel += list(s.niching(FU, np.array(sel, int), fr[k], s.N - len(sel)))
        s.X, s.F = XU[sel], FU[sel]

def ut_15():
    W = das_dennis(3, 4); assert len(W) == 15 and np.allclose(W.sum(1), 1)                # C(6,2)=15 reference points on the s
    ref, d = associate(0.5 * W[1:] + 1e-9, W[1:]); assert (ref == np.arange(14)).all() and d.max() < 1e-6   # a point on a ray
A2(15, cls_runner(NSGA3), unit=ut_15)
```

```
[15] NSGA-III: unit test passed
```



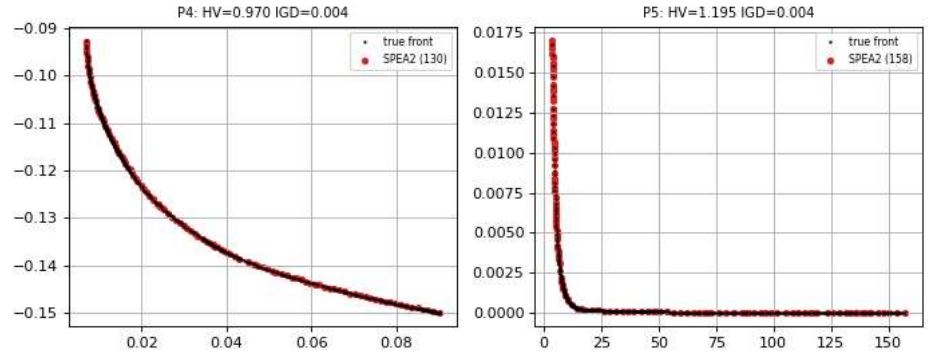
A2.16 SPEA2

Strength S(i) = #dominated by i; raw fitness R(i) = Σ strengths of dominators; density D = 1/(σ$_k$+2) with k = √(N+$\bar{N}$); fixed-size archive truncated by nearest-neighbour distance.

```
def spea2_fit(F):
    D = dom_matrix(F); S = D.sum(1); R = (D * S[:, None]).sum(0); k = max(1, int(np.sqrt(len(F))))
    return R + 1 / (np.sort(cdist(F, F), 1)[:, min(k, len(F) - 1)] + 2)

def spea2_select(F, n):
    fit = spea2_fit(F); nd = np.where(fit < 1)[0]
    if len(nd) < n: return np.argsort(fit)[:n]
    if len(nd) == n: return nd
    idx = list(range(len(nd))); Dm = cdist(F[nd], F[nd]); np.fill_diagonal(Dm, np.inf)
    while len(idx) > n: sub = np.sort(Dm[np.ix_(idx, idx)], 1); idx.pop(np.lexsort(sub.T[::-1])[0])
    return nd[idx]

class SPEA2(MOEA):
    """Steps: 1) union population+archive  2) fitness R+D  3) environmental selection (non-dominated, truncate/fill) -> archive
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.AX = np.empty((0, s.p.n)); s.AF = np.empty((0, s.p.m))
    def step(s):
        XU, FU = np.vstack([s.X, s.AX]) if len(s.AX) else s.X, np.vstack([s.F, s.AF]) if len(s.AF) else s.F
```

```
            sel = spea2_select(FU, s.N); s.AX, s.AF = XU[sel], FU[sel]; fit = spea2_fit(s.AF)
            par = s.tourn(len(s.AX), s.N, lambda i, j: fit[i] < fit[j]); s.X = s.offspring(s.AX[par]); s.F = s.evaluate(s.X)
        def final(s): return s.dec(np.vstack([s.AX, s.X]))

    def ut_16():
        F = np.array([[1, 4], [2, 2], [4, 1], [3, 3.], [5, 5.]]); fit = spea2_fit(F)
        assert (fit[:3] < 1).all() and (fit[3:] >= 1).all()                                  # non-dominated <1, dominated >=1
        assert len(spea2_select(np.random.default_rng(0).random((30, 2)), 10)) == 10
    A2(16, cls_runner(SPEA2), unit=ut_16)
```

```
[16] SPEA2: unit test passed
```

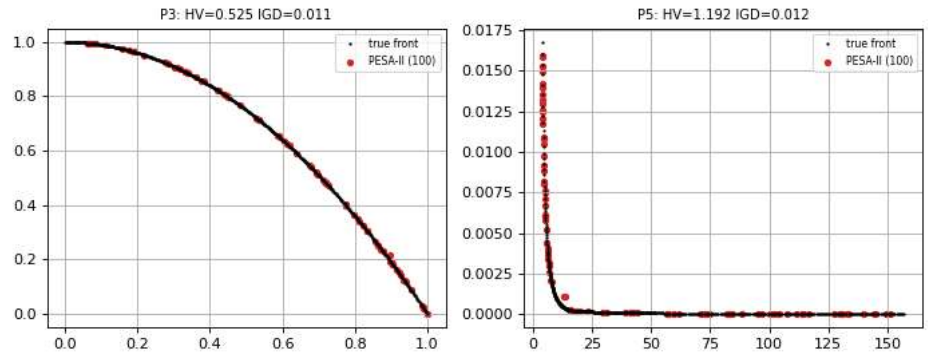


A2.17 PESA-II

External archive divided into grid regions; parents are chosen by region (less crowded region wins), then a random member of that region.

```
    class PESA2(MOEA):
        """Steps: 1) small internal population  2) archive of non-dominated points with grid  3) region tournament + random member
        Ndef = 20; cap = 100; div = 10
        def initialize(s):
            s.A = Archive(s.cap, s.div, s.rng); s.X = s.newpop(s.N); F = s.evaluate(s.X); s.A.add(s.X, F)
        def step(s):
            par = s.A.X[s.A.region_pick(s.N, s.rng)]; s.X = s.offspring(par); s.A.add(s.X, s.evaluate(s.X))
        def final(s): return s.dec(s.A.X)

    def ut_17():
        A = Archive(cap=2, div=4, rng=np.random.default_rng(0)); F = np.array([[0.0, 1.0], [0.01, 0.99], [1.0, 0.0]]); A.add(F, F)
        assert len(A.F) == 2 and (A.F[:, 0] == 1.0).any() and (A.F[:, 0] == 1.0).sum() == 1   # the lone point survives; the crowde
        assert A.counts(np.array([[0, 0], [0.01, 0.01], [1, 1.]])).tolist() == [2, 2, 1]
    A2(17, cls_runner(PESA2), unit=ut_17)
```

```
[17] PESA-II: unit test passed
```



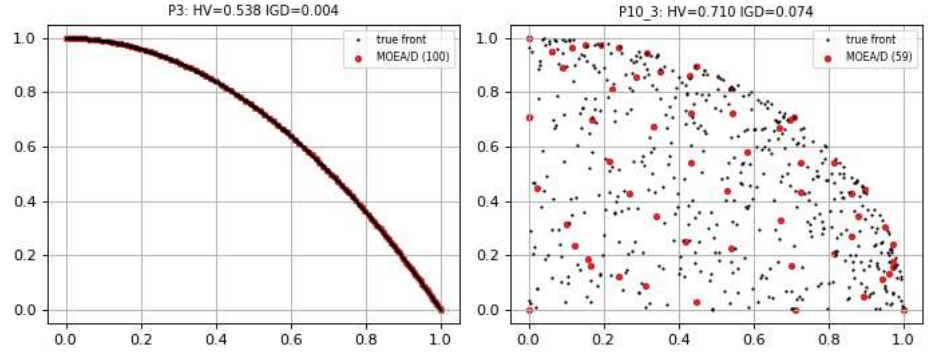
A2.18 MOEA/D and A2.19 MOEA/D-DE

The problem is split into N Tchebycheff sub-problems (weight vectors); each sub-problem mates only with its T nearest neighbours and may replace up to `nr` neighbours. The -DE variant uses DE/rand/1 variation.

```
    class MOEAD(MOEA):
        """Steps: 1) weight vectors + T-neighbourhoods  2) pick parents from neighbourhood (prob delta)  3) variation  4) update id
        T = 20; delta = 0.9; nr = 2; de = False; Fde = 0.5; CR = 1.0
        def initialize(s):
            M = s.p.m; s.W = np.c_[np.linspace(0, 1, s.N), 1 - np.linspace(0, 1, s.N)] if M == 2 else das_dennis(M, {3: 12, 5: 6}.g
            s.B = np.argsort(cdist(s.W, s.W), 1)[:, :min(s.T, s.N)]; s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.z = s.F.min(0)
```

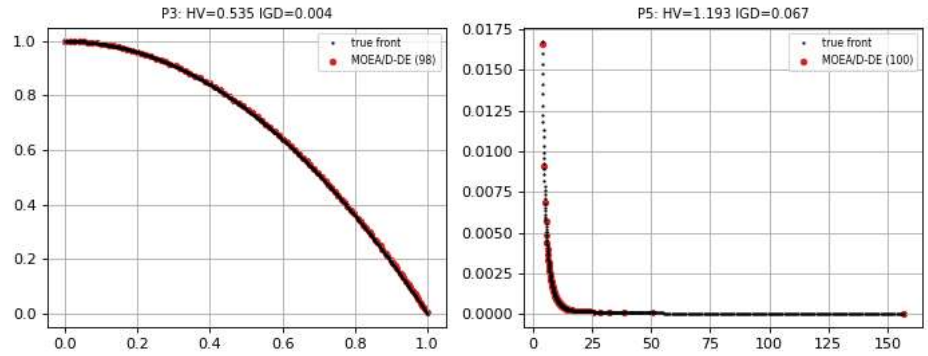
```
    def step(s):
        for i in s.rng.permutation(s.N):
            pool = s.B[i] if s.rng.random() < s.delta else np.arange(s.N)
            if s.de:
                r = s.rng.choice(pool, 3, replace=len(pool) < 3); v = s.X[r[0]] + s.Fde * (s.X[r[1]] - s.X[r[2]])
                m = s.rng.random(s.p.n) < s.CR; m[s.rng.integers(s.p.n)] = True; c = s.mutate(s.repair(np.where(m, v, s.X[i])[N
            else:
                a, b = s.rng.choice(pool, 2, replace=False); c = s.offspring(s.X[[a, b]])[:1]
            fc = s.evaluate(c)[0]; s.z = np.minimum(s.z, fc); cnt = 0
            for j in s.rng.permutation(pool):
                if tcheb(fc, s.W[j], s.z) <= tcheb(s.F[j], s.W[j], s.z): s.X[j], s.F[j] = c[0], fc; cnt += 1
                if cnt >= s.nr: break
            if s.evals >= s.budget: break

def ut_18():
    z = np.zeros(2); assert tcheb(np.array([.2, .2]), np.array([.5, .5]), z) < tcheb(np.array([.4, .4]), np.array([.5, .5]), z)
    m = MOEAD(P3, budget=50, seed=0, N=20, T=5); assert all(i in m.B[i] for i in range(m.N)) and m.B.shape == (20, 5)   # each
A2(18, cls_runner(MOEAD), unit=ut_18)
```

```
[18] MOEA/D: unit test passed
```




```
class MOEADDE(MOEAD):
    """MOEA/D with DE/rand/1 variation: v = x_r1 + F (x_r2 - x_r3), binomial crossover CR=1, then polynomial mutation. F=0.5.""
    de = True
def ut_19():
    m = MOEADDE(P3, budget=300, seed=0, N=30, T=6).run(); assert (m.X >= P3.lb - 1e-12).all() and (m.X <= P3.ub + 1e-12).all()
A2(19, cls_runner(MOEADDE), unit=ut_19)
```

```
[19] MOEA/D-DE: unit test passed
```



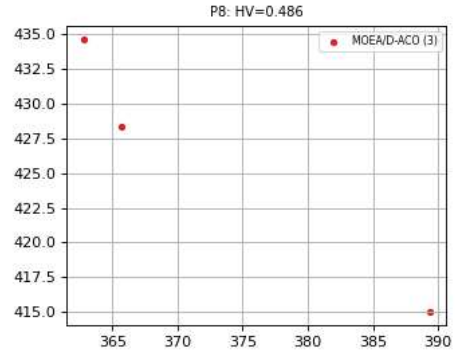
A2.20 MOEA/D-ACO

Each sub-problem owns a pheromone matrix; an ant builds a tour using τ^α·η^β where η = 1/(λ₁d + λ₂t); solutions update neighbouring sub-problems (Tchebycheff) and deposit pheromone on their neighbours' matrices.

```
def build_tour(rng, W):
    """Probabilistic route construction from an attractiveness matrix."""
    n = len(W); cur = rng.integers(n); t = [cur]; vis = np.zeros(n, bool); vis[cur] = True
    for _ in range(n - 1):
        w = np.where(vis, 0, W[cur]); cur = rng.choice(n, p=w / w.sum()); t.append(cur); vis[cur] = True
    return np.array(t)

class MOEAD_ACO(MOEA):
    """Steps: 1) weights, neighbourhoods, one pheromone matrix per sub-problem  2) ant builds tour  3) update neighbours (Tcheby
    Ndef = 20; T = 4; rho = 0.1; alpha = 1.0; beta = 2.0; nr = 2
    def initialize(s):
```

```
        p = s.p; n = p.n; t = np.linspace(0, 1, s.N); s.W = np.c_[t, 1 - t]; s.B = np.argsort(cdist(s.W, s.W), 1)[:, :s.T]
        dn, tn = p.D / p.D.mean(), p.Tm / p.Tm.mean(); s.eta = np.array([1 / np.maximum(w[0] * dn + w[1] * tn, 1e-9) for w in s.
        for e in s.eta: np.fill_diagonal(e, 1e-12)
        s.tau = np.ones((s.N, n, n)); s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.z = s.F.min(0)
    def step(s):
        for i in range(s.N):
            t = build_tour(s.rng, (s.tau[i] ** s.alpha) * (s.eta[i] ** s.beta)); f = s.evaluate(t[None])[0]; s.z = np.minimum(s.
            for j in s.rng.permutation(s.B[i]):
                if tcheb(f, s.W[j], s.z) <= tcheb(s.F[j], s.W[j], s.z): s.X[j], s.F[j] = t, f; c += 1
                if c >= s.nr: break
        for i in range(s.N):
            s.tau[i] *= (1 - s.rho)
            for j in s.B[i]:
                t = s.X[j]; nx = np.roll(t, -1); d = 1 / (tcheb(s.F[j], s.W[i], s.z) + 0.05); s.tau[i][t, nx] += d; s.tau[i][nx,

def ut_20():
    a = MOEAD_ACO(P8, budget=100, seed=0, N=6, T=3); assert all(sorted(x) == list(range(15)) for x in a.X)      # every route vi
    a.run(); assert all(sorted(x) == list(range(15)) for x in a.X)
A2(20, cls_runner(MOEAD_ACO), unit=ut_20)
```

```
[20] MOEA/D-ACO: unit test passed
```

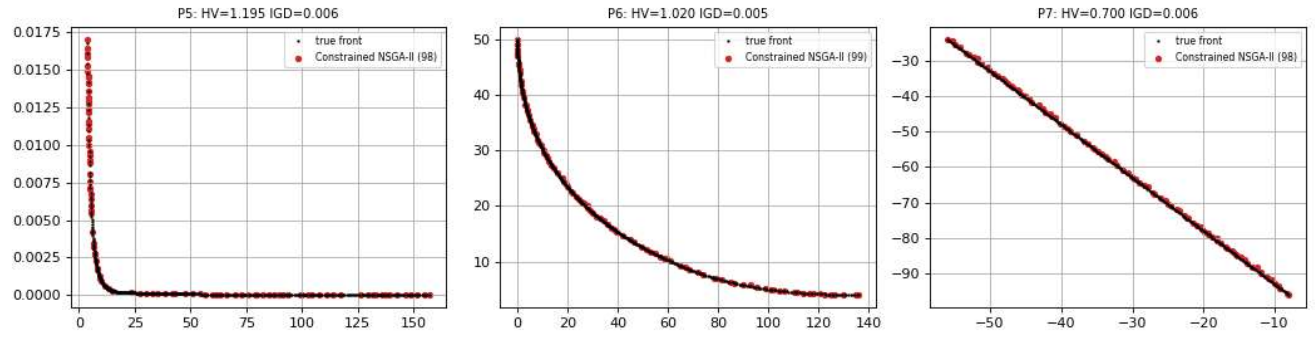


A2.21 Constrained NSGA-II

NSGA-II with Deb's feasibility rules inside dominance: (1) feasible beats infeasible, (2) both feasible → usual dominance, (3) both infeasible → smaller violation wins.

```
class CNSGA2(NSGA2):
    """NSGA-II steps with constrained-dominance (Deb's three rules) used by sorting, crowded comparison and survivor selection.
    use_cv = True

def ut_21():
    sel, rank, _ = env_select(np.array([[5., 5.], [1., 1.], [2., 2.]]), np.array([0, 1., .5]), 1); assert sel[0] == 0       # f
    sel, rank, _ = env_select(np.array([[5., 5.], [1., 1.], [2., 2.]]), np.array([0, 1., .5]), 3); assert list(sel) == [0, 2, 1
A2(21, cls_runner(CNSGA2), unit=ut_21)
```

```
[21] Constrained NSGA-II: unit test passed
```

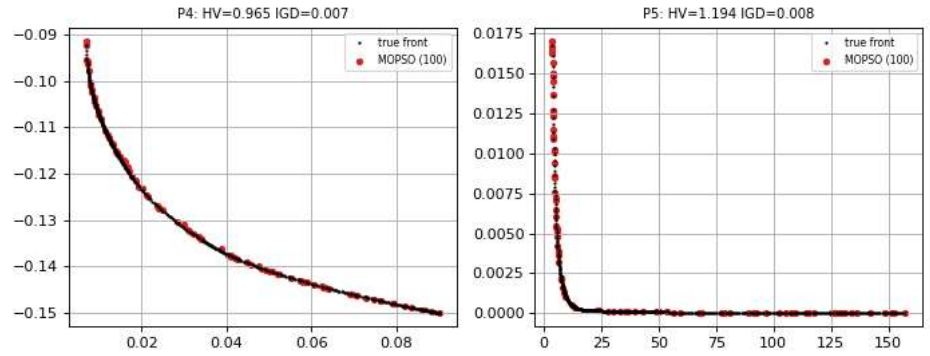


A2.22 MOPSO

Particles move with `v = w·v + c₁r₁(pbest−x) + c₂r₂(gbest−x)`; the global best is drawn from an external archive (sparse grid cells preferred).

```
class MOPSO(MOEA):
    """Steps: 1) init swarm, archive  2) leader from archive (grid roulette)  3) velocity+position update  4) mutation (10%)  5
    Ndef = 50; swarm = True; w = 0.4; c1 = 1.5; c2 = 1.5; cap = 100; pm = 0.1
    def initialize(s):
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.V = np.zeros_like(s.X); s.PB, s.PF = s.X.copy(), s.F.copy()
        s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X, s.F)
    def step(s):
        L = s.A.X[s.A.pick(s.N, s.rng)]; r1, r2 = s.rng.random((2,) + s.X.shape); rg = s.p.ub - s.p.lb
        s.V = np.clip(s.w * s.V + s.c1 * r1 * (s.PB - s.X) + s.c2 * r2 * (L - s.X), -.5 * rg, .5 * rg); s.X = np.clip(s.X + s.V
        mm = s.rng.random(s.N) < s.pm; s.X[mm] = polymut(s.rng, s.X[mm], s.p.lb, s.p.ub, pm=0.5) if mm.any() else s.X[mm]
        s.F = s.evaluate(s.X); nd = lambda a, b: (a <= b).all(1) & (a < b).any(1)
        upd = nd(s.F, s.PF) | (~nd(s.PF, s.F) & (s.rng.random(s.N) < .5)); s.PB[upd], s.PF[upd] = s.X[upd], s.F[upd]; s.A.add(s
    def final(s): return s.dec(s.A.X)

def ut_22():
    m = MOPSO(P4, budget=10 ** 6, seed=0, N=10, w=0, c1=0, c2=0, pm=0); X0 = m.X.copy(); m.step(); assert np.allclose(m.X, X0)
    m = MOPSO(P4, budget=10 ** 6, seed=0, N=20); [m.step() for _ in range(5)]; assert len(nds(m.A.F)[0]) == len(m.A.F)
A2(22, cls_runner(MOPSO), unit=ut_22)
```

```
[22] MOPSO: unit test passed
```

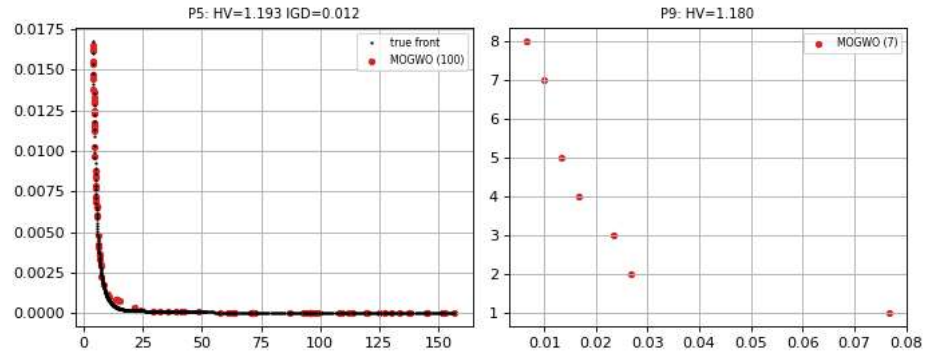


A2.23 MOGWO

Three leaders α, β, δ are taken from the archive; every wolf moves to the average of three encircling moves `X_k − A·|C·X_k − X|` with `a` shrinking from 2 to 0.

```
class MOGWO(MOEA):
    """Steps: 1) init pack + archive  2) pick alpha,beta,delta from archive  3) a=2(1-t/T); A=2a r1-a; C=2 r2  4) X=(X1+X2+X3)/3
    Ndef = 50; swarm = True; cap = 100
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X, s.F)
    def step(s):
        a = 2 * (1 - s.evals / s.budget); L = s.A.X[s.A.pick(3, s.rng)]; Xn = 0
        for k in range(3):
            A = 2 * a * s.rng.random(s.X.shape) - a; C = 2 * s.rng.random(s.X.shape); Xn = Xn + (L[k] - A * np.abs(C * L[k] - s.X
        s.X = np.clip(Xn, s.p.lb, s.p.ub); s.F = s.evaluate(s.X); s.A.add(s.X, s.F)
    def final(s): return s.dec(s.A.X)

def ut_23():
    m = MOGWO(P5, budget=10 ** 6, seed=0, N=20); m.step(); assert (m.X >= P5.lb).all() and (m.X <= P5.ub).all() and len(nds(m.A.F
    assert 2 * (1 - 0 / 100) == 2 and 2 * (1 - 100 / 100) == 0                          # coefficient a shrinks 2 -> 0 (explore
A2(23, cls_runner(MOGWO), unit=ut_23)
```

```
[23] MOGWO: unit test passed
```

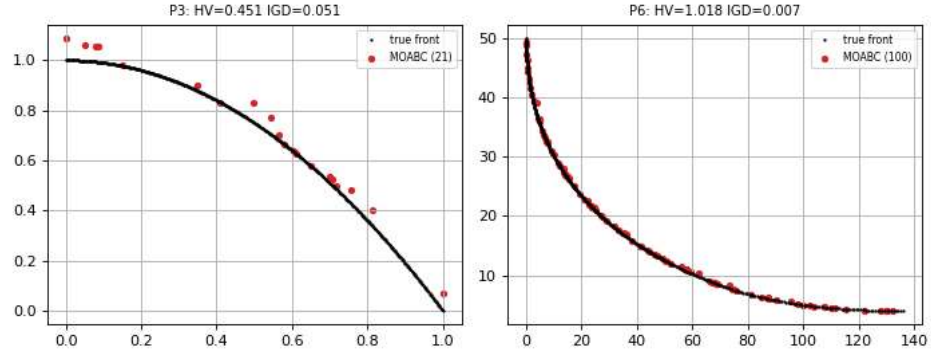


A2.24 MOABC

Employed bees perturb one dimension toward a random neighbour; onlookers choose sources by Pareto rank; greedy replacement by dominance; scouts replace sources unimproved for `limit` trials.

```
class MOABC(MOEA):
    """Steps: 1) food sources  2) employed phase v=x+phi(x-x_k) (one dim), keep if v dominates x  3) onlookers (prob ~ 1/(1+ran
    Ndef = 40; swarm = True; limit = 10; cap = 100
    def initialize(s):
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.trial = np.zeros(s.N, int); s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X,
    def attempt(s, idx):
        n = len(idx); k = (idx + s.rng.integers(1, s.N, n)) % s.N; j = s.rng.integers(0, s.p.n, n); V = s.X[idx].copy(); r = np
        V[r, j] = s.X[idx, j] + s.rng.uniform(-1, 1, n) * (s.X[idx, j] - s.X[k, j]); V = np.clip(V, s.p.lb, s.p.ub); Fv = s.eva
        for t, i in enumerate(idx):
            if (Fv[t] <= s.F[i]).all() and (Fv[t] < s.F[i]).any(): s.X[i], s.F[i], s.trial[i] = V[t], Fv[t], 0
            else: s.trial[i] += 1
    def step(s):
        s.attempt(np.arange(s.N)); rank = np.zeros(s.N)
        for k, f in enumerate(nds(s.F)): rank[f] = k
        p = 1 / (1 + rank); s.attempt(s.rng.choice(s.N, s.N, p=p / p.sum())); sc = np.where(s.trial > s.limit)[0]
        if len(sc): s.X[sc] = s.newpop(len(sc)); s.F[sc] = s.evaluate(s.X[sc]); s.trial[sc] = 0; s.A.add(s.X[sc], s.F[sc])
    def final(s): return s.dec(s.A.X)

def ut_24():
    m = MOABC(P3, budget=10 ** 6, seed=0, N=20, limit=2); Fb = m.F.copy(); m.attempt(np.arange(20))
    assert all(((m.F[i] <= Fb[i]).all()) for i in range(20))                              # greedy: a source only ever moves to
    m.trial[:] = 5; m.step(); assert (m.trial <= 5).all() and m.trial.min() == 0         # scouts reset exhausted sources
A2(24, cls_runner(MOABC), unit=ut_24)
```

```
[24] MOABC: unit test passed
```

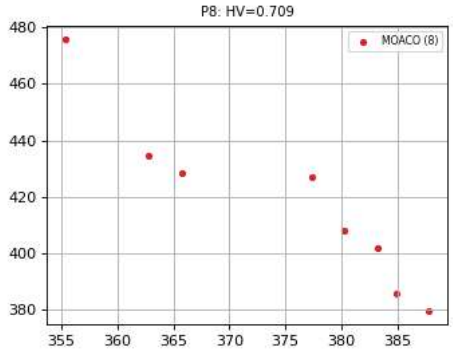


A2.25 MOACO

Two pheromone matrices (distance, time); each ant draws a random weight λ and builds a tour with `(τ₁^λ τ₂^(1−λ))^α (η₁^λ η₂^(1−λ))^β`; archive members deposit pheromone.

```
class MOACO(MOEA):
    """Steps: 1) tau1, tau2, eta1=1/d, eta2=1/t  2) each ant: random lambda, probabilistic tour  3) evaluate, archive  4) evapo
    Ndef = 20; rho = 0.1; alpha = 1.0; beta = 2.0; cap = 100
    def initialize(s):
        n = s.p.n; s.t1, s.t2 = np.ones((n, n)), np.ones((n, n)); s.e1, s.e2 = 1 / s.p.D, 1 / s.p.Tm
        for e in (s.e1, s.e2): np.fill_diagonal(e, 1e-12)
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X, s.F)
    def step(s):
        T = [build_tour(s.rng, ((s.t1 ** l) * (s.t2 ** (1 - l))) ** s.alpha * ((s.e1 ** l) * (s.e2 ** (1 - l))) ** s.beta) for
        s.X = np.array(T); s.F = s.evaluate(s.X); s.A.add(s.X, s.F); s.t1 *= 1 - s.rho; s.t2 *= 1 - s.rho
        for t, f in zip(s.A.X, s.A.F):
            nx = np.roll(t, -1); d1, d2 = 1 / (f[0] + .05), 1 / (f[1] + .05); s.t1[t, nx] += d1; s.t1[nx, t] += d1; s.t2[t, nx]
    def final(s): return s.A.X

def ut_25():
    m = MOACO(P8, budget=10 ** 6, seed=0, N=8); t1 = m.t1.copy(); m.step()
    assert all(sorted(x) == list(range(15)) for x in m.X) and np.allclose(m.t1, m.t1.T) and m.t1.max() > 1 and m.t1.min() < 1
A2(25, cls_runner(MOACO), unit=ut_25)
```

```
[25] MOACO: unit test passed
```

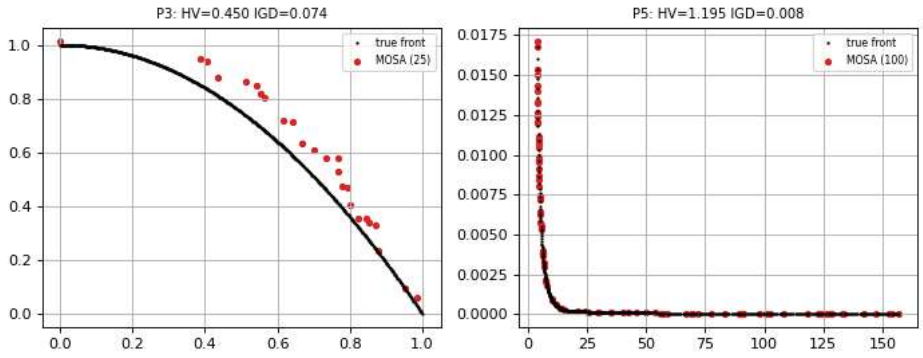


A2.26 MOSA

A set of annealing chains with different weight vectors; neighbour moves are accepted if they dominate or by the Boltzmann rule `exp(-ΔE/T)` on the weighted energy; geometric cooling.

```
class MOSA(MOEA):
    """Steps: 1) chains with weight vectors, T0  2) neighbour move  3) accept if dominates / dE<=0 / rand<exp(-dE/T)  4) archiv
    Ndef = 30; T0 = 0.5; Tend = 1e-3; cap = 100
    def initialize(s):
        M = s.p.m; t = np.linspace(0, 1, s.N); s.w = np.c_[t, 1 - t] if M == 2 else s.rng.dirichlet(np.ones(M), s.N)
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.T = s.T0; s.cool = (s.Tend / s.T0) ** (1 / max(1, s.budget // s.N))
        s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X, s.F)
    def step(s):
        Xn = s.neighbor(s.X); Fn = s.evaluate(Xn); dE = (s.w * (Fn - s.F)).sum(1); dom = (Fn <= s.F).all(1) & (Fn < s.F).any(1)
        acc = dom | (dE <= 0) | (s.rng.random(s.N) < np.exp(-np.maximum(dE, 0) / s.T)); s.X[acc], s.F[acc] = Xn[acc], Fn[acc]
        s.A.add(Xn, Fn); s.T *= s.cool
    def final(s): return s.dec(s.A.X)

def ut_26():
    assert np.exp(-0.1 / 0.01) < np.exp(-0.1 / 1.0)                                    # worse moves are accepted less often
    m = MOSA(P3, budget=3000, seed=0, N=30).run(); assert m.T < m.T0 * 0.5
A2(26, cls_runner(MOSA), unit=ut_26)
```

```
[26] MOSA: unit test passed
```

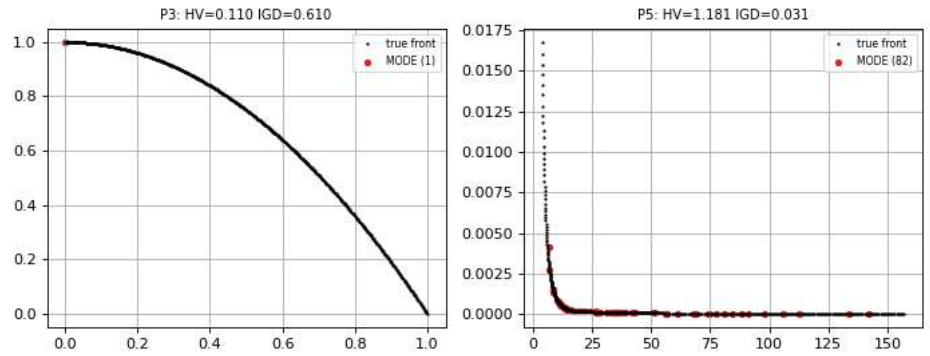


A2.27 MODE

DE/rand/1 mutation `v = x_r1 + F(x_r2 − x_r3)`, binomial crossover; the trial replaces its target **only if it dominates it**.

```
class MODE(MOEA):
    """Steps: 1) population  2) v=x_r1+F(x_r2-x_r3)  3) binomial crossover CR  4) trial replaces target only if it dominates. N
    swarm = True; Fm = 0.5; CR = 0.9
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def step(s):
        N = s.N; r = np.array([s.rng.choice(N, 3, replace=False) for _ in range(N)]); V = s.X[r[:, 0]] + s.Fm * (s.X[r[:, 1]] -
        m = s.rng.random(s.X.shape) < s.CR; m[np.arange(N), s.rng.integers(0, s.p.n, N)] = True; U = np.clip(np.where(m, V, s.X
        Fu = s.evaluate(U); b = (Fu <= s.F).all(1) & (Fu < s.F).any(1); s.X[b], s.F[b] = U[b], Fu[b]

def ut_27():
    m = MODE(P3, budget=10 ** 6, seed=0, N=30); Fb = m.F.copy(); m.step(); assert (m.F <= Fb + 1e-12).all()      # a target nev
A2(27, cls_runner(MODE), unit=ut_27)
```

```
[27] MODE: unit test passed
```

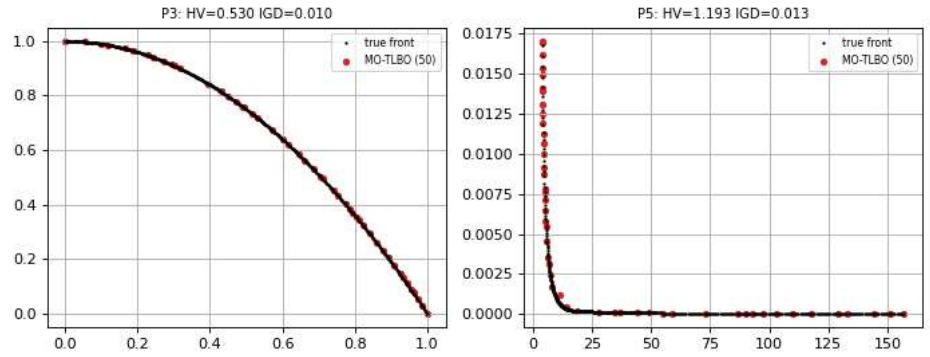


A2.28 MO-TLBO

Teacher phase moves learners toward a non-dominated teacher using a shrinking teaching factor; learner phase moves toward/away from a random partner; survivors by non-dominated sorting + crowding.

```
class MOTLBO(MOEA):
    """Steps: 1) class of learners  2) teacher = random non-dominated learner; Xn=X+r(T-TF*mean), TF=2-t/T  3) learner phase wi
    Ndef = 50; swarm = True
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def merge(s, Xn):
        Xn = np.clip(Xn, s.p.lb, s.p.ub); Fn = s.evaluate(Xn); XU, FU = np.vstack([s.X, Xn]), np.vstack([s.F, Fn]); sel, _, _ =
    def step(s):
        T = s.X[s.rng.choice(nds(s.F)[0])]; TF = 2 - s.evals / s.budget; s.merge(s.X + s.rng.random(s.X.shape) * (T - TF * s.X.
        j = s.rng.integers(0, s.N, s.N); d = (s.F <= s.F[j]).all(1) & (s.F < s.F[j]).any(1); sg = np.where(d, 1.0, -1.0)[:, Non
        s.merge(s.X + s.rng.random(s.X.shape) * sg * (s.X - s.X[j]))

def ut_28():
    m = MOTLBO(P3, budget=10 ** 6, seed=0, N=30); h0 = hv(m.F[nds(m.F)[0]], P3.ref); m.step()
    assert len(m.X) == 30 and hv(m.F[nds(m.F)[0]], P3.ref) >= h0 - 1e-12                  # elitist merge never loses hypervolu
    assert 1 <= 2 - 0.5 <= 2                                                              # teaching factor stays in [1,2] and
A2(28, cls_runner(MOTLBO), unit=ut_28)
```

```
[28] MO-TLBO: unit test passed
```



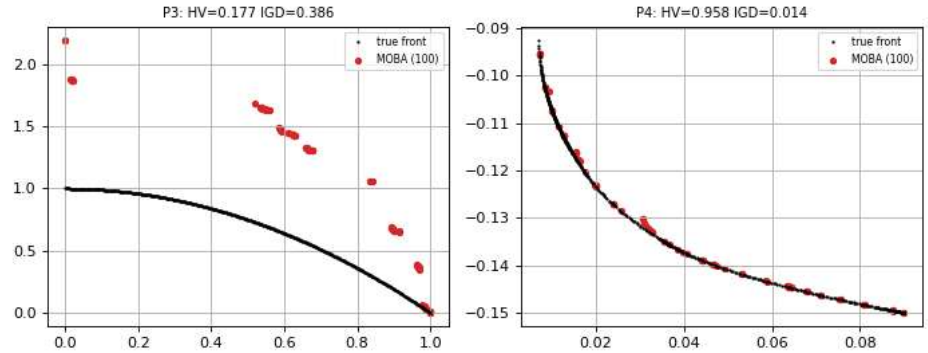
A2.29 MOBA

Bats fly with frequency-driven velocity toward an archive leader; a random local walk around the leader is used when `rand > pulse rate`; a move is accepted with probability = loudness, then loudness decreases and pulse rate increases.

```
class MOBA(MOEA):
    """Steps: 1) bats, loudness A=1, pulse r0  2) f=fmin+(fmax-fmin)b; v+=(x-leader)f; x+=v  3) local walk if rand>r  4) accept
    Ndef = 40; swarm = True; fmin = 0.0; fmax = 2.0; alpha = 0.9; gamma = 0.9; r0 = 0.5; cap = 100
    def initialize(s):
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.V = np.zeros_like(s.X); s.A_ = np.ones(s.N); s.r_ = np.zeros(s.N)
        s.A = Archive(s.cap, 10, s.rng); s.A.add(s.X, s.F)
    def step(s):
        rg = s.p.ub - s.p.lb; L = s.A.X[s.A.pick(s.N, s.rng)]; f = s.fmin + (s.fmax - s.fmin) * s.rng.random((s.N, 1))
        s.V = np.clip(s.V + (s.X - L) * f, -.3 * rg, .3 * rg); Xn = np.clip(s.X + s.V, s.p.lb, s.p.ub); loc = s.rng.random(s.N)
        Xn[loc] = np.clip(L[loc] + 0.02 * s.rng.normal(size=Xn[loc].shape) * rg * s.A_.mean(), s.p.lb, s.p.ub); Fn = s.evaluate
        acc = (s.rng.random(s.N) < s.A_) & ~((s.F <= Fn).all(1) & (s.F < Fn).any(1)); s.X[acc], s.F[acc] = Xn[acc], Fn[acc]
        s.A_[acc] *= s.alpha; s.r_[acc] = s.r0 * (1 - np.exp(-s.gamma * (s.gen + 1))); s.A.add(Xn, Fn)
```

```
    def final(s): return s.dec(s.A.X)

def ut_29():
    m = MOBA(P3, budget=10 ** 6, seed=0, N=20); A0 = m.A_.copy(); m.step()
    assert (m.A_ <= A0 + 1e-12).all() and (m.A_ < A0).any() and (m.r_ <= m.r0 + 1e-12).all() and (m.r_ > 0).any()   # loudness
A2(29, cls_runner(MOBA), unit=ut_29)
```

```
[29] MOBA: unit test passed
```

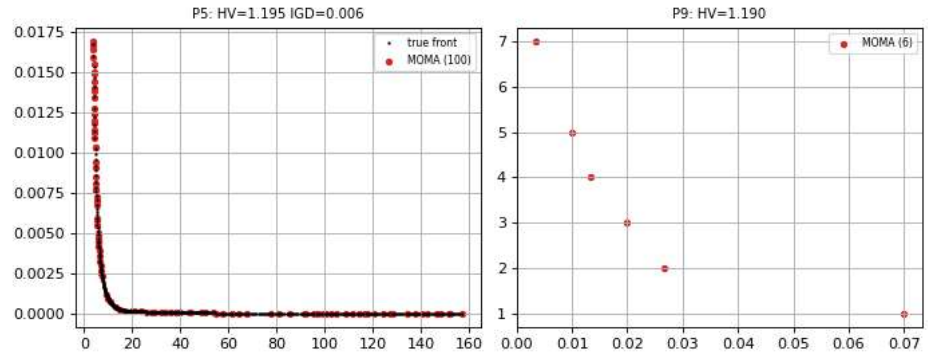


A2.30 MOMA

NSGA-II global search plus hill-climbing on a fraction of the population: a neighbour replaces a member only if it dominates it.

```
class MOMA(NSGA2):
    """Steps: NSGA-II generation, then local search: pick ls_frac of the population, ls_steps neighbour moves, replace if the n
    ls_frac = 0.1; ls_steps = 3
    def local_search(s):
        idx = s.rng.choice(len(s.X), max(1, int(s.ls_frac * len(s.X))), replace=False)
        for _ in range(s.ls_steps):
            Xn = s.neighbor(s.X[idx]); Fn = s.evaluate(Xn); cv = s.last_cv; b = (Fn <= s.F[idx]).all(1) & (Fn < s.F[idx]).any(1
            s.X[idx[b]], s.F[idx[b]], s.CV[idx[b]] = Xn[b], Fn[b], cv[b]
    def step(s): NSGA2.step(s); s.local_search()

def ut_30():
    m = MOMA(P3, budget=10 ** 6, seed=0, N=20, ls_frac=1.0); Fb = m.F.copy(); m.local_search(); assert (m.F <= Fb + 1e-12).all(
A2(30, cls_runner(MOMA), unit=ut_30)
```

```
[30] MOMA: unit test passed
```



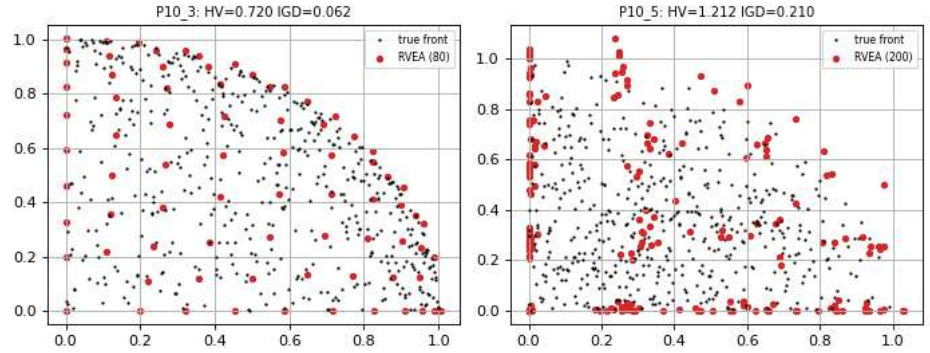
A2.31 RVEA

Reference vectors guide selection: each solution is assigned to its nearest vector (by angle) and the one with the smallest angle-penalized distance (APD) survives per vector; vectors are re-scaled periodically.

```
class RVEA(MOEA):
    """Steps: 1) reference vectors (Das-Dennis)  2) random mating  3) translate by ideal, assign by angle  4) APD=(1+M(t/T)^a *
    alpha = 2.0; fr = 0.1; div = None
    def initialize(s):
        M = s.p.m; W = das_dennis(M, s.div or {2: 99, 3: 12, 5: 6}.get(M, 4)); s.W0 = W / np.linalg.norm(W, axis=1, keepdims=Tr
        s.X = s.newpop(s.N); s.F = s.evaluate(s.X); s.Tg = max(1, s.budget // s.N)
    def apd_select(s, F):
        Z = F - F.min(0); nr = np.linalg.norm(Z, axis=1) + 1e-12; ang = np.arccos(np.clip((Z @ s.Wv.T) / nr[:, None], -1, 1));
        cw = s.Wv @ s.Wv.T; np.fill_diagonal(cw, -1); gam = np.arccos(np.clip(cw.max(1), -1, 1)) + 1e-12; sel = []
        for v in np.unique(asg):
```

```
            m = np.where(asg == v)[0]; sel.append(m[np.argmin((1 + F.shape[1] * (s.gen / s.Tg) ** s.alpha * ang[m, v] / gam[v])
        return np.array(sel)
    def step(s):
        off = s.offspring(s.X[s.rng.integers(0, len(s.X), s.N)]); XU, FU = np.vstack([s.X, off]), np.vstack([s.F, s.evaluate(of
        if s.gen > 0 and s.gen % max(1, int(s.fr * s.Tg)) == 0:
            v = s.W0 * (s.F.max(0) - s.F.min(0) + 1e-12); s.Wv = v / np.linalg.norm(v, axis=1, keepdims=True)

def ut_31():
    m = RVEA(P10_3, budget=10 ** 6, seed=0); sel = m.apd_select(np.array([[1., 0, 0], [2., 0, 0], [0, 1., 0]]))
    assert set(sel) == {0, 2}                                                    # same vector -> smaller distance win
A2(31, cls_runner(RVEA), unit=ut_31)
```

```
[31] RVEA: unit test passed
```

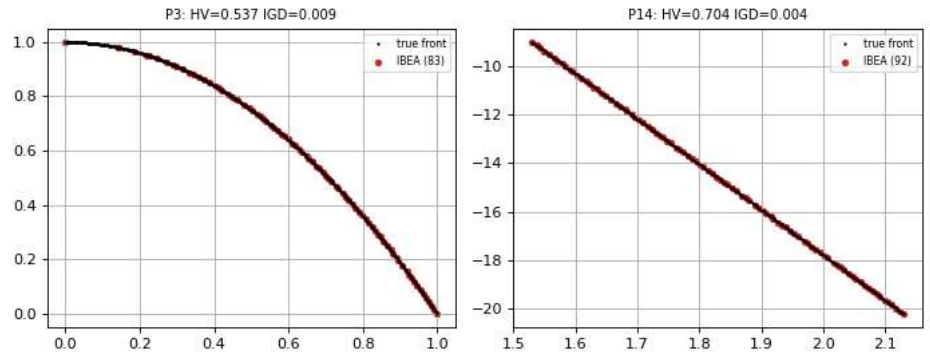


A2.32 IBEA

Fitness from the additive ε-indicator `I(x,y)=max_k f_k(x)−f_k(y)`: `F(x)=−Σ_y exp(−I(y,x)/(c κ))`. No explicit diversity step; the worst individual is removed one at a time and fitness is updated.

```
class IBEA(MOEA):
    """Steps: 1) pairwise eps-indicator  2) fitness F(x)=-sum exp(-I(y,x)/(c*kappa))  3) tournament on fitness  4) merge, repea
    kappa = 0.05
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def fitness(s, F):
        I = (F[:, None, :] - F[None, :, :]).max(2); E = np.exp(-I / ((np.abs(I).max() + 1e-12) * s.kappa)); np.fill_diagonal(E,
    def select(s, F, n):
        fit, E = s.fitness(F); act = list(range(len(F)))
        while len(act) > n: w = act[int(np.argmin(fit[act]))]; act.remove(w); fit = fit + E[w]
        return np.array(act)
    def step(s):
        fit, _ = s.fitness(s.F); par = s.tourn(len(s.X), s.N, lambda i, j: fit[i] > fit[j]); off = s.offspring(s.X[par])
        XU, FU = np.vstack([s.X, off]), np.vstack([s.F, s.evaluate(off)]); sel = s.select(FU, s.N); s.X, s.F = XU[sel], FU[sel]

def ut_32():
    m = IBEA(P3, budget=10 ** 6, seed=0, N=10); fit, _ = m.fitness(np.array([[0, 0.], [1, 1.]])); assert fit[1] < fit[0]      #
    F = np.array([[0, 1.], [1, 0.], [.5, .5], [1, 1.]]); assert 3 not in m.select(F, 3)                                       #
A2(32, cls_runner(IBEA), unit=ut_32)
```

```
[32] IBEA: unit test passed
```

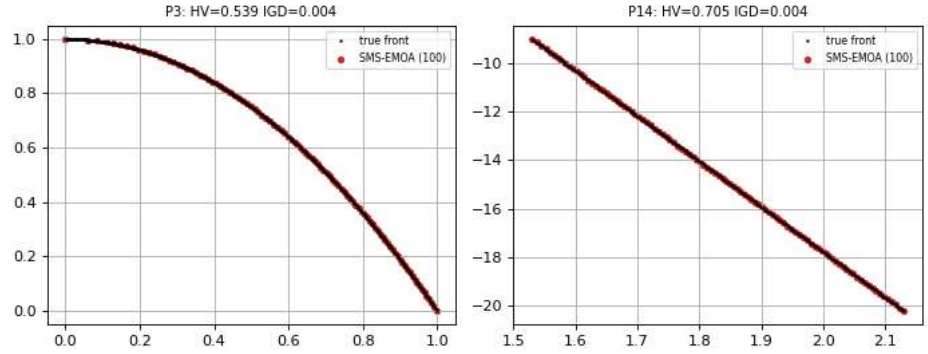


A2.33 SMS-EMOA

Steady state: one offspring per step; after non-dominated sorting the individual of the last front with the smallest hypervolume contribution is discarded.

```
def contrib2d(F, ref):
    """Exact hypervolume contribution of every point of a 2-D non-dominated set."""
    o = np.argsort(F[:, 0]); P = np.minimum(F[o], ref); c = np.zeros(len(F))
    for k in range(len(P)):
        right = P[k + 1, 0] if k + 1 < len(P) else ref[0]; up = P[k - 1, 1] if k > 0 else ref[1]; c[o[k]] = max(0.0, (right - P
    return c

def contrib_mc(F, ref, rng, S=4000):
    """Monte-Carlo contribution = volume dominated by this point only (3+ objectives)."""
    lo = np.minimum(F.min(0), ref - 1e-6); smp = rng.uniform(lo, ref, (S, F.shape[1])); dom = (F[None] <= smp[:, None]).all(2)
    return (dom & (dom.sum(1) == 1)[:, None]).sum(0) / S * np.prod(ref - lo)

class SMSEMOA(MOEA):
    """Steps: 1) population  2) one offspring (2 random parents)  3) sort union  4) delete the least-HV-contributor of the last
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def step(s):
        i, j = s.rng.choice(len(s.X), 2, replace=False); off = s.offspring(s.X[[i, j]])[:1]; XU, FU = np.vstack([s.X, off]), np
        last = nds(FU)[-1]; c = contrib2d(FU[last], s.p.ref) if s.p.m == 2 else contrib_mc(FU[last], s.p.ref, s.rng)
        keep = np.ones(len(FU), bool); keep[last[np.argmin(c)]] = False; s.X, s.F = XU[keep], FU[keep]

def ut_33():
    F = np.array([[1, 3], [2, 2], [3, 1.]]); c = contrib2d(F, np.array([4., 4.])); assert np.allclose(c, [1, 1, 1])   # stairca
    assert abs(contrib2d(np.array([[1, 3], [2, 2.], [2.1, 2.]]), np.array([4., 4.]))[1]) < 0.3
A2(33, cls_runner(SMSEMOA), unit=ut_33)
```

```
[33] SMS-EMOA: unit test passed
```

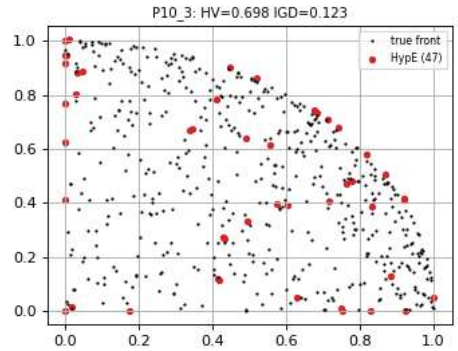


A2.34 HypE

Hypervolume contributions are estimated by Monte-Carlo sampling; a sample dominated by `c` individuals gives each of them credit `α_c = (1/c)·Π_{j<c}(k−j)/(|P|−j)` where k is the number of individuals still to remove.

```
class HypE(MOEA):
    """Steps: 1) mating, offspring  2) sample S points in the box [min F, ref]  3) HypE fitness with weights alpha_c  4) remove
    Ndef = 50; S = 2000; chunk = 10
    def initialize(s): s.X = s.newpop(s.N); s.F = s.evaluate(s.X)
    def fitness(s, F, k):
        ref = s.p.ref; lo = np.minimum(F.min(0), ref - 1e-6); smp = s.rng.uniform(lo, ref, (s.S, F.shape[1])); dom = (F[None] <
        for c in range(1, P + 1):
            a = 1.0
            for j in range(1, c): a *= max(k - j, 0) / (P - j)
            al[c] = a / c
        return (dom * al[dom.sum(1)][:, None]).sum(0) * np.prod(ref - lo) / s.S
    def select(s, F, n):
        act = np.arange(len(F))
        while len(act) > n:
            k = len(act) - n; fit = s.fitness(F[act], k); act = np.delete(act, np.argsort(fit)[:min(s.chunk, k)])
        return act
    def step(s):
        off = s.offspring(s.X[s.rng.permutation(len(s.X))]); XU, FU = np.vstack([s.X, off]), np.vstack([s.F, s.evaluate(off)]);

def ut_34():
    m = HypE(P10_3, budget=10 ** 6, seed=0, N=10, S=20000); F = np.array([[.1, .1, .1], [.5, .5, .5], [1., 1., 1.]]); fit = m.f
    assert fit[1] == 0 and fit[0] > 0                                                    # k=1: only exclusive volume counts,
A2(34, cls_runner(HypE), unit=ut_34)
```

```
[34] HypE: unit test passed
```

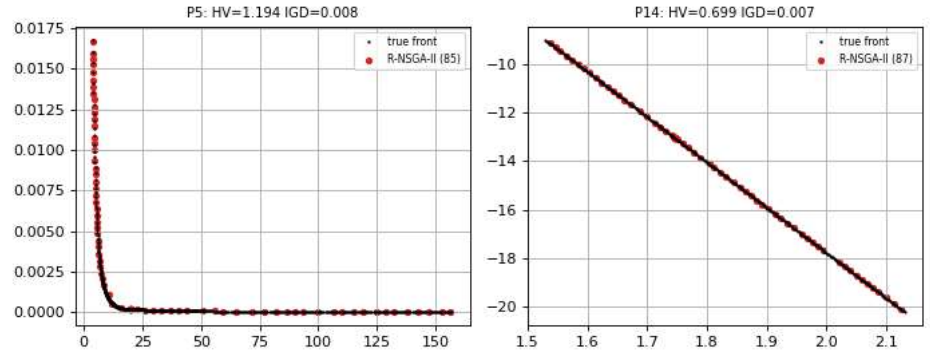


A2.35 R-NSGA-II

NSGA-II where crowding distance is replaced by closeness to the decision-maker's reference point; ε-clearing keeps near-duplicates out of the preferred region.

```
def eps_clear(F, order, eps):
    """Walk the preference order; points closer than eps to an already kept point are moved to the back."""
    left, out, later = list(order), [], []
    while left:
        a = left.pop(0); out.append(a); keep = []
        for b in left: (later if np.linalg.norm(F[a] - F[b]) < eps else keep).append(b)
        left = keep
    return np.array(out + later)

class RNSGA2(NSGA2):
    """NSGA-II with preference: survivors of the last front are ordered by distance to the reference point (normalized), eps-cl
    eps = 0.02; ref_pt = None
    def initialize(s):
        s.zn = s.p.norm(np.asarray(s.ref_pt if s.ref_pt is not None else s.p.pref, float)[None])[0]; NSGA2.initialize(s)
    def survive(s, F, CV, n):
        fr = nds(F); rank = np.zeros(len(F), int); crowd = np.zeros(len(F)); sel = []
        for k, f in enumerate(fr):
            order = eps_clear(F[f], np.argsort(np.linalg.norm(F[f] - s.zn, axis=1)), s.eps); pr = np.empty(len(f)); pr[order] =
            if len(sel) < n: sel += list(f[order][:n - len(sel)])
        return np.array(sel), rank, crowd

def ut_35():
    F = np.array([[0, 0], [0.001, 0], [1, 1.]]); assert list(eps_clear(F, [0, 1, 2], 0.01)) == [0, 2, 1]       # near-duplicate
def run_35(P, s, b):
    a = RNSGA2(P, budget=b, seed=s).run(); Fn = a.result(); d = np.linalg.norm(Fn - a.zn, axis=1); print(f"  {P.name}: closest
A2(35, run_35, unit=ut_35)
```

```
[35] R-NSGA-II: unit test passed
  P5: closest front point to reference is 0.018 (normalized)
  P14: closest front point to reference is 0.128 (normalized)
```



P13 — multi-task learning set-up (used by algorithms 36 and 37)

(a) toy: $f_1=\|x-a\|^2$, $f_2=\|x-b\|^2$, a=(3,3), b=(−3,−3), start (0,5). (b) shared-body network with a classification head and a regression head (forward/backward by hand).

```
def min_norm(G):
    """Min-norm point of the convex hull of the rows of G (Frank-Wolfe); alpha on the simplex."""
    M = len(G)
    if M == 2:
        d = G[0] - G[1]; a = np.clip(((G[1] - G[0]) @ G[1]) / (d @ d + 1e-18), 0, 1); return np.array([a, 1 - a])
    a = np.ones(M) / M; Gm = G @ G.T
    for _ in range(3000):
        t = np.argmin(Gm @ a); dd = np.eye(M)[t] - a; den = dd @ Gm @ dd
        if den <= 1e-18: break
        a = a + np.clip(-(dd @ Gm @ a) / den, 0, 1) * dd
    return a

def gradnorm_update(w, L, L0, gn, alpha=1.5, eta=0.01):
    """GradNorm: push each task's weighted gradient norm G_i=w_i*||g_i|| toward mean(G)*r_i^alpha (r_i = relative inverse train
    G = w * gn; r = (L / L0) / (L / L0).mean(); tgt = G.mean() * r ** alpha; w = np.clip(w * np.exp(-eta * np.sign(G - tgt)), 0

A_, B_ = np.array([3., 3.]), np.array([-3., -3.])
def toy_run(method, x0=(0., 5.), lr=0.05, iters=150):
    x = np.array(x0); w = np.ones(2); path = [x.copy()]; L0 = np.array([((x - A_) ** 2).sum(), ((x - B_) ** 2).sum()])
    for _ in range(iters):
        g = np.array([2 * (x - A_), 2 * (x - B_)]); L = np.array([((x - A_) ** 2).sum(), ((x - B_) ** 2).sum()])
        if method == "equal": d = g.sum(0)
        elif method == "mgda": d = 2 * (min_norm(g) @ g)
        else: w = gradnorm_update(w, L, L0, np.linalg.norm(g, axis=1)); d = w @ g
        x = x - lr * d; path.append(x.copy())
    return np.array(path)

def make_mtl_data(rng, n=300):
    X = rng.normal(size=(n, 5)); return X, (X[:, 0] + X[:, 1] > 0).astype(float), 20 * (X[:, 2] - .5 * X[:, 3])   # regression

def init_net(rng, h=12): return dict(W1=rng.normal(0, .3, (5, h)), b1=np.zeros(h), wc=rng.normal(0, .3, h), bc=np.zeros(1), wr=

def net_grads(P, X, yc, yr):
    """Forward + manual backward: returns losses (cls, reg), shared-body gradients per task, head gradients per task."""
    n = len(X); H = np.tanh(X @ P["W1"] + P["b1"]); pc = 1 / (1 + np.exp(-(H @ P["wc"] + P["bc"][0]))); pr = H @ P["wr"] + P["b
    Lc = -np.mean(yc * np.log(pc + 1e-12) + (1 - yc) * np.log(1 - pc + 1e-12)); Lr = np.mean((pr - yr) ** 2); dzc, dpr = (pc -
    sh = lambda dz, w: (lambda dA: np.r_[(X.T @ dA).ravel(), dA.sum(0)])(np.outer(dz, w) * (1 - H ** 2))
    return (Lc, Lr), (sh(dzc, P["wc"]), sh(dpr, P["wr"])), (dict(wc=H.T @ dzc, bc=np.array([dzc.sum()])), dict(wr=H.T @ dpr, br

def train_mtl(method, seed=0, epochs=400, lr=0.002):
    rng = np.random.default_rng(seed); X, yc, yr = make_mtl_data(rng); P = init_net(rng); w = np.ones(2); L0 = None; hist = [];
    for _ in range(epochs):
        L, (gc, gr), (hc, hr) = net_grads(P, X, yc, yr); L = np.array(L); L0 = L.copy() if L0 is None else L0; hist.append(L)
        if method == "equal": d = gc + gr
        elif method == "mgda": d = 2 * (min_norm(np.array([gc, gr])) @ np.array([gc, gr]))
        else: w = gradnorm_update(w, L, L0, np.array([np.linalg.norm(gc), np.linalg.norm(gr)])); d = w[0] * gc + w[1] * gr
        P["W1"] -= lr * d[:5 * h].reshape(5, h); P["b1"] -= lr * d[5 * h:]
        sc = w if method == "gradnorm" else np.ones(2)
        for k, v in hc.items(): P[k] -= lr * sc[0] * v
        for k, v in hr.items(): P[k] -= lr * sc[1] * v
    return np.array(hist), w
```

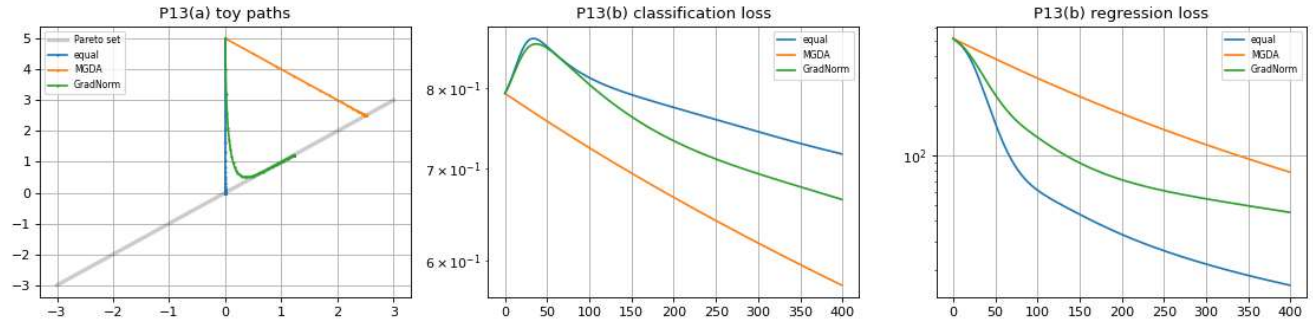
A2.36 MGDA

The min-norm point $d = \Sigma\ \alpha_i\ g_i$ of the convex hull of the task gradients is a common descent direction ($d \cdot g_i \geq \|d\|^2$ for every i); it vanishes at Pareto-stationary points.

```
def ut_36():
    assert np.allclose(min_norm(np.array([[1., 0], [0, 1.]])), [.5, .5])                    # orthogonal equal-length gradients
    g = np.random.default_rng(0).normal(size=(3, 5)); a = min_norm(g); d = a @ g
    assert abs(a.sum() - 1) < 1e-9 and (g @ d >= d @ d - 1e-3).all()                        # common descent: every task improve
    G2 = np.array([[1., 1.], [-1., -1.]]); assert np.allclose(min_norm(G2) @ G2, 0)          # opposite gradients -> d=0 (Paret
ut_36(); print("[36] MGDA: unit test passed")
pe, pm_, pg = toy_run("equal"), toy_run("mgda"), toy_run("gradnorm")
print("toy final x — equal:", pe[-1].round(3), " MGDA:", pm_[-1].round(3), " GradNorm:", pg[-1].round(3), "(Pareto set: x1 = x2
hE, _ = train_mtl("equal"); hM, _ = train_mtl("mgda"); hG, wG = train_mtl("gradnorm")
fig, ax = plt.subplots(1, 3, figsize=(14, 3.6))
ax[0].plot([-3, 3], [-3, 3], "k-", lw=3, alpha=.2, label="Pareto set")
for pth, lab in ((pe, "equal"), (pm_, "MGDA"), (pg, "GradNorm")): ax[0].plot(pth[:, 0], pth[:, 1], ".-", ms=2, label=lab)
ax[0].set_title("P13(a) toy paths"); ax[0].legend(fontsize=7)
for a_, k, t in ((ax[1], 0, "classification loss"), (ax[2], 1, "regression loss")):
    for h_, lab in ((hE, "equal"), (hM, "MGDA"), (hG, "GradNorm")): a_.semilogy(h_[:, k], label=lab)
    a_.set_title("P13(b) " + t); a_.legend(fontsize=7)
plt.tight_layout(); plt.show()
```

```
LOG.append(dict(idx=36, name="MGDA", prob="P13", budget=400, hv=np.nan, sp=np.nan, igd=np.nan, rt=0, status="works", n=0))
print("final losses (cls, reg): equal", hE[-1].round(3), "| MGDA", hM[-1].round(3), "| GradNorm", hG[-1].round(3))
```

```
[36] MGDA: unit test passed
toy final x — equal: [0. 0.]  MGDA: [2.5 2.5]  GradNorm: [1.224 1.224] (Pareto set: x1 = x2, -3<=x1<=3)
```




```
final losses (cls, reg): equal [ 0.717 16.184] | MGDA [ 0.576 79.13 ] | GradNorm [ 0.665 44.984]
```

A2.37 GradNorm

Task weights w_i are trained so that the weighted gradient norms $w_i\|\nabla L_i\|$ match $\text{mean}(G)\cdot r_i^{\alpha}$, balancing tasks with very different gradient scales (α=1.5).

```
def ut_37():
    w = gradnorm_update(np.ones(2), np.array([1., 1.]), np.array([1., 1.]), np.array([10., 1.]))
    assert w[0] < 1 < w[1] and abs(w.sum() - 2) < 1e-9                                    # large-gradient task is down-weighte
ut_37(); print("[37] GradNorm: unit test passed")
print("learned GradNorm weights (cls, reg) after training:", wG.round(3), "-> the regression task (huge gradients) is down-weigh
LOG.append(dict(idx=37, name="GradNorm", prob="P13", budget=400, hv=np.nan, sp=np.nan, igd=np.nan, rt=0, status="works", n=0))
```

```
[37] GradNorm: unit test passed
learned GradNorm weights (cls, reg) after training: [1.703 0.297] -> the regression task (huge gradients) is down-weighted
```

A3. Implementation log

One row per algorithm and test problem (fixed seed). HV uses the normalized reference point 1.1; IGD only where the true front is known; `n` = size of the obtained front.

```
def show_log():
    rows = [(r["idx"], r["name"], r["prob"], r["budget"], f"{r['hv']:.4f}", f"{r['igd']:.4f}", f"{r['sp']:.4f}", f"{r['rt']:.2f
            for r in sorted(LOG, key=lambda r: (r["idx"], r["prob"]))]
    table(rows, ["#", "Algorithm", "Problem", "Eval budget", "HV", "IGD", "Spacing", "Runtime", "Status", "n"])
show_log()
done = sorted({r["idx"] for r in LOG}); print("algorithms logged:", len(done), "of 37; missing:", sorted(set(range(1, 38)) - se
```

#	Algorithm	Problem	Eval budget	HV	IGD	Spacing	Runtime	Status	n
1	Weighted Sum	P1	10000	1.0065	0.0405	0.0176	0.06s	works	11
1	Weighted Sum	P3	10000	0.2100	0.3543	nan	0.18s	works	2
1	Weighted Sum	P4	10000	0.9046	0.0544	0.1101	0.27s	works	9
2	ε-Constraint	P2	10000	1.0557	0.1114	0.1817	0.48s	works	11
2	ε-Constraint	P3	10000	0.4950	0.0369	0.0388	0.60s	works	11
2	ε-Constraint	P5	10000	1.1370	0.0685	0.0238	2.30s	works	11
3	Benson's Method	P1	10000	0.9844	0.0613	0.0686	0.25s	works	12
3	Benson's Method	P7	10000	0.6416	0.0422	0.0729	0.26s	works	12
4	Weight Metric (p-norm, Tchebycheff)	P1	10000	1.0073	0.0560	0.0599	0.36s	works	11
4	Weight Metric (p-norm, Tchebycheff)	P3	10000	0.4953	0.0357	0.0206	1.07s	works	11
5	Value Function	P14	10000	0.5382	0.1183	0.0002	0.22s	works	3
5	Value Function	P4	10000	0.8317	0.1896	0.0051	0.19s	works	3
6	Goal Programming	P11	10000	1.2100	nan	nan	0.01s	works	1
6	Goal Programming	P7	10000	0.3407	0.3819	nan	0.01s	works	1
7	Lexicographic	P11	10000	1.2100	nan	nan	0.01s	works	1
7	Lexicographic	P7	10000	0.2100	0.3524	nan	0.01s	works	2
8	TOPSIS	P12/P4	0	nan	nan	nan	0.00s	works	5
9	VIKOR	P12/P4	0	nan	nan	nan	0.00s	works	5
10	VEGA	P2	10000	1.0015	0.0896	0.0055	0.05s	works	91
10	VEGA	P3	10000	0.1090	0.6101	nan	0.09s	works	1
11	MOGA	P3	10000	0.5077	0.0237	0.0186	0.50s	works	35
11	MOGA	P5	10000	1.1911	0.0172	0.0242	0.48s	works	53
12	NSGA	P5	10000	1.1913	0.0140	0.0183	0.46s	works	65
12	NSGA	P7	10000	0.6267	0.0441	0.0464	0.39s	works	23
13	NPGA	P3	10000	0.4217	0.0635	0.0309	0.97s	works	23
13	NPGA	P5	10000	1.1915	0.0145	0.0499	1.09s	works	46
14	NSGA-II	P3	10000	0.5376	0.0045	0.0042	0.87s	works	100
14	NSGA-II	P5	10000	1.1950	0.0062	0.0074	0.87s	works	98
14	NSGA-II	P6	10000	1.0200	0.0053	0.0056	0.84s	works	100

Task B — Master comparison table with evidence

Rows are grouped by family. `Evidence` is filled automatically from the A2 runs (HV / IGD / Spacing / runtime) — cite these numbers (and the figure of the A2 cell) next to at least two advantages and two disadvantages when you write the report.

```
B = {  # idx: (what it does, [4 advantages], [4 disadvantages], best choice, avoid)
 1: ("Minimize w·f for a sweep of weights", ["simple, any solver", "one cheap scalar solve per point", "exact points on convex
    ["misses non-convex regions", "equal weight steps give uneven spacing", "needs objective scaling", "no diversity control,
 2: ("Minimize one objective with the others bounded by ε, sweep ε", ["reaches non-convex fronts", "ε grid controls spacing", "
    ["needs ε ranges (ideal/nadir)", "infeasible ε fail", "grid grows as k^(M-1)", "may return weakly Pareto points"], "non-cc
 3: ("Maximize improvement over a feasible reference point", ["always dominates the start point", "finds non-convex points", "n
    ["result depends on the start point", "extra variables/constraints", "points cluster near x0", "needs many starts for cove
 4: ("Minimize weighted distance to the ideal point (p=1,2,∞)", ["p=∞ reaches non-convex fronts", "p tunes the compromise", "we
    ["needs the ideal point", "p=∞ is non-smooth (epigraph)", "p=1 equals weighted sum", "Tchebycheff may give weak Pareto poi
 5: ("Maximize the decision-maker's utility", ["one preferred answer", "single fast solve", "encodes trade-offs directly", "any
    ["utility hard to elicit", "wrong utility gives wrong answer silently", "no picture of the trade-off", "local solvers can
 6: ("Targets with deviation variables d⁻, d⁺, minimize weighted deviations", ["natural for target-driven DMs", "an LP for line
    ["depends on targets and weights", "deviations need normalization", "can return dominated solutions", "no front"], "target
 7: ("Optimize objectives in strict priority order", ["simple logic", "sequence of cheap LPs", "no weights", "ties broken by la
    ["lower priorities ignored unless ties", "very tolerance-sensitive", "single solution", "strict priority is rarely realist
 8: ("Rank alternatives by closeness to ideal and anti-ideal", ["no optimizer needed", "cheap", "easy to explain", "mixed min/m
    ["not an optimizer", "rank reversal possible", "compensatory: bad criterion can be masked", "weight- and normalization-dep
 9: ("Compromise ranking from S, R, Q with parameter v", ["explicit compromise via v", "regret term protects against a very bad
    ["not an optimizer", "depends on weights and v", "Q depends on the candidate set", "compromise may be non-unique"], "risk-
 10: ("Sub-populations each selected by one objective", ["simplest MOEA", "no sorting needed", "binary/real/perm encodings", "i
    ["collapses to the extremes", "no diversity mechanism", "non-elitist", "middle of the front under-represented"], "teaching
 11: ("Rank = dominators+1, fitness sharing", ["Pareto ranking reaches non-convex fronts", "sharing preserves spread", "simple
    ["σ_share must be tuned", "non-elitist", "O(N²) sharing", "rank ties flatten selection pressure"], "small bi-objective prc
 12: ("Non-dominated fronts with sharing inside each front", ["explicit front layering", "sharing inside fronts", "non-convex f
    ["O(MN³) sorting", "σ_share tuning", "non-elitist", "slower than NSGA-II"], "baseline comparison", "large populations"),
 13: ("Pareto tournament vs a comparison set, niche-count tiebreak", ["cheap O(N·t_dom) selection", "no explicit ranking", "non
```

```
        ["needs t_dom and σ", "noisy tournaments", "non-elitist", "weak with many objectives"], "low-cost bi-objective runs", "man
    14: ("Fast non-dominated sort, crowding distance, elitism", ["elitist", "O(MN²) sorting", "parameter-free diversity", "strong
        ["crowding fails for 4+ objectives", "most points non-dominated when M is large", "ignores local geometry", "favours bound
    15: ("Reference points replace crowding distance", ["scales to many objectives", "uniform spread", "elitist", "adaptive normal
        ["population tied to reference points", "empty niches on irregular fronts", "more complex code", "sensitive to divisions p
    16: ("Strength + raw fitness + k-th neighbour density, fixed archive", ["fine-grained fitness", "truncation keeps boundaries",
        ["O(N² log N) per generation", "k-th neighbour cost", "archive size parameter", "weaker for many objectives"], "bi/tri-obj
    17: ("Grid regions of the archive, region-based selection", ["cheap grid density", "small internal population", "region select
        ["grid divisions to tune", "moving grid bounds", "cells grow as div^M", "depends on archive size"], "low-dimensional objec
    18: ("Many Tchebycheff sub-problems with neighbour cooperation", ["cheap O(N·T) per generation", "neighbours share information
        ["weights must match front shape", "T, δ, nr sensitivity", "poor on disconnected fronts", "needs ideal point z*"], "regula
    19: ("MOEA/D with DE/rand/1 variation", ["DE handles complicated Pareto sets", "few operators", "good with variable linkage",
        ["real-valued only", "F, CR to tune", "diversity loss if nr large", "needs bound repair"], "continuous problems with linka
    20: ("MOEA/D with ant-colony route construction", ["constructive search suits routing", "neighbour pheromone sharing", "native
        ["N·n² pheromone memory", "many ACO parameters", "slow per evaluation", "only constructive problems"], "TSP/routing (P8)",
    21: ("NSGA-II with Deb's feasibility rules", ["no penalty parameters", "cheap rules", "elitist, spread preserved", "works with
        ["infeasible diversity lost", "needs a CV scale", "can stall on disconnected feasible regions", "infeasible-only starts dr
    22: ("Particles guided by personal best and archive leader", ["fast convergence", "few parameters", "archive leader selection"
        ["premature convergence", "grid parameters", "needs relaxation for discrete", "needs velocity clamping"], "smooth continuc
    23: ("α, β, δ leaders from the archive, encircling", ["few parameters", "archive-guided leaders", "exploration→exploitation by
        ["diversity depends on leader choice", "late exploitation only", "continuous only", "can cluster"], "quick continuous prob
    24: ("Employed, onlooker, scout bees with dominance greedy choice", ["few parameters", "scouts restore diversity", "one-dimens
        ["slow convergence", "greedy rule discards non-dominated moves", "2N evaluations/iteration", "limit tuning"], "multimodal
    25: ("Two pheromone matrices, probabilistic route construction", ["natural for routing", "two matrices encode objectives", "ar
        ["O(n²) memory", "slow iterations", "α, β, ρ tuning", "pheromone stagnation"], "routing/graph problems (P8)", "continuous
    26: ("Neighbour moves, Boltzmann acceptance, cooling", ["very low memory", "simple", "escapes local optima", "any encoding"],
        ["cooling schedule tuning", "chains cooperate weakly", "slow in many variables", "needs weights for energy"], "memory-limi
    27: ("DE mutation/crossover; trial replaces target only if it dominates", ["few parameters", "strong on continuous", "self-sca
        ["no diversity operator", "F, CR to tune", "continuous only", "stalls if no trial dominates"], "continuous problems with s
    28: ("Teacher and learner phases, shrinking teaching factor", ["no algorithm-specific parameters", "two complementary phases",
        ["2N evaluations/generation", "teacher choice biases", "continuous only", "mean move can collapse diversity"], "parameter-
    29: ("Frequency-driven velocity, loudness and pulse-rate schedules", ["adaptive explore→exploit", "simple updates", "archive g
        ["many parameters", "leader dependence", "may stagnate", "needs mapping for discrete"], "smooth continuous (P3, P4)", "dis
    30: ("NSGA-II global search plus hill-climbing", ["local search sharpens convergence", "NSGA-II diversity", "any encoding with
        ["extra evaluations", "more parameters", "premature convergence risk", "neighbour step scale matters"], "mixed global/loca
    31: ("Angle-penalized distance with adaptive reference vectors", ["balances convergence/diversity over time", "scales to many
        ["population can shrink", "needs total generations", "α, fr parameters", "scaling issues on irregular fronts"], "many-obje
    32: ("ε-indicator fitness, no diversity step", ["no diversity operator", "any indicator plugs in", "simple fitness", "good for
        ["κ sensitive", "O(N²) fitness", "indicator bias", "needs scaled objectives"], "bi/many-objective with scaled objectives (
    33: ("Steady-state, discard the least hypervolume contributor", ["excellent spread and convergence", "steady-state", "monotone
        ["HV costly for 4+ objectives", "reference-point sensitive", "serial evaluations", "slow wall-clock"], "2-3 objectives, qu
    34: ("Monte-Carlo hypervolume-contribution fitness", ["scales HV to many objectives", "time controllable via samples", "no div
        ["sampling noise", "S tuning", "chunk removal is approximate", "slow for small M"], "many objectives with indicator qualit
    35: ("NSGA-II with reference-point preference and ε-clearing", ["focuses on the DM region", "saves evaluations", "ε-clearing k
        ["wrong reference misguides search", "ε needs tuning", "no global front", "DM must provide a point"], "DM with a target (P
    36: ("Min-norm combination of gradients = common descent direction", ["no weights", "guaranteed common descent", "stops at Par
        ["needs gradients", "local only", "QP cost grows with tasks", "favours small-gradient tasks"], "differentiable multi-task
    37: ("Adaptive task weights from gradient norms", ["balances gradient scales", "learns weights online", "cheap", "works in dee
        ["needs shared-layer gradient norms", "α and weight-lr to tune", "weights can hit the floor", "heuristic, no Pareto guaran
}
assert len(B) == 37 and all(len(v[1]) >= 4 and len(v[2]) >= 4 for v in B.values())

def evidence(idx):
    rs = [r for r in LOG if r["idx"] == idx]
    if not rs: return "-"
    return "; ".join(f"{r['prob']}: HV {r['hv']:.3f}" + ("" if np.isnan(r['igd']) else f", IGD {r['igd']:.3f}") + ("" if np.isr
for fam in dict.fromkeys(FAMILY.values()):
    display(Markdown(f"**{fam}**"))
    table([(i, NAME[i], B[i][0], "; ".join(B[i][1]), "; ".join(B[i][2]), B[i][3], B[i][4], evidence(i)) for i in range(1, 38) i
           ["#", "Algorithm", "What it does", "Advantages", "Disadvantages", "Best choice", "Avoid", "Evidence (own runs)"])
```

Classical

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
1	Weighted Sum	Minimize w·f for a sweep of weights	simple, any solver; one cheap scalar solve per point; exact points on convex fronts; scales to many objectives per solve	misses non-convex regions; equal weight steps give uneven spacing; needs objective scaling; no diversity control, many solves	convex/linear problems (P1, P4, P7)	non-convex or disconnected fronts (P3)	P1: HV 1.006, IGD 0.040, Sp 0.018, 0.1s; P3: HV 0.210, IGD 0.354, 0.2s; P4: HV 0.905, IGD 0.054, Sp 0.110, 0.3s
2	ε-Constraint	Minimize one objective with the others bounded by ε, sweep ε	reaches non-convex fronts; ε grid controls spacing; handles constraints naturally; one objective per solve	needs ε ranges (ideal/nadir); infeasible ε fail; grid grows as k^(M-1); may return weakly Pareto points	non-convex bi-objective with known ranges (P3, P5)	many objectives (grid explosion)	P2: HV 1.056, IGD 0.111, Sp 0.182, 0.5s; P3: HV 0.495, IGD 0.037, Sp 0.039, 0.6s; P5: HV 1.137, IGD 0.069, Sp 0.024, 2.3s
3	Benson's Method	Maximize improvement over a feasible reference point	always dominates the start point; finds non-convex points; no weights; any feasible start works	result depends on the start point; extra variables/constraints; points cluster near x0; needs many starts for coverage	improving an existing design	when an even front is needed	P1: HV 0.984, IGD 0.061, Sp 0.069, 0.3s; P7: HV 0.642, IGD 0.042, Sp 0.073, 0.3s
4	Weight Metric (p-norm, Tchebycheff)	Minimize weighted distance to the ideal point (p=1,2,∞)	p=∞ reaches non-convex fronts; p tunes the compromise; weights give DM control; every Pareto point reachable (p=∞)	needs the ideal point; p=∞ is non-smooth (epigraph); p=1 equals weighted sum; Tchebycheff may give weak Pareto points	non-convex fronts with DM weights (P3)	when no ideal point is available	P1: HV 1.007, IGD 0.056, Sp 0.060, 0.4s; P3: HV 0.495, IGD 0.036, Sp 0.021, 1.1s
5	Value Function	Maximize the decision-maker's utility	one preferred answer; single fast solve; encodes trade-offs directly; any smooth solver	utility hard to elicit; wrong utility gives wrong answer silently; no picture of the trade-off; local solvers can be trapped	DM with a known utility (P4, P14)	exploration with unknown preferences	P4: HV 0.832, IGD 0.190, Sp 0.005, 0.2s; P14: HV 0.538, IGD 0.118, Sp 0.000, 0.2s
6	Goal Programming	Targets with deviation variables d^-, d^+, minimize weighted deviations	natural for target-driven DMs; an LP for linear goals; reports shortfalls explicitly; handles unattainable goals	depends on targets and weights; deviations need normalization; can return dominated solutions; no front	target planning (P11, P7)	when targets are unknown	P11: HV 1.210, 0.0s; P7: HV 0.341, IGD 0.382, 0.0s
7	Lexicographic	Optimize objectives in strict priority order	simple logic; sequence of cheap LPs; no weights; ties broken by later priorities	lower priorities ignored unless ties; very tolerance-sensitive; single solution; strict priority is rarely realistic	clear priority order (P11)	balanced trade-offs	P11: HV 1.210, 0.0s; P7: HV 0.210, IGD 0.352, 0.0s

Ranking/MCDM

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
8	TOPSIS	Rank alternatives by closeness to ideal and anti-ideal	no optimizer needed; cheap; easy to explain; mixed min/max criteria	not an optimizer; rank reversal possible; compensatory: bad criterion can be masked; weight- and normalization-dependent	ranking a finite front or supplier list (P12)	generating solutions	
9	VIKOR	Compromise ranking from S, R, Q with parameter v	explicit compromise via v; regret term protects against a very bad criterion; acceptability checks; cheap	not an optimizer; depends on weights and v; Q depends on the candidate set; compromise may be non-unique	risk-averse supplier choice (P12)	generating solutions	

Non-elitist EA

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
10	VEGA	Sub-populations each selected by one objective	simplest MOEA; no sorting needed; binary/real/perm encodings; illustrates population search	collapses to the extremes; no diversity mechanism; non-elitist; middle of the front under-represented	teaching / baseline	any serious front approximation	P2: HV 1.001, IGD 0.090, Sp 0.005, 0.1s; P3: HV 0.109, IGD 0.610, 0.1s
11	MOGA	Rank = dominators+1, fitness sharing	Pareto ranking reaches non-convex fronts; sharing preserves spread; simple fitness; any encoding	σ_share must be tuned; non-elitist; O(N²) sharing; rank ties flatten selection pressure	small bi-objective problems	many objectives, tight budgets	P3: HV 0.508, IGD 0.024, Sp 0.019, 0.5s; P5: HV 1.191, IGD 0.017, Sp 0.024, 0.5s
12	NSGA	Non-dominated fronts with sharing inside each front	explicit front layering; sharing inside fronts; non-convex fronts; historic	O(MN³) sorting; σ_share tuning; non-elitist; slower than NSGA-II	baseline comparison	large populations	P5: HV 1.191, IGD 0.014, Sp 0.018, 0.5s; P7: HV 0.627, IGD 0.044, Sp

		inside each front	baseline	than NSGA-II			IGD 0.044, Sp 0.046, 0.4s
13	NPGA	Pareto tournament vs a comparison set, niche-count tiebreak	cheap O(N·t_dom) selection; no explicit ranking; non-convex capable; niching via counts	needs t_dom and σ; noisy tournaments; non-elitist; weak with many objectives	low-cost bi-objective runs	many objectives	P3: HV 0.422, IGD 0.064, Sp 0.031, 1.0s; P5: HV 1.192, IGD 0.015, Sp 0.050, 1.1s

Elitist EA

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
14	NSGA-II	Fast non-dominated sort, crowding distance, elitism	elitist; $O(MN^2)$ sorting; parameter-free diversity; strong default for 2-3 objectives	crowding fails for 4+ objectives; most points non-dominated when M is large; ignores local geometry; favours boundaries	default for $M \le 3$ (P3, P5, P6)	$M \ge 5$	P3: HV 0.538, IGD 0.005, Sp 0.004, 0.9s; P5: HV 1.195, IGD 0.006, Sp 0.007, 0.9s; P6: HV 1.020, IGD 0.005, Sp 0.006, 0.8s
16	SPEA2	Strength + raw fitness + k-th neighbour density, fixed archive	fine-grained fitness; truncation keeps boundaries; elitist; good spread	$O(N^2 \log N)$ per generation; k-th neighbour cost; archive size parameter; weaker for many objectives	bi/tri-objective with spread priority (P4, P5)	large many-objective	P4: HV 0.970, IGD 0.004, Sp 0.005, 4.0s; P5: HV 1.195, IGD 0.004, Sp 0.007, 4.9s
17	PESA-II	Grid regions of the archive, region-based selection	cheap grid density; small internal population; region selection spreads search; elitist archive	grid divisions to tune; moving grid bounds; cells grow as div^M; depends on archive size	low-dimensional objective spaces	$M \ge 5$	P3: HV 0.525, IGD 0.011, Sp 0.008, 1.7s; P5: HV 1.192, IGD 0.012, Sp 0.008, 1.7s

Elitist many-obj

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
15	NSGA-III	Reference points replace crowding distance	scales to many objectives; uniform spread; elitist; adaptive normalization	population tied to reference points; empty niches on irregular fronts; more complex code; sensitive to divisions p	many-objective with regular fronts (P10)	irregular/degenerate fronts	P10_3: HV 0.733, IGD 0.057, Sp 0.034, 1.6s; P10_5: HV 1.152, IGD 0.220, Sp 0.068, 2.4s

Decomposition

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
18	MOEA/D	Many Tchebycheff sub-problems with neighbour cooperation	cheap $O(N \cdot T)$ per generation; neighbours share information; uniform spread with good weights; scales to many objectives	weights must match front shape; T, δ, nr sensitivity; poor on disconnected fronts; needs ideal point z^*	regular fronts, many objectives (P3, P10)	disconnected or degenerate fronts	P3: HV 0.538, IGD 0.004, Sp 0.002, 7.2s; P10_3: HV 0.710, IGD 0.074, Sp 0.054, 7.1s
19	MOEA/D-DE	MOEA/D with DE/rand/1 variation	DE handles complicated Pareto sets; few operators; good with variable linkage; keeps decomposition efficiency	real-valued only; F, CR to tune; diversity loss if nr large; needs bound repair	continuous problems with linkage (P5)	discrete encodings	P3: HV 0.535, IGD 0.004, Sp 0.003, 5.9s; P5: HV 1.193, IGD 0.067, Sp 0.083, 5.7s
20	MOEA/D-ACO	MOEA/D with ant-colony route construction	constructive search suits routing; neighbour pheromone sharing; native permutations; no crossover repair	$N \cdot n^2$ pheromone memory; many ACO parameters; slow per evaluation; only constructive problems	TSP/routing (P8)	continuous problems	P8: HV 0.486, Sp 0.159, 7.5s

Constrained

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
21	Constrained NSGA-II	NSGA-II with Deb's feasibility rules	no penalty parameters; cheap rules; elitist, spread preserved; works with narrow feasible sets	infeasible diversity lost; needs a CV scale; can stall on disconnected feasible regions; infeasible-only starts drift to least violation	constrained design (P5, P6)	unconstrained problems (no benefit)	P5: HV 1.195, IGD 0.006, Sp 0.007, 0.5s; P6: HV 1.020, IGD 0.005, Sp 0.005, 0.5s; P7: HV 0.700, IGD 0.006, Sp 0.005, 0.6s

Swarm

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
22	MOPSO	Particles guided by personal best and archive leader	fast convergence; few parameters; archive leader selection; easy real coding	premature convergence; grid parameters; needs relaxation for discrete; needs velocity clamping	smooth continuous problems (P4)	discrete or deceptive problems	P4: HV 0.965, IGD 0.007, Sp 0.007, 0.8s; P5: HV 1.194, IGD 0.008, Sp 0.009, 0.8s
		α, β, δ leaders	few parameters; archive-guided	diversity depends on leader		quick	P5: HV 1.193, IGD 0.013, Sp

23	MOGWO	α, β, δ leaders from the archive, encircling	leaders; exploration→exploitation by a; simple	diversity depends on leader choice; late exploitation only; continuous only; can cluster	quick continuous problems (P5)	discrete problems	IGD 0.012, Sp 0.009, 0.6s; P9: HV 1.180, Sp 0.033, 19.2s
24	MOABC	Employed, onlooker, scout bees with dominance greedy choice	few parameters; scouts restore diversity; one-dimension local search; greedy dominance	slow convergence; greedy rule discards non-dominated moves; 2N evaluations/iteration; limit tuning	multimodal continuous (P3, P6)	tight budgets	P3: HV 0.451, IGD 0.051, Sp 0.079, 0.3s; P6: HV 1.018, IGD 0.007, Sp 0.008, 0.8s
25	MOACO	Two pheromone matrices, probabilistic route construction	natural for routing; two matrices encode objectives; archive deposits; no crossover repair	O(n²) memory; slow iterations; α, β, ρ tuning; pheromone stagnation	routing/graph problems (P8)	continuous problems	P8: HV 0.709, Sp 0.120, 5.7s
29	MOBA	Frequency-driven velocity, loudness and pulse-rate schedules	adaptive explore→exploit; simple updates; archive guidance; fast convergence	many parameters; leader dependence; may stagnate; needs mapping for discrete	smooth continuous (P3, P4)	discrete problems	P3: HV 0.177, IGD 0.386, Sp 0.004, 0.4s; P4: HV 0.958, IGD 0.014, Sp 0.007, 1.4s

Physics-inspired

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
26	MOSA	Neighbour moves, Boltzmann acceptance, cooling	very low memory; simple; escapes local optima; any encoding	cooling schedule tuning; chains cooperate weakly; slow in many variables; needs weights for energy	memory-limited, rugged problems	tight budgets	P3: HV 0.450, IGD 0.074, Sp 0.072, 0.2s; P5: HV 1.195, IGD 0.008, Sp 0.009, 0.7s

Evolutionary

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
27	MODE	DE mutation/crossover; trial replaces target only if it dominates	few parameters; strong on continuous; self-scaling steps; simple replacement	no diversity operator; F, CR to tune; continuous only; stalls if no trial dominates	continuous problems with smooth fronts	need for even spread	P3: HV 0.110, IGD 0.610, 0.1s; P5: HV 1.181, IGD 0.031, Sp 0.016, 0.2s

Human-inspired

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
28	MO-TLBO	Teacher and learner phases, shrinking teaching factor	no algorithm-specific parameters; two complementary phases; elitist merge; simple	2N evaluations/generation; teacher choice biases; continuous only; mean move can collapse diversity	parameter-light continuous runs	discrete problems	P3: HV 0.530, IGD 0.010, Sp 0.011, 0.3s; P5: HV 1.193, IGD 0.013, Sp 0.015, 0.3s

Hybrid

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
30	MOMA	NSGA-II global search plus hill-climbing	local search sharpens convergence; NSGA-II diversity; any encoding with a neighbour move; fewer generations needed	extra evaluations; more parameters; premature convergence risk; neighbour step scale matters	mixed global/local landscapes (P5, P9)	very tight budgets	P5: HV 1.195, IGD 0.006, Sp 0.007, 0.4s; P9: HV 1.190, Sp 0.029, 20.3s

Many-objective

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
31	RVEA	Angle-penalized distance with adaptive reference vectors	balances convergence/diversity over time; scales to many objectives; adaptive vectors; elitist	population can shrink; needs total generations; α, fr parameters; scaling issues on irregular fronts	many-objective (P10)	irregular fronts	P10_3: HV 0.720, IGD 0.062, Sp 0.024, 0.2s; P10_5: HV 1.212, IGD 0.210, Sp 0.049, 0.2s

Indicator-based

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
32	IBEA	ε-indicator fitness, no diversity step	no diversity operator; any indicator plugs in; simple fitness; good for many objectives	κ sensitive; O(N²) fitness; indicator bias; needs scaled objectives	bi/many-objective with scaled objectives (P14)	unscaled objectives	P3: HV 0.537, IGD 0.009, Sp 0.015, 0.5s; P14: HV 0.704, IGD 0.004, Sp 0.003, 0.5s
33	SMS-EMOA	Steady-state, discard the least hypervolume contributor	excellent spread and convergence; steady-state; monotone hypervolume; great for 2-3 objectives	HV costly for 4+ objectives; reference-point sensitive; serial evaluations; slow wall-clock	2-3 objectives, quality first (P3, P14)	M ≥ 5	P3: HV 0.539, IGD 0.004, Sp 0.005, 12.8s; P14: HV 0.705, IGD 0.004, Sp 0.001, 12.0s
			scales HV to many	sampling noise; S tuning;			

#	Algorithm	What it does	Advantages	Disadvantages	Best choice	Avoid	Evidence (own runs)
34	HypE	Monte-Carlo hypervolume contribution fitness	controllable via samples; no diversity operator; works for M ≥ 5	objectives; time	sampling noise; S tuning; chunk removal is approximate; slow for small M	many objectives with indicator quality (P10)	tiny budgets	P10_3: HV 0.698, IGD 0.123, Sp 0.076, 6.8s
Preference-based							
35	R-NSGA-II	NSGA-II with reference-point preference and ε-clearing	focuses on the DM region; saves evaluations; ε-clearing keeps spread; reuses NSGA-II	wrong reference misguides search; ε needs tuning; no global front; DM must provide	DM with a target (P5, P14)	no preference information	P5: HV 1.194, IGD 0.008, Sp 0.010, 2.7s; P14: HV 0.699, IGD 0.007, Sp 0.011, 2.2s
Gradient-based							

How the families differ. *Classical* methods turn the problem into single-objective solves: they are fast and exact when their assumptions hold (weighted sum needs convexity, ε-constraint and Tchebycheff do not) but produce one point per solve, so diversity is whatever the weight/ε grid gives. *Non-elitist EAs* (VEGA, MOGA, NSGA, NPGA) use a population and obtain diversity from sharing or niching, but without elitism they lose good points and converge slowly; VEGA has no diversity at all and drifts to the extremes. *Elitist Pareto EAs* (NSGA-II, SPEA2, PESA-II) keep the best set and use crowding, density or grid counts for diversity; they handle non-convex fronts but their diversity measures degrade as M grows. *Decomposition* (MOEA/D family) gets diversity from the weight vectors and is cheap, but depends on the weights matching the front shape. *Many-objective and indicator methods* (NSGA-III, RVEA, IBEA, SMS-EMOA, HypE) replace crowding by reference points/vectors or indicator contributions, which keeps selection pressure when most solutions are mutually non-dominated. *Metaheuristics* (SwarmO, SA, DE, TLBO, EA) mainly differ in the variation operator; diversity comes from an archive (grid) and constraints from penalties unless Deb's rules are used (alg. 21). *Gradient methods* (MGDA, GradNorm) need differentiability and are local, but are the only ones that scale to millions of parameters. For constraints: Deb's rules (21) avoid penalty tuning; for non-convex fronts, Pareto-based, ε-constraint and Tchebycheff approaches work while weighted sum cannot.

Task C — Deep dive on your three assigned algorithms

Set `ROLL` to the last two digits of your registration number. The three algorithms are `((R-1) mod 37)+1`, `((R+11) mod 37)+1`, `((R+23) mod 37)+1`. Each needs: C1 study card (write it in your own words — template below), C2 hand trace reproduced by code, C3 sensitivity study (3 values × seeds).

```
ROLL = 44                                          # <-- set your roll number (00-99); R=7 gives algorithms 7, 19, 31
DEEP = [((ROLL - 1) % 37) + 1, ((ROLL + 11) % 37) + 1, ((ROLL + 23) % 37) + 1]
assert [((7 - 1) % 37) + 1, ((7 + 11) % 37) + 1, ((7 + 23) % 37) + 1] == [7, 19, 31]       # example from the PDF
assert [((0 - 1) % 37) + 1, ((0 + 11) % 37) + 1, ((0 + 23) % 37) + 1] == [37, 12, 24]      # R=00 example
assert [((99 - 1) % 37) + 1, ((99 + 11) % 37) + 1, ((99 + 23) % 37) + 1] == [25, 37, 12]   # R=99 example
print("Deep-dive algorithms:", [(i, NAME[i]) for i in DEEP])

CLS = {10: VEGA, 11: MOGA, 12: NSGA, 13: NPGA, 14: NSGA2, 15: NSGA3, 16: SPEA2, 17: PESA2, 18: MOEAD, 19: MOEADDE, 20: MOEAD_ACO
       22: MOPSO, 23: MOGWO, 24: MOABC, 25: MOACO, 26: MOSA, 27: MODE, 28: MOTLBO, 29: MOBA, 30: MOMA, 31: RVEA, 32: IBEA, 33: SM
for i in DEEP:                                     # docstrings = steps + default parameters (reuse in the study card)
    print(f"\n[{i}] {NAME[i]} — {FAMILY[i]}"); print("   ", (CLS[i].__doc__ if i in CLS else "classical / gradient method: see i
```

```
Deep-dive algorithms: [(7, 'Lexicographic'), (19, 'MOEA/D-DE'), (31, 'RVEA')]

[7] Lexicographic — Classical
    classical / gradient method: see its cell above

[19] MOEA/D-DE — Decomposition
    MOEA/D with DE/rand/1 variation: v = x_r1 + F (x_r2 - x_r3), binomial crossover CR=1, then polynomial mutation. F=0.5.

[31] RVEA — Many-objective
    Steps: 1) reference vectors (Das-Dennis)  2) random mating  3) translate by ideal, assign by angle  4) APD=(1+M(t/T)^a * the
```

C1. Study card (fill in your own words)

Field	Your entry
Full form, year, authors, problem it fixed	
Core idea (≤ 5 lines) + flow diagram	
Numbered steps with formulas	
Parameters: effect if too small / too large	
Convergence · diversity · constraints · non-convex fronts · > 3 objectives	
Time complexity per generation / per solve	

C2. Hand trace

The code prints two full iterations on a tiny instance (population 4 on P1, or two weight/ε settings for classical methods). Do the same two iterations by hand and compare every number.

```
def hand_trace(idx):
    np.set_printoptions(precision=4, suppress=True); print(f"===== Hand trace of [{idx}] {NAME[idx]} =====")
    if idx in CLS:
        P = P8 if idx in (20, 25) else P1; a = CLS[idx](P, budget=10 ** 6, seed=0, N=4) if idx not in (15, 31) else CLS[idx](P,
        for g in range(3):
            if g: a.step(); a.gen += 1
            print(f"-- generation {g} ({'initial' if g == 0 else 'after step'}): x ->  objectives (normalized)")
            for x, f in zip(a.X[:6], a.F[:6]): print("   ", np.asarray(x).round(3).ravel() if P is P1 else list(map(int, x))[:8
    else:
```

```
            tr = {1: lambda: weighted_sum(P1, 0, 2), 2: lambda: eps_constraint(P1, 0, 2), 3: lambda: benson(P1, 0, 2), 4: lambda: w
                  5: lambda: value_function(P1, 0, (0.3, 0.7))}
            if idx in tr:
                X = tr[idx](); print("two settings -> x:", X.ravel(), " objectives:", P1.f(X)[0].round(4).tolist())
            elif idx == 6: print("goal programming (targets 2400, 600):", [goal_programming(P11.lin, [2400, 600], w)[0] for w in ([
            elif idx == 7: print("lexicographic orders:", lexicographic(P11.lin, [0, 1]), lexicographic(P11.lin, [1, 0]))
            elif idx in (8, 9):
                M = np.array([[1., 10.], [2., 5.], [3., 1.]]); w = np.array([.5, .5]); bn = np.array([False, True]); print("3 alter
                print("TOPSIS closeness:" if idx == 8 else "VIKOR Q:", (topsis(M, w, bn) if idx == 8 else vikor(M, w, bn)).round(4)
            else:
                print("toy path, first 3 steps:", toy_run("mgda" if idx == 36 else "gradnorm")[:4].tolist())
    for i in DEEP: hand_trace(i)
```

```
    ===== Hand trace of [7] Lexicographic =====
    lexicographic orders: [300. 200.] [300. 200.]
```

```
    ===== Hand trace of [19] MOEA/D-DE =====
    -- generation 0 (initial): x ->  objectives (normalized)
        [6.733] -> [0.8708 0.0045]
        [3.428] -> [0.0115 0.7974]
        [1.369] -> [0.1663 1.9819]
        [1.149] -> [0.2142 2.1398]
    -- generation 1 (after step): x ->  objectives (normalized)
        [7.041] -> [1.0205 0.0001]
        [6.001] -> [0.5628 0.0624]
        [3.921] -> [0.053  0.5926]
        [3.921] -> [0.053  0.5926]
    -- generation 2 (after step): x ->  objectives (normalized)
        [7.041] -> [1.0205 0.0001]
        [5.127] -> [0.2828 0.2192]
        [5.127] -> [0.2828 0.2192]
        [3.921] -> [0.053  0.5926]
    ===== Hand trace of [31] RVEA =====
    -- generation 0 (initial): x ->  objectives (normalized)
        [6.733] -> [0.8708 0.0045]
        [3.428] -> [0.0115 0.7974]
        [1.369] -> [0.1663 1.9819]
        [1.149] -> [0.2142 2.1398]
    -- generation 1 (after step): x ->  objectives (normalized)
        [3.428] -> [0.0115 0.7974]
        [6.733] -> [0.8708 0.0045]
    -- generation 2 (after step): x ->  objectives (normalized)
        [3.629] -> [0.0247 0.7104]
        [6.733] -> [0.8708 0.0045]
```

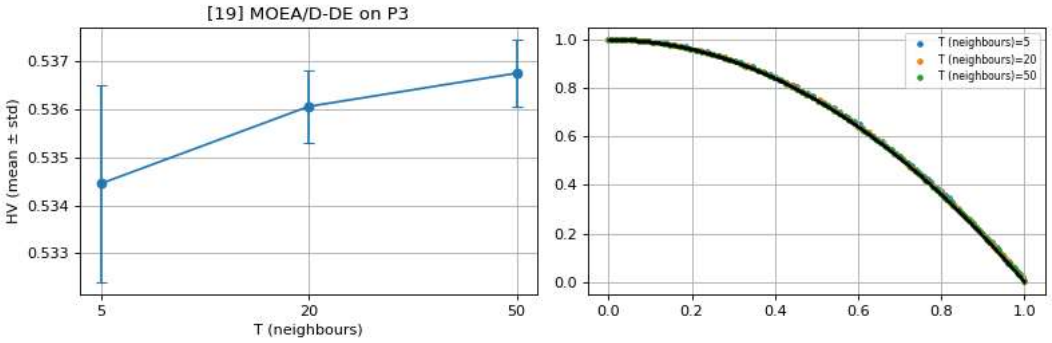
C3. Sensitivity study

Most important parameter of each deep-dive algorithm at three values, evaluated over the seeds in `RUNS` (10 in FULL mode). Mean ± std of HV on the algorithm's test problem.

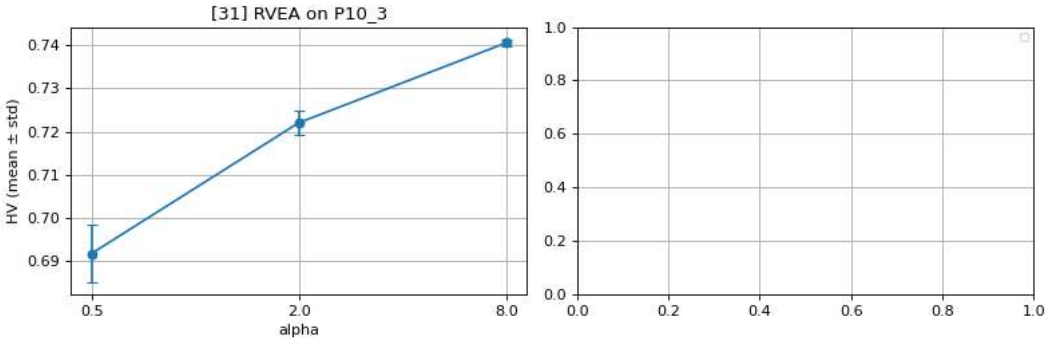
```
def mk(cls, **kw): return lambda P, s, b: cls(P, budget=b, seed=s, **kw).run().result()
SENS = {  # idx: (parameter, values, factory(value)->runner, problem)
 1: ("n_weights", [3, 6, 11], lambda v: (lambda P, s, b: finalize(P, weighted_sum(P, s, v))), "P4"),
 2: ("n_eps", [3, 6, 11], lambda v: (lambda P, s, b: finalize(P, eps_constraint(P, s, v))), "P5"),
 3: ("n_start", [3, 6, 12], lambda v: (lambda P, s, b: finalize(P, benson(P, s, v))), "P7"),
 4: ("n_weights", [3, 6, 11], lambda v: (lambda P, s, b: finalize(P, weight_metric(P, s, np.inf, v))), "P3"),
 10: ("N", [20, 50, 100], lambda v: mk(VEGA, N=v), "P3"), 11: ("sigma", [0.02, 0.1, 0.5], lambda v: mk(MOGA, sigma=v), "P3"),
 12: ("sigma", [0.02, 0.1, 0.5], lambda v: mk(NSGA, sigma=v), "P5"), 13: ("sigma", [0.02, 0.1, 0.5], lambda v: mk(NPGA, sigma=v)
 14: ("N", [20, 50, 100], lambda v: mk(NSGA2, N=v), "P3"), 15: ("divisions", [19, 49, 99], lambda v: mk(NSGA3, div=v), "P3"),
 16: ("N", [20, 50, 100], lambda v: mk(SPEA2, N=v), "P4"), 17: ("grid div", [5, 10, 20], lambda v: mk(PESA2, div=v), "P3"),
 18: ("T (neighbours)", [5, 20, 50], lambda v: mk(MOEAD, T=v), "P3"), 19: ("T (neighbours)", [5, 20, 50], lambda v: mk(MOEADDE,
 20: ("rho", [0.02, 0.1, 0.5], lambda v: mk(MOEAD_ACO, rho=v), "P8"), 21: ("N", [20, 50, 100], lambda v: mk(CNSGA2, N=v), "P5"),
 22: ("inertia w", [0.1, 0.4, 0.9], lambda v: mk(MOPSO, w=v), "P5"), 23: ("N", [20, 50, 100], lambda v: mk(MOGWO, N=v), "P5"),
 24: ("limit", [3, 10, 30], lambda v: mk(MOABC, limit=v), "P3"), 25: ("rho", [0.02, 0.1, 0.5], lambda v: mk(MOACO, rho=v), "P8")
 26: ("T0", [0.05, 0.5, 5.0], lambda v: mk(MOSA, T0=v), "P3"), 27: ("F", [0.2, 0.5, 0.9], lambda v: mk(MODE, Fm=v), "P3"),
 28: ("N", [20, 50, 100], lambda v: mk(MOTLBO, N=v), "P3"), 29: ("alpha", [0.5, 0.9, 0.99], lambda v: mk(MOBA, alpha=v), "P3"),
 30: ("ls_frac", [0.0, 0.1, 0.5], lambda v: mk(MOMA, ls_frac=v), "P5"), 31: ("alpha", [0.5, 2.0, 8.0], lambda v: mk(RVEA, alpha=
 32: ("kappa", [0.01, 0.05, 0.5], lambda v: mk(IBEA, kappa=v), "P3"), 33: ("N", [20, 50, 100], lambda v: mk(SMSEMOA, N=v), "P3")
 34: ("samples S", [200, 1000, 3000], lambda v: mk(HypE, S=v), "P10_3"), 35: ("eps", [0.001, 0.02, 0.2], lambda v: mk(RNSGA2, eps

def sensitivity(idx):
    """Run 3 parameter values x RUNS seeds; plot mean±std HV and one front per value."""
    if idx in (5, 6, 7, 8, 9, 36, 37):                    # deterministic or non-front methods: vary the input and report the an
        print(f"[{idx}] {NAME[idx]} is a deterministic method: vary its input parameter and compare the outputs")
        if idx == 5: [print("  a =", a, "->", P4.f(value_function(P4, 0, (a,)))[0].round(4).tolist()) for a in (0.2, 0.5, 0.8)]
        elif idx == 6: [print("  weights", w, "-> x =", goal_programming(P11.lin, [2400, 600], w)[0].round(2)) for w in ([1, 1],
        elif idx == 7: [print("  tolerance", t, "-> x =", lexicographic(P7.lin, [0, 1], t).round(2)) for t in (0.0, 0.05, 0.2)]
        elif idx in (8, 9): [print("  weight shift", d, "->", rank((topsis if idx == 8 else vikor)(Mx, np.array([.3 + d, .3 - d,
        else:
```

```
            for lr in (0.0005, 0.002, 0.005): h, _ = train_mtl("mgda" if idx == 36 else "gradnorm", 0, 300, lr); print(f"  lr={lr
        return
    par, vals, fac, pn = SENS[idx]; P = PROBS[pn]; b = BUDGET_MANY if P.m > 2 else BUDGET; res = {}; fig, ax = plt.subplots(1, 2
    for v in vals:
        H, S_, G = [], [], []
        for r in range(RUNS):
            Fn = fac(v)(P, SEED0 + r, b); h, sp, ig = metrics(P, Fn); H.append(h); S_.append(sp); G.append(ig)
            if r == 0 and P.m == 2: R = P.denorm(Fn); ax[1].scatter(R[:, 0], R[:, 1], s=10, label=f"{par}={v}")
        res[v] = (np.mean(H), np.std(H), np.nanmean(S_), np.nanmean(G))
    ax[0].errorbar(range(len(vals)), [res[v][0] for v in vals], [res[v][1] for v in vals], fmt="o-", capsize=4); ax[0].set_xtick
    ax[0].set_xlabel(par); ax[0].set_ylabel("HV (mean ± std)"); ax[0].set_title(f"[{idx}] {NAME[idx]} on {pn}")
    if P.pf is not None and P.m == 2: ax[1].plot(P.pf[:, 0], P.pf[:, 1], "k.", ms=2)
    ax[1].legend(fontsize=7); plt.tight_layout(); plt.show()
    table([(v, f"{res[v][0]:.4f} ± {res[v][1]:.4f}", f"{res[v][2]:.4f}", "-" if np.isnan(res[v][3]) else f"{res[v][3]:.4f}") for
for i in DEEP: sensitivity(i)
```

```
[7] Lexicographic is a deterministic method: vary its input parameter and compare the outputs
  tolerance 0.0 -> x = [70. 20.]
  tolerance 0.05 -> x = [66.5 23.5]
  tolerance 0.2 -> x = [56. 34.]
```



T (neighbours)	HV	Spacing	IGD
5	0.5344 ± 0.0020	0.0034	0.0049
20	0.5361 ± 0.0008	0.0031	0.0044
50	0.5368 ± 0.0007	0.0032	0.0042



alpha	HV	Spacing	IGD
0.5	0.6916 ± 0.0066	0.0255	0.0735
2.0	0.7220 ± 0.0028	0.0252	0.0612
8.0	0.7406 ± 0.0008	0.0271	0.0563

C3 interpretation (write 3–4 lines per algorithm): state which value was best, whether the change is monotone, and link it to the parameter's role in the algorithm steps (too small → ..., too large → ...).

Task D — Head-to-head comparison

Equal evaluation budget, `RUNS` seeds each, mean ± std of HV and Spacing (IGD where the true front is known). Classical methods 1–4 use a ≥ 10-point sweep; 5–7 and 8–9 and 36–37 are handled as the PDF specifies below.

```
RES = {}
def run_many(idx, runner, pn, budget=None):
    """Run one algorithm on one problem for all seeds; store metrics and the first front in RES[(pn, idx)]."""
```

```
        P = PROBS[pn]; b = budget or (BUDGET_MANY if P.m > 2 else BUDGET); H, S_, G, rt = [], [], [], []; front = None
        for r in range(RUNS):
            t = time.time(); Fn = runner(P, SEED0 + r, b); rt.append(time.time() - t); h, sp, ig = metrics(P, Fn); H.append(h); S_.
            front = Fn if front is None else front
        RES[(pn, idx)] = dict(hv=H, sp=S_, igd=G, front=front, rt=np.mean(rt))
    def rtab(pn, ids, title):
        """Ranked table (by mean HV, ties by IGD) for the algorithms of one experiment."""
        rows = sorted(ids, key=lambda i: (-round(np.mean(RES[(pn, i)]["hv"]), 3), np.nan_to_num(np.nanmean(RES[(pn, i)]["igd"]), na

        for k, i in enumerate(rows, 1):
            r = RES[(pn, i)]; ig = np.nanmean(r["igd"]); out.append((k, i, NAME[i], f"{np.mean(r['hv']):.4f} ± {np.std(r['hv']):.4f
                                                                    "-" if np.isnan(ig) else f"{ig:.4f} ± {np.nanstd(r['igd']):.4
        display(Markdown(f"**{title}**")); table(out, ["Rank", "#", "Algorithm", "HV (↑)", "Spacing (↓)", "IGD (↓)", "Runtime"]); r
    def plot_fronts(pn, ids, title, ncol=4):
        P = PROBS[pn]; n = len(ids); nr = int(np.ceil(n / ncol)); fig, axs = plt.subplots(nr, ncol, figsize=(3.4 * ncol, 2.8 * nr),
        for ax, i in zip(axs.ravel(), ids):
            F = RES[(pn, i)]["front"]; R = P.denorm(F) if len(F) else F
            if P.pf is not None: ax.plot(P.pf[:, 0], P.pf[:, 1], "k.", ms=1)
            if len(R): ax.scatter(R[:, 0], R[:, 1], s=6, c="C3")
            ax.set_title(f"[{i}] {NAME[i]}", fontsize=8)
        for ax in axs.ravel()[n:]: ax.axis("off")
        fig.suptitle(title); plt.tight_layout(); plt.show()

    def runner_for(i, pn):
        P = PROBS[pn]
        if i == 35: pt = P.pref if P.pref is not None else np.array([0.3, 0.7]); return lambda P_, s, b: RNSGA2(P_, budget=b, seed=
        if i >= 10: return cls_runner(CLS[i])
        sweep = {1: weighted_sum, 2: eps_constraint, 3: benson}
        if i == 4: return lambda P_, s, b: finalize(P_, weight_metric(P_, s, np.inf, 11))
        return lambda P_, s, b: finalize(P_, sweep[i](P_, s))
```

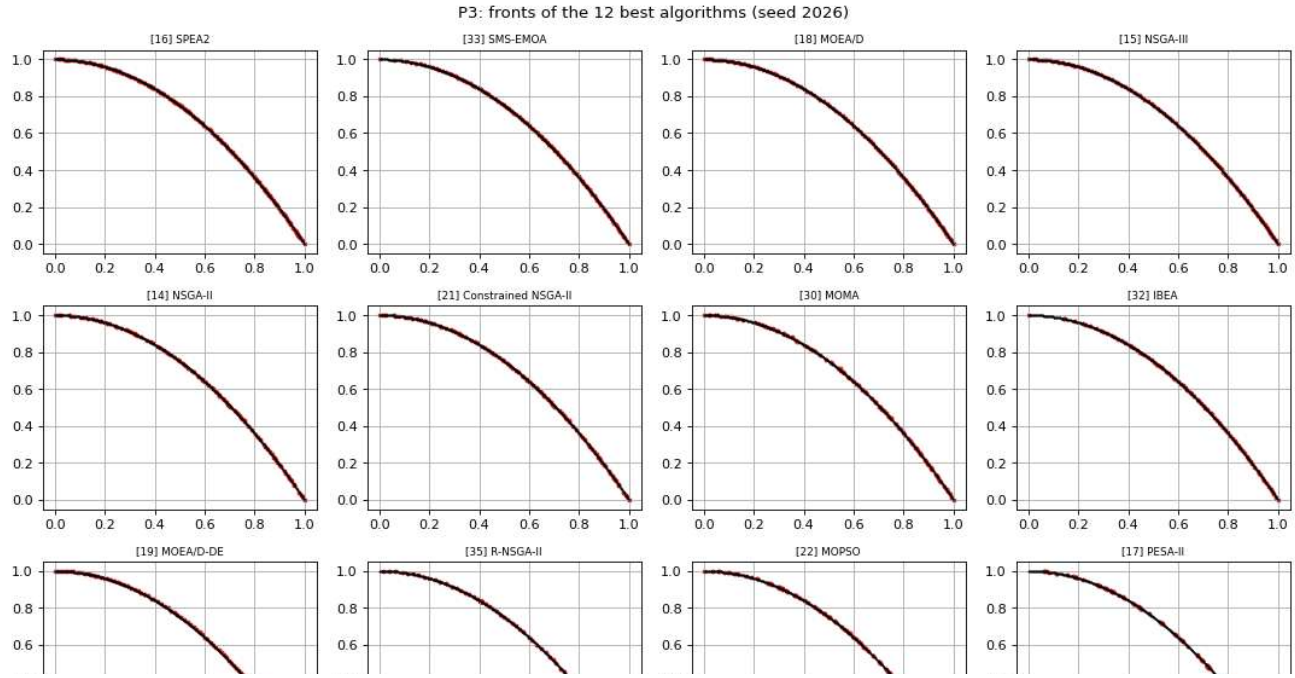
D1. Continuous bi-objective: P3 (unconstrained) and P5 (constrained)

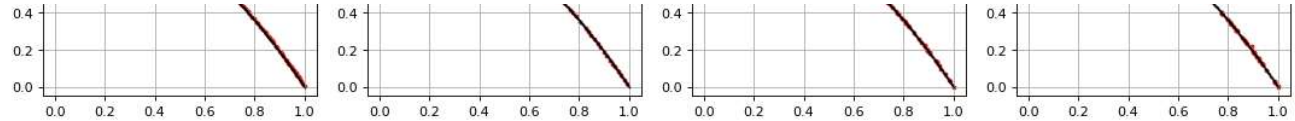
```
D1_IDS = [1, 2, 3, 4] + [i for i in range(10, 36) if i not in (20, 25)]
for pn in ("P3", "P5"):
    t0 = time.time()
    for i in D1_IDS: run_many(i, runner_for(i, pn), pn)
    rows = rtab(pn, D1_IDS, f"Ranking on {pn} ({'constrained' if pn == 'P5' else 'unconstrained'}), budget {BUDGET}, {RUNS} see
    plot_fronts(pn, rows[:12], f"{pn}: fronts of the 12 best algorithms (seed {SEED0})")
```

Ranking on P3 (unconstrained), budget 10000, 10 seeds

Rank	#	Algorithm	HV (↑)	Spacing (↓)	IGD (↓)	Runtime
1	16	SPEA2	0.5391 ± 0.0001	0.0051 ± 0.0002	0.0033 ± 0.0001	2.9s
2	33	SMS-EMOA	0.5388 ± 0.0000	0.0046 ± 0.0004	0.0043 ± 0.0001	12.3s
3	18	MOEA/D	0.5384 ± 0.0001	0.0025 ± 0.0002	0.0038 ± 0.0001	6.8s
4	15	NSGA-III	0.5382 ± 0.0001	0.0025 ± 0.0002	0.0038 ± 0.0000	0.7s
5	14	NSGA-II	0.5372 ± 0.0003	0.0053 ± 0.0007	0.0048 ± 0.0002	0.5s
6	21	Constrained NSGA-II	0.5372 ± 0.0003	0.0053 ± 0.0007	0.0048 ± 0.0002	0.5s
7	30	MOMA	0.5369 ± 0.0004	0.0055 ± 0.0006	0.0050 ± 0.0003	0.4s
8	32	IBEA	0.5372 ± 0.0002	0.0125 ± 0.0028	0.0082 ± 0.0007	0.5s
9	19	MOEA/D-DE	0.5361 ± 0.0008	0.0031 ± 0.0004	0.0044 ± 0.0002	5.5s
10	35	R-NSGA-II	0.5342 ± 0.0009	0.0108 ± 0.0004	0.0063 ± 0.0001	1.9s
11	22	MOPSO	0.5329 ± 0.0017	0.0073 ± 0.0007	0.0072 ± 0.0010	0.9s
12	17	PESA-II	0.5269 ± 0.0090	0.0082 ± 0.0015	0.0107 ± 0.0032	1.6s
13	12	NSGA	0.5213 ± 0.0043	0.0164 ± 0.0053	0.0164 ± 0.0038	0.2s
14	31	RVEA	0.5166 ± 0.0038	0.0127 ± 0.0053	0.0167 ± 0.0033	0.2s
15	11	MOGA	0.5046 ± 0.0047	0.0243 ± 0.0084	0.0257 ± 0.0048	0.2s
16	4	Weight Metric (p-norm, Tchebycheff)	0.4954 ± 0.0003	0.0204 ± 0.0006	0.0357 ± 0.0001	0.4s
17	2	ε-Constraint	0.4950 ± 0.0000	0.0388 ± 0.0000	0.0369 ± 0.0000	0.2s
18	24	MOABC	0.4559 ± 0.0339	0.0452 ± 0.0180	0.0488 ± 0.0152	0.3s
19	26	MOSA	0.4469 ± 0.0249	0.0385 ± 0.0172	0.0666 ± 0.0210	0.2s
20	13	NPGA	0.4449 ± 0.0158	0.0377 ± 0.0118	0.0548 ± 0.0087	0.4s
21	23	MOGWO	0.4028 ± 0.1917	0.0080 ± 0.0010	0.1906 ± 0.2743	0.8s
22	28	MO-TLBO	0.4007 ± 0.1705	0.0094 ± 0.0045	0.1589 ± 0.2111	0.3s
23	1	Weighted Sum	0.2100 ± 0.0000	0.6667 ± 0.0000	0.3543 ± 0.0000	0.2s
24	10	VEGA	0.1141 ± 0.0144	0.0636 ± 0.0377	0.5527 ± 0.0771	0.1s
25	3	Benson's Method	0.1100 ± 0.0000	nan ± nan	0.6097 ± 0.0000	0.1s
26	27	MODE	0.1098 ± 0.0002	nan ± nan	0.6098 ± 0.0001	0.1s
27	29	MOBA	0.0668 ± 0.1287	0.0074 ± 0.0072	0.7264 ± 0.3912	0.3s
28	34	HypE	0.0000 ± 0.0000	0.2133 ± 0.0841	2.7976 ± 0.2687	6.0s

```
P3 done in 445s
```

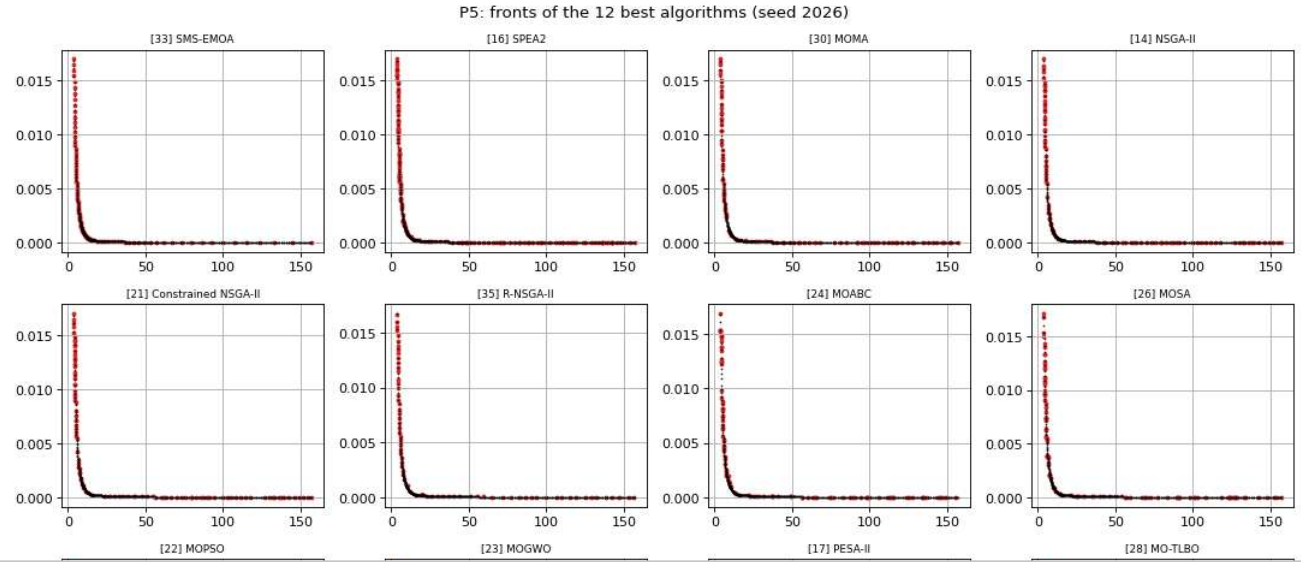


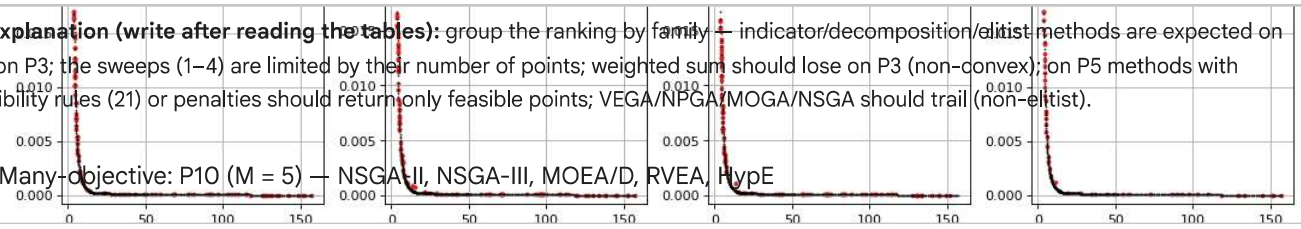


Ranking on P5 (constrained), budget 10000, 10 seeds

Rank	#	Algorithm	HV (↑)	Spacing (↓)	IGD (↓)	Runtime
1	33	SMS-EMOA	1.1956 ± 0.0000	0.0148 ± 0.0006	0.0062 ± 0.0002	11.6s
2	16	SPEA2	1.1954 ± 0.0001	0.0066 ± 0.0003	0.0040 ± 0.0002	4.4s
3	30	MOMA	1.1948 ± 0.0006	0.0072 ± 0.0004	0.0067 ± 0.0015	0.4s
4	14	NSGA-II	1.1947 ± 0.0002	0.0069 ± 0.0007	0.0069 ± 0.0006	0.4s
5	21	Constrained NSGA-II	1.1947 ± 0.0002	0.0069 ± 0.0007	0.0069 ± 0.0006	0.5s
6	35	R-NSGA-II	1.1948 ± 0.0002	0.0098 ± 0.0008	0.0072 ± 0.0003	2.6s
7	24	MOABC	1.1946 ± 0.0001	0.0117 ± 0.0024	0.0073 ± 0.0004	0.6s
8	26	MOSA	1.1947 ± 0.0001	0.0095 ± 0.0017	0.0074 ± 0.0004	0.7s
9	22	MOPSO	1.1944 ± 0.0003	0.0088 ± 0.0007	0.0085 ± 0.0009	0.8s
10	23	MOGWO	1.1928 ± 0.0009	0.0090 ± 0.0007	0.0113 ± 0.0020	0.6s
11	17	PESA-II	1.1926 ± 0.0008	0.0089 ± 0.0007	0.0117 ± 0.0014	1.6s
12	28	MO-TLBO	1.1933 ± 0.0008	0.0125 ± 0.0011	0.0125 ± 0.0015	0.3s
13	29	MOBA	1.1935 ± 0.0005	0.0088 ± 0.0011	0.0140 ± 0.0021	0.8s
14	12	NSGA	1.1927 ± 0.0009	0.0189 ± 0.0032	0.0153 ± 0.0026	0.2s
15	19	MOEA/D-DE	1.1926 ± 0.0001	0.0818 ± 0.0031	0.0648 ± 0.0044	5.6s
16	18	MOEA/D	1.1926 ± 0.0001	0.0829 ± 0.0006	0.0664 ± 0.0002	6.1s
17	15	NSGA-III	1.1926 ± 0.0000	0.0831 ± 0.0003	0.0664 ± 0.0003	0.7s
18	32	IBEA	1.1934 ± 0.0004	0.0295 ± 0.0130	0.1097 ± 0.0093	0.5s
19	34	HypE	1.1917 ± 0.0009	0.0162 ± 0.0063	0.1092 ± 0.0291	6.2s
20	11	MOGA	1.1914 ± 0.0012	0.0244 ± 0.0052	0.0173 ± 0.0027	0.2s
21	31	RVEA	1.1906 ± 0.0010	0.0230 ± 0.0033	0.1797 ± 0.0129	0.2s
22	27	MODE	1.1884 ± 0.0037	0.0405 ± 0.0228	0.0300 ± 0.0103	0.1s
23	13	NPGA	1.1874 ± 0.0088	0.0429 ± 0.0206	0.0187 ± 0.0047	0.4s
24	4	Weight Metric (p-norm, Tchebycheff)	1.1861 ± 0.0000	0.0735 ± 0.0000	0.1726 ± 0.0000	0.2s
25	1	Weighted Sum	1.1728 ± 0.0000	0.3477 ± 0.0001	0.1370 ± 0.0000	0.2s
26	3	Benson's Method	1.1593 ± 0.0159	0.0976 ± 0.0725	0.1171 ± 0.0241	0.1s
27	10	VEGA	1.1504 ± 0.0084	0.0141 ± 0.0068	0.0802 ± 0.0243	0.0s
28	2	ε-Constraint	1.1246 ± 0.0141	0.1174 ± 0.1141	0.0668 ± 0.0024	0.4s

```
P5 done in 464s
```





D1 explanation (write after reading the tables): group the ranking by family — indicator/decomposition/elitist methods are expected on top on P3; the sweeps (1–4) are limited by their number of points; weighted sum should lose on P3 (non-convex); on P5 methods with feasibility rules (21) or penalties should return only feasible points; VEGA/NPGA/MOGA/NSGA should trail (non-elitist).

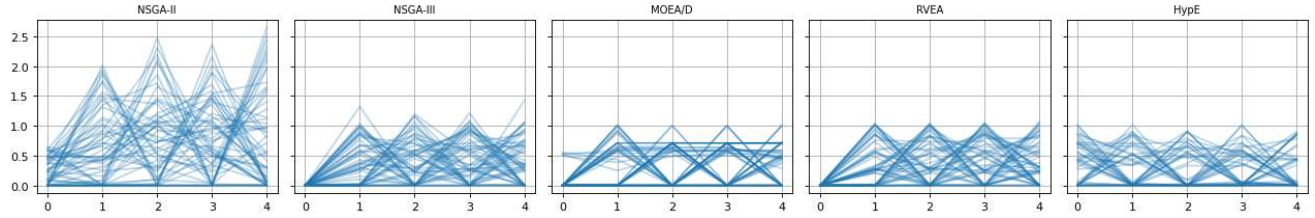
D2. Many-objective: P10 (M = 5) — NSGA-II, NSGA-III, MOEA/D, RVEA, HypE

```
D2_IDS = [14, 15, 18, 31, 34]
for i in D2_IDS: run_many(i, cls_runner(CLS[i]), "P10_5")
rtab("P10_5", D2_IDS, f"Ranking on P10 (M=5), budget {BUDGET_MANY}, {RUNS} seeds")
# why NSGA-II degrades: share of non-dominated individuals in a random population and in the NSGA-II population
rr = np.random.default_rng(0)
for Pm in (P10_3, P10_5): Fr = Pm.f(Pm.init(rr, 200))[0]; print(f"{Pm.name}: non-dominated share of 200 random points = {len(nd
Fn5 = RES[("P10_5", 14)]["front"]; print("NSGA-II final population on M=5: mean distance of its front to the true front (IGD) =
                                        "vs NSGA-III", f"{np.nanmean(RES[('P10_5', 15)]['igd']):.3f}")
# parallel-coordinate view of the M=5 fronts
fig, axs = plt.subplots(1, 5, figsize=(16, 2.8), sharey=True)
for ax, i in zip(axs, D2_IDS):
    F = RES[("P10_5", i)]["front"]
    for f in F[:60]: ax.plot(range(5), f, c="C0", alpha=.3)
    ax.set_title(NAME[i], fontsize=9)
plt.tight_layout(); plt.show()
```

Ranking on P10 (M=5), budget 10000, 10 seeds

Rank	#	Algorithm	HV (↑)	Spacing (↓)	IGD (↓)	Runtime
1	31	RVEA	1.2141 ± 0.0060	0.0504 ± 0.0025	0.2090 ± 0.0021	0.2s
2	15	NSGA-III	1.1622 ± 0.0131	0.0725 ± 0.0054	0.2219 ± 0.0036	1.1s
3	34	HypE	1.1489 ± 0.0100	0.1447 ± 0.0084	0.3449 ± 0.0104	7.4s
4	18	MOEA/D	1.1425 ± 0.0064	0.1124 ± 0.0203	0.3411 ± 0.0088	7.8s
5	14	NSGA-II	0.1233 ± 0.1076	0.1817 ± 0.0233	0.6099 ± 0.1149	0.5s

```
P10_3: non-dominated share of 200 random points = 0.29
P10_5: non-dominated share of 200 random points = 0.54
NSGA-II final population on M=5: mean distance of its front to the true front (IGD) = 0.610 vs NSGA-III 0.222
```



What degrades NSGA-II: with M = 5 almost every individual is non-dominated, so sorting stops discriminating and the crowding distance (sum of 1-D gaps) is dominated by boundary points and is a poor density measure in 5-D; selection pressure collapses and the population spreads without converging. NSGA-III/RVEA/MOEA/D replace it with reference points/vectors/weights that keep spread explicit; HypE keeps pressure through hypervolume credit.

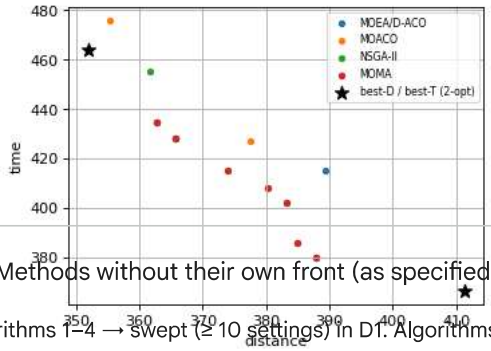
D3. Discrete: P8 — MOEA/D-ACO, MOACO, NSGA-II and MOMA (permutation operators)

```
D3_IDS = [20, 25, 14, 30]
for i in D3_IDS: run_many(i, cls_runner(CLS[i]), "P8")
rtab("P8", D3_IDS, f"Ranking on P8 (15-city bi-objective TSP), budget {BUDGET}, {RUNS} seeds")
print("Operators used by NSGA-II / MOMA on permutations: order crossover (OX) + segment-inversion mutation; MOMA adds inversion
fig, ax = plt.subplots(figsize=(5, 3.6))
for i in D3_IDS: R = P8.denorm(RES[("P8", i)]["front"]); ax.scatter(R[:, 0], R[:, 1], s=14, label=NAME[i])
ax.scatter(*np.array([[_e1[0], _e1[1]], [_e2[0], _e2[1]]]).T, c="k", marker="*", s=90, label="best-D / best-T (2-opt)"); ax.set
print("shortest-distance tour:", _e1.round(1), "| shortest-time tour:", _e2.round(1))
```

Ranking on P8 (15-city bi-objective TSP), budget 10000, 10 seeds

Rank	#	Algorithm	HV (↑)	Spacing (↓)	IGD (↓)	Runtime
1	30	MOMA	0.7557 ± 0.0009	0.1131 ± 0.0119	-	0.8s
2	14	NSGA-II	0.7542 ± 0.0038	0.1097 ± 0.0247	-	1.0s
3	25	MOACO	0.7166 ± 0.0336	0.1070 ± 0.0092	-	5.6s
4	20	MOEA/D-ACO	0.6379 ± 0.1362	0.1223 ± 0.0339	-	7.3s

```
Operators used by NSGA-II / MOMA on permutations: order crossover (OX) + segment-inversion mutation; MOMA adds inversion hill-cl
```




```
shortest-distance tour: [351.8 463.7] | shortest-time tour: [411.4 386.9]
```

D4. Methods without their own front (as specified in the PDF)

Algorithms 1–4 → swept (≥ 10 settings) in D1. Algorithms 5–7 → solutions and the inputs that produced them (printed in A2.5–A2.7, summarised here). Algorithms 8–9 → applied to the NSGA-II front below. Algorithms 36–37 → both task losses vs. fixed equal weights.

```
# --- 5-7: solution and the input that produced it ---
table([("5 Value Function", f"a = {a}", P4.f(value_function(P4, 0, (a,)))[0].round(4).tolist()) for a in (0.2, 0.5, 0.8)] +
      [("6 Goal Programming", f"targets (2400, 600), weights {w}", goal_programming(P11.lin, [2400, 600], w)[0].round(1).tolist
      [("7 Lexicographic", f"priority {o}", lexicographic(P11.lin, o).round(1).tolist()) for o in ([0, 1], [1, 0])], ["Method",
# --- 8-9: top-3 on the NSGA-II front of P5, then shift one weight by 0.1 ---
Fr5 = P5.denorm(RES[("P5", 14)]["front"]); Fr5 = Fr5[np.argsort(Fr5[:, 0])]; bn = np.array([False, False]); w_a, w_b = np.array
def top3(sc, hi): o = np.argsort(-sc if hi else sc)[:3]; return o.tolist()
rows = []
for nm, f_, hi in (("TOPSIS", lambda w: topsis(Fr5, w, bn), True), ("VIKOR v=0.5", lambda w: vikor(Fr5, w, bn, 0.5), False)):
    a, b_ = top3(f_(w_a), hi), top3(f_(w_b), hi); rows.append((nm, a, b_, "flips" if a != b_ else "stable"))
table(rows, ["Method", "Top-3 indices, w=(0.5,0.5)", "Top-3 indices, w=(0.6,0.4)", "Ranking"]); print("Top-1 designs (mass kg,
# --- 36-37: both task losses against fixed equal weights ---
table([("Equal weights (fixed)", *hE[-1].round(3)), ("36 MGDA", *hM[-1].round(3)), ("37 GradNorm", *hG[-1].round(3))], ["Methoc
print("Regression gradients are ~20x larger: equal weights let that task dominate; MGDA/GradNorm rebalance by gradient scale (s
```

Method	Input	Solution found
5 Value Function	a = 0.2	[[0.0512, -0.1413]]
5 Value Function	a = 0.5	[[0.0301, -0.1317]]
5 Value Function	a = 0.8	[[0.0158, -0.118]]
6 Goal Programming	targets (2400, 600), weights [1, 1]	[300.0, 200.0]
6 Goal Programming	targets (2400, 600), weights [0.0004166666666666667, 0.0016666666666666668]	[300.0, 200.0]
7 Lexicographic	priority [0, 1]	[300.0, 200.0]
7 Lexicographic	priority [1, 0]	[300.0, 200.0]

Method	Top-3 indices, w=(0.5,0.5)	Top-3 indices, w=(0.6,0.4)	Ranking
TOPSIS	[47, 48, 46]	[46, 47, 45]	flips
VIKOR v=0.5	[47, 48, 46]	[46, 47, 45]	flips